

Study Note

# AIS 31 version 3.0

RASE 류지은

# Content

|          |                                                          |          |
|----------|----------------------------------------------------------|----------|
| <b>4</b> | <b>Mathematical Background</b>                           | <b>3</b> |
| 4.2      | Probability, Stochastics, and Random Variables . . . . . | 3        |
| 4.2.1    | Definitions and Basic Concepts . . . . .                 | 3        |
| 4.2.2    | Useful Theorems and Facts . . . . .                      | 44       |
| 4.3      | Entropy and Guesswork . . . . .                          | 71       |
| 4.3.1    | Entropy . . . . .                                        | 71       |

## 4. Mathematical Background

### 4.2 Probability, Stochastics, and Random Variables

#### 4.2.1 Definitions and Basic Concepts

488

In the following,  $\Omega$  denotes a non-empty set.

489

In this document,  $\Omega$  usually represents the admissible 의도하여 허용되는 values of random numbers, random experiments, or measurements. Usually,  $\Omega$  is finite (typically,  $\Omega = \{0, 1\}^k$  or  $\Omega = \mathbb{Z}_n$ ) or it equals  $\mathbb{R}^m$  or a subset of  $\mathbb{R}^m$  ( $m \geq 1$ ).

Note 1: Experiments with finite  $\Omega$  are, for example, coin tosses and dice rolls. In the context of RNGs, random numbers are important examples that assume values in a finite or in a countable set  $\Omega$ , e.g.,  $\Omega = \{0, 1\}$  and  $\Omega = \mathbb{N}_0$ .

Note 2: Examples for  $\Omega \subseteq \mathbb{R}^m$  are timing measurements and voltage measurements.

490

$\mathcal{P}(\Omega)$  denotes the power set of  $\Omega$ . The power set contains all subsets of  $\Omega$ . If  $\Omega$  is finite then  $|\mathcal{P}(\Omega)| = 2^{|\Omega|}$ .

491

Paragraphs 492 to 501 contain basic definitions and facts from probability and measure theory, which will be needed below for proper definitions of independence or stationary stochastic processes, for example. However, these concepts are rather "technical". Paragraphs 503 to 506 provide a "light version" thereof, which should suffice to understand the subsequent definitions and concepts.



492

A  $\sigma$ -algebra  $\mathcal{A}$  **structural** over  $\Omega$  is a set of subsets of  $\Omega$ , i.e.,  $\mathcal{A} \subseteq \mathcal{P}(\Omega)$ , that fulfills the following conditions: **axiom**

- a  $\Omega \in \mathcal{A}$
- b If  $A \in \mathcal{A}$ , then also its complement  $A^c := \Omega \setminus A \in \mathcal{A}$
- c If  $A_1, A_2, \dots \in \mathcal{A}$  then  $\bigcup_{n \geq 1} A_n \in \mathcal{A}$

493

Note: Par. 492, Condition (c), includes finite sequences  $A_1, A_2, \dots, A_k$ . Note that such a finite sequence can formally be extended by  $A_{k+1} = A_{k+2} = \dots = \{\}$  to an infinite sequence with the same union set.

It has completeness in itself.  $\bigcap_{n \geq 1} A_n \in \mathcal{A}$  with De Morgan's laws by it.

494

Example:

- i  $\mathcal{P}(\Omega)$  is a  $\sigma$ -algebra over  $\Omega$ . **trivial**
- ii The Borel  $\sigma$ -algebra  $\mathcal{B}(\mathbb{R})$  over  $\mathbb{R}$  is the smallest  $\sigma$ -algebra that contains the open intervals (equivalently, that contains the open subsets of  $\mathbb{R}$ ).
- iii More generally, for  $m \geq 1$ , the Borel  $\sigma$ -algebra  $\mathcal{B}(\mathbb{R}^m)$  over  $\mathbb{R}^m$  is the smallest  $\sigma$ -algebra that contains the open subsets of  $\mathbb{R}^m$ .

Most of the sets known at the undergraduate level are the Borel  $\sigma$ -algebra.

495

A **probability measure** 측도: 확률, 길이, 부피 등의 기준에 따른 정도를 부여하는 것  $\nu$  on a  $\sigma$ -algebra  $\mathcal{A}$  is a **mapping**  $\nu : \mathcal{A} \rightarrow [0, 1]$  with the following properties: **axiom**

- 1.  $\nu(\Omega) = 1$
- 2. If the sets  $A_1, A_2, \dots \in \mathcal{A}$  are mutually disjoint,  $\nu\left(\bigcup_{n \geq 1} A_n\right) = \sum_{n \geq 1} \nu(A_n)$ .  
(The sequence  $A_1, A_2, \dots$  may be finite or countable.)

Since  $\mathcal{A}$  is a  $\sigma$ -algebra, the countable( $\infty$ ) union  $\bigcup_{n \geq 1} A_n$  belongs to  $\mathcal{A}$ , making the term  $\nu(\bigcup_{n \geq 1} A_n)$  well-defined.



496

More generally, if a mapping  $\nu : \mathcal{A} \rightarrow [0, \infty]$  fulfills Condition (b) from par. 495, and if  $\nu(\Omega) < \infty$ , we refer to  $\nu$  as a finite measure; otherwise,  $\nu$  is an infinite measure. If there exists a countable sequence  $C_1 \subseteq C_2 \subseteq C_3 \cdots \in \mathcal{A}$  such that  $\nu(C_n) < \infty$  for all  $n \in \mathbb{N}$  and  $\bigcup_{n \geq 1} C_n = \Omega$ , then  $\nu$  is a  $\sigma$ -finite measure.

497

Any  $A \in \mathcal{A}$  is said to be an event or a measurable 확률 공간에서는 event만 measure 가능함. 즉, measurable= 사건으로 볼 수 있는 set. A pair  $(\Omega, \mathcal{A})$  is denoted as a measurable space 아직 measure가 할당되지 않은 space, while the triple  $(\Omega, \mathcal{A}, \nu)$  is called a **measure space**. If  $\nu$  is a probability measure, the triple  $(\Omega, \mathcal{A}, \nu)$  is a **probability space**.

Probability space  $(\Omega, \mathcal{F}, \Pr) : (\text{sample space, event set, probability function})$

498

Example:

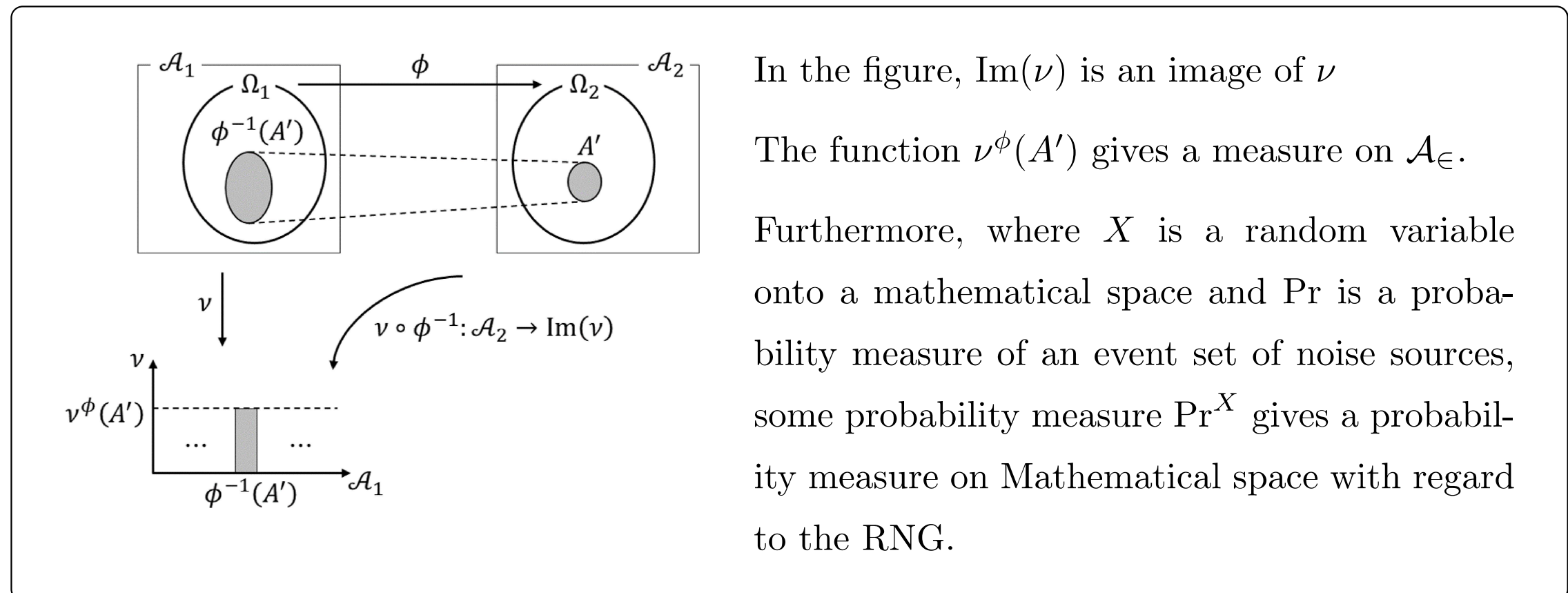
1. Let  $B(n, p)$  denote a binomial distribution with parameters  $n$  and  $p$ . Then  $B(n, p)$  is a probability measure on  $\mathcal{P}(\{0, \dots, n\})$ .
2. The Lebesgue measure  $\lambda$  is a  $\sigma$ -finite measure on  $\mathcal{B}(\mathbb{R})$ .  
Note: The Lebesgue measure corresponds to the "geometric" measure Euclid space의 subset에 길이/부피/너비 등을 할당하는 map on  $\mathbb{R}$ , i.e.,  $\lambda([a, b)) = b - a$  if  $a \leq b$ .
3. The standard normal distribution (standard Gaussian distribution)  $N(0, 1)$  is a probability measure on  $\mathcal{B}(\mathbb{R})$ .
4. The Lebesgue measure  $\lambda_m$  on  $\mathbb{R}^m$  is a  $\sigma$ -finite measure.

499

If there is no ambiguity 상황에 따라 판단 about the  $\sigma$ -algebra  $\mathcal{A}$ , we often loosely speak of "measures on  $\Omega$ ". Unless otherwise stated, in this document  $\mathcal{A} = \mathcal{P}(\Omega)$  for countable  $\Omega$  (finite or infinite), and for  $\mathbb{R}$ ,  $\mathbb{R}^m$ , and measurable subsets  $\Omega \subseteq \mathbb{R}^m$ , we use the Borel  $\sigma$ -algebras  $\mathcal{A} = \mathcal{B}(\mathbb{R})$ ,  $\mathcal{A} = \mathcal{B}(\mathbb{R}^m)$ , or  $\mathcal{A} = \mathcal{B}(\Omega)$ , respectively.

500

Assume that  $(\Omega_1, \mathcal{A}_1, \nu)$  is a probability space and that  $(\Omega_2, \mathcal{A}_2)$  is a measurable space. Furthermore, let  $\phi : \Omega_1 \rightarrow \Omega_2$  be a mapping. We call  $\phi$  **measurable** (or more precisely,  $(\mathcal{A}_1, \mathcal{A}_2)$ -measurable) if for each  $A' \in \mathcal{A}_2$  the pre-image  $\phi^{-1}(A') \in \mathcal{A}_1$ . If  $\nu$  is a measure on  $\mathcal{A}_1$ , then  $\nu^\phi(A') := \nu(\phi^{-1}(A'))$  for all  $A' \in \mathcal{A}_2$  defines a measure on  $\mathcal{A}_2$ . We denote  $\nu^\phi$  the image measure (or: transformed measure) of  $\nu$  under  $\phi$ .



501

Assume that  $\mathcal{A}_1$  and  $\mathcal{A}_2$  are  $\sigma$ -algebras over  $\Omega_1$  and  $\Omega_2$ . A **random variable**  $X$  is a measurable mapping  $X : \Omega_1 \rightarrow \Omega_2$ . In our context, usually  $\Omega_2$  is finite, countable, or a subset of  $\mathbb{R}^m$ .

502

Outside of mathematical proofs, the probability space of a random variable is usually not explicitly stated. We point out that a random variable  $X : \Omega_1 \rightarrow \Omega_2$  with probability space  $(\Omega_1, \mathcal{A}_1, \nu)$  can also be interpreted as a random variable on the measure space  $(\Omega_2, \mathcal{A}_2, \nu^X)$ . Here,  $\nu^X$  denotes the image measure (or: transformed measure) of  $X$ , i.e.,  $\nu^X(A_2) = \nu(X^{-1}(A_2))$  for all  $A_2 \in \mathcal{A}_2$ .

The last sentence of Par. 502 "Furthermore,  $\text{Prob}(X \in A_1) = \nu(A_1)$  quantifies the probability that the random variable  $X$  assumes a value in  $A_1$ ." seems to need revision for the equation to be "Prob( $X \in A_2$ ) = Prob( $X^{-1} \in A_1$ ) =  $\nu(A_1)$ ."

503

[”light version” of pars. 492 to 501] As already mentioned above, these definitions and concepts are needed for mathematically precise definitions in the following. Fortunately, in the context of RNG evaluations, problems concerning measurability hardly occur. The paragraphs 504 to 506 thus provide a ”light version”. This light version should suffice for at least an intuitive understanding of the following definitions and concepts and to apply them correctly. This, in particular, refers to the material collected in Subsection 4.2.2.

504

[”light version” of pars. 492 to 501 ctd.] Some of the following definitions and conditions refer to ”measurable subsets” of some space  $\Omega$  (equivalently, to elements of a  $\sigma$ -algebra on  $\Omega$ ). If  $\Omega$  is finite or countable, all subsets of  $\Omega$  are measurable (to be precise: with regard to the  $\sigma$ -algebra  $\mathcal{P}(\Omega)$ ). If  $\Omega \subseteq \mathbb{R}^m$  one may think of ”regular” subsets as (depending on the dimension  $m$ ) intervals, rectangles, circles, cuboids, balls, etc. and countable unions thereof. (There exist further measurable and non-measurable subsets, but this should be of little importance for RNG evaluations.)

505

[”light version” of pars. 492 to 501 ctd.] In this document and, more generally, in the context of the evaluation of RNGs, random variables usually assume values in finite or countable sets or in subsets of  $\mathbb{R}$  or  $\mathbb{R}^m$ . We may speak of random variables on finite or countable set  $\Omega$  (e.g.,  $\Omega = \{0, 1\}$ ), or random variables on  $\mathbb{R}$  (also: ”real-valued random variables”), random variables on  $\mathbb{R}^m$ , or random variables on  $\Omega$ .

506

[”light version” of pars. 492 to 501 ctd.] The expression  $\text{Prob}(X \in A)$  quantifies the probability that the random variable  $X$  assumes a value in the set  $A \subseteq \Omega$ .

507

$X \sim \nu$  means that the random variable  $X$  has distribution  $\nu$ , i.e., that  $\text{Prob}(X \in A) = \nu(A)$ . The term  $\text{Prob}(X \in A)$  quantifies the probability that  $X$  assumes a value in the set  $A$ . Values that are assumed (or: taken on) by a random variable  $X$  are called realizations 시행, 측정(evaluation)을 통해 얻은 결과 of  $X$ .



508

[Notation] In this document, we denote random variables by capital letters and their realizations usually by the corresponding small letters.

509

Example: Assume that the random variable  $X$  models the tossing of a fair coin. Then  $\text{Prob}(X = 0) = \text{Prob}(X = 1) = 0.5$  if we identify "head" and "tail" with 1 and 0. These probabilities quantify the knowledge on the outcome of a future coin toss (and on a past experiment to a person who does not know its outcome). Possible realizations of  $X$  are 0 and 1.

510

In this document, we model non-deterministic phenomena by random variables. Their realizations are observable as random numbers, voltage, or timing, for example.

511

Definition: The term  $B(n, p)$  denotes the **binomial distribution** with parameters  $n$  and  $p$ , which is given by

$$\text{Prob}(X = k) = \binom{n}{k} p^k (1 - p)^{n-k} \quad \text{for } k = 0, \dots, n. \quad (4.1)$$

Definition: The Poisson distribution with parameter  $\tau > 0$  is given by

$$\text{Prob}(X = k) = \frac{\tau^k}{k!} e^{-\tau} \quad \text{for } k \in \mathbb{N}_0. \quad (4.2)$$

Note: The parameter  $\tau > 0$  equals the mean number of events per time interval of length 1.

The Poisson distribution is a special case of the binomial distribution with  $p = \frac{\tau}{n}$  where  $n \rightarrow \infty$ , for the number of repetitions  $k$ , i.e.,  $\lim_{n \rightarrow \infty} \binom{n}{k} \left(\frac{\tau}{n}\right)^k \left(1 - \frac{\tau}{n}\right)^{n-k}$ .

513

Definition: The geometric대수학의 geometric mean 등과 관련됨 distribution  $\mathcal{G}_p$  with parameter  $p \in (0, 1]$  is a discrete distribution on  $\mathbb{N}$ . More precisely,

$$\mathcal{G}_p(k) := p(1-p)^{k-1} \quad \text{for } k \in \mathbb{N}. \quad (4.3)$$

The term  $\mathcal{G}_p(k)$  equals the probability that a sequence of iid Bernoulli trials with individual success probability  $p$  is successful for the first time in the  $k^{th}$  trial.

Note: In the literature, there also exists an alternative definition of the geometric distribution that only counts the number of failures, i.e.,  $k-1$  in place of  $k$ .

514

Definition: The letters  $\lambda$  and  $\lambda_m$  denote the Lebesgue measures on  $\mathbb{R}$  or  $\mathbb{R}^m$ , respectively. It is  $\lambda([a, b)) = b-a$  if  $a \leq b$ . Accordingly,  $\lambda_m\left(\prod_{j=1}^m [a_j, b_j)\right) = \prod_{j=1}^m (b_j - a_j)$  if  $a_j \leq b_j$  for  $1 \leq j \leq m$ .

Note: The Lebesgue measure  $\lambda$  corresponds to the “geometricEuclid geometry와 관련됨” measure on  $\mathbb{R}$  (not to be mixed up with the geometric distribution defined in par. 513).

515

Definition: The term  $N(\mu, \sigma^2)$  denotes the **normal (Gaussian) distribution** with expectation  $\mu$  and variance  $\sigma^2$ . It has the density

$$\phi(x) := \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}. \quad (4.4)$$

Induction of the density

- Preliminary 1

$n! \approx \sqrt{2\pi n} \left(\frac{n}{e}\right)^n$  by Stirling's approximation for large  $n$ .

- Preliminary 2

$$\begin{aligned} \ln \frac{n}{2} \left(1 + \frac{2\delta}{n}\right) &= \ln \frac{n}{2} + \ln \left(1 + \frac{2\delta}{n}\right) \\ &\approx \ln \frac{n}{2} + \left(\frac{2\delta}{n} - \frac{1}{2} \left(\frac{2\delta}{n}\right)^2\right) \end{aligned} \quad \text{by Taylor Series approx.}$$

Similarly,  $\ln \frac{n}{2} \left(1 + \frac{2\delta}{n}\right) \approx \ln \frac{n}{2} - \left(\frac{2\delta}{n} + \frac{1}{2} \left(\frac{2\delta}{n}\right)^2\right)$ .



- Preliminary 3

For given  $R = \frac{n^n}{\left(\frac{n}{2} + \delta\right)^{n/2 + \delta} \left(\frac{n}{2} - \delta\right)^{n/2 - \delta} 2^n}$ ,

$$\begin{aligned}
 \ln R &= n \ln n - \left(\frac{n}{2} + \delta\right) \ln \left(\frac{n}{2} + \delta\right) \\
 &\quad - \left(\frac{n}{2} - \delta\right) \ln \left(\frac{n}{2} - \delta\right) - n \ln 2 \\
 &\approx n \ln n - \left(\frac{n}{2} + \delta\right) \left( \ln \frac{n}{2} + \frac{2\delta}{n} - \frac{1}{2} \left(\frac{2\delta}{n}\right)^2 \right) \\
 &\quad - \left(\frac{n}{2} - \delta\right) \left( \ln \frac{n}{2} - \frac{2\delta}{n} - \frac{1}{2} \left(\frac{2\delta}{n}\right)^2 \right) - n \ln 2 \quad \text{by Pre.2} \\
 &= n \ln \frac{n}{2} - \frac{n}{2} \ln \frac{n}{2} + \frac{n}{4} \left(\frac{2\delta}{n}\right)^2 - \delta \left(\frac{2\delta}{n}\right) \\
 &\quad - \frac{n}{2} \ln \frac{n}{2} + \frac{n}{4} \left(\frac{2\delta}{n}\right)^2 - \delta \left(\frac{2\delta}{n}\right) \\
 &= \frac{n}{2} \left(\frac{2\delta}{n}\right)^2 - \frac{(2\delta)^2}{n} \\
 &= -\frac{(2\delta)^2}{2n}.
 \end{aligned}$$

$$\therefore R \approx \exp(-2\delta^2/n).$$

For the fair coin tossing,  $\Pr[X = x] = \binom{n}{x} \left(\frac{1}{2}\right)^n = \frac{n!}{x!(n-x)!} \left(\frac{1}{2}\right)^n$ .

Let  $\mu = \mathbb{E}X = \frac{n}{2}$  and  $\delta := x - \frac{n}{2}$ .

Then  $x = \frac{n}{2} + \delta$ ,  $n - x = \frac{n}{2} - \delta$ .

$$\begin{aligned}
 \Pr[X = x] &= \frac{n!}{x!(n-x)!} \left(\frac{1}{2}\right)^n \\
 &= \frac{\sqrt{2\pi n} \left(\frac{n}{e}\right)^n}{\sqrt{2\pi x} \left(\frac{x}{e}\right)^x \sqrt{2\pi(n-x)} \left(\frac{n-x}{e}\right)^{(n-x)}} \left(\frac{1}{2}\right)^n \quad \text{by Pre.1} \\
 &= \frac{\sqrt{n} \cdot n^n}{\sqrt{2\pi x} \cdot x^x \sqrt{n-x} \cdot (n-x)^{(n-x)}} \left(\frac{1}{2}\right)^n \\
 &= \frac{\sqrt{n}}{\sqrt{2\pi x(n-x)}} \cdot \frac{n^n}{x^x \cdot (n-x)^{(n-x)} \cdot 2^n} \\
 &= \frac{\sqrt{n}}{\sqrt{2\pi \left(\frac{n}{2} + \delta\right) \left(\frac{n}{2} - \delta\right)}} \cdot \frac{n^n}{\left(\frac{n}{2} + \delta\right)^{n/2 + \delta} \cdot \left(\frac{n}{2} - \delta\right)^{n/2 - \delta} \cdot 2^n} \\
 &= \frac{\sqrt{n}}{\sqrt{\frac{n^2}{2} \pi \left(1 + \frac{2\delta}{n}\right) \left(1 - \frac{2\delta}{n}\right)}} \cdot \frac{n^n}{\left(\frac{n}{2} + \delta\right)^{n/2 + \delta} \cdot \left(\frac{n}{2} - \delta\right)^{n/2 - \delta} \cdot 2^n}
 \end{aligned}$$

$$\begin{aligned} &\approx \frac{1}{\sqrt{\frac{n\pi}{2} \left(1 - \left(\frac{2\delta}{n}\right)^2\right)}} \cdot \exp(-2\delta^2/n) && \text{by Pre.3} \\ &\approx \sqrt{\frac{2}{n\pi}} \cdot \exp(-2\delta^2/n) && \text{for } \delta \ll n \end{aligned}$$

Let  $\sigma^2 = \frac{n}{4}$ .

Since  $\delta = x - \frac{n}{2} = x - \mu$ ,

$$\begin{aligned} \Pr[X = x] &= \sqrt{\frac{2}{n\pi}} \cdot \exp(-2\delta^2/n) \\ &= \sqrt{\frac{2}{4 \cdot \frac{n}{4}\pi}} \cdot \exp(-2(x - \mu)^2/(4 \cdot n/4)) \\ &= \sqrt{\frac{1}{2\sigma^2\pi}} \cdot \exp(-(x - \mu)^2/(2\sigma^2)) \end{aligned}$$

In particular,  $N(0, 1)$  is called **standard normal distribution**. Its cumulative distribution function  $\Phi(\cdot)$  is given by

$$\Phi(x) := \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x e^{-\frac{t^2}{2}} dt. \quad (4.5)$$

Note 1: To be precise, “density” means “Lebesgue density”. In this document, densities with respect to other measures than the (one-dimensional or multidimensional) Lebesgue measure are not relevant. For this reason, we briefly speak of “density” in place of “Lebesgue density” in the following.

Note 2: Normal distributions exist in  $\mathbb{R}^k$  for each  $k \geq 1$  (multivariate normal distributions); cf. par. 542.

516

Definition: The **Gamma distribution** with the shape parameter  $\alpha > 0$  and rate parameter  $\beta > 0$  has the density

$$\gamma_{\alpha,\beta}(x) := \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x} \quad \text{for } x > 0. \quad (4.6)$$

Gamma function  $\Gamma(\alpha) := \int_0^\infty x^{\alpha-1} e^{-x} dx$  ( $\alpha > 0$ ) is defined analytic continuation 2!과 3!의 사이인 2.5! 등의 of this integral function. It offers a useful property from its integral, which takes the form of (polynomial) \* (exponential).

The gamma distribution is a Conjugate prior distribution.

Note: Occasionally, the Gamma distribution is not characterized by a shape parameter and a rate parameter but by a shape parameter and a scale parameter. Thus, caution is advised when results from different books and papers are applied. The scale parameter is the reciprocal value of the shape parameter.

517

The random variable  $X$  is called discrete if  $\Omega$  is countable (finite or infinite). If  $\Omega$  is finite, we also call  $X$  a finite random variable. Examples are binomially distributed random variables and Poisson-distributed random variables. Section 4.4 treats given random mappings. There, the realizations of the random variables are mappings between sets.

518

Let  $X$  be a random variable that assumes values in a finite set  $\Omega$ . We say that  $X$  is **uniformly distributed** (or equivalently: unbiased, equidistributed) if it assumes all  $\omega \in \Omega$  with the same probability, namely  $\text{Prob}(X = \omega) = |\Omega|^{-1}$ . Otherwise,  $X$  is said to be biased.

Note: Precisely formulated, it should actually read  $\text{Prob}(X = \{\omega\})$  instead of  $\text{Prob}(X = \omega)$ . However, the shorter expression “ $\text{Prob}(X = \omega)$ ” is common for finite and countable  $\Omega$ .

519

A random variable  $X$  has density  $f : \Omega \rightarrow [0, \infty]$  with respect to a measure  $\tau$  if  $\text{Prob}(X \in A) = \int_A f(\omega) d\tau(\omega)$  for all measurable sets  $A$ . Equivalently, a measure  $\nu$  has density  $f : \Omega \rightarrow [0, \infty]$  with respect to a measure  $\tau$  if  $\nu(A) = \int_A f(\omega) d\tau(\omega)$  for all measurable sets  $A$ .

Note: Densities do not exist for each pair of measures  $(\nu, \tau)$ .  $\nu$ 는 point 기준,  $\tau$ 는 length 기준일 때  $p_\nu(x) = 1 \neq p_\tau(x) = 0$



520

In our context, usually  $\Omega \subseteq \mathbb{R}^m$  with  $m \geq 1$ , and  $\tau = \lambda_m$ . Then  $\lambda(dx) = \lambda(x-x') = -x-x' = -dx$   
 이므로

$$\text{Prob}(X \in A) = \int_A f(x) \lambda_m(dx) = \int_A f(x) dx. \quad (4.7)$$

521

Let  $X$  denote a random variable that assumes values in  $\mathbb{R}^m$  and has distribution  $\nu$ . If the integral

$$E(X) := \int_{\Omega} x \nu(dx) \quad (4.8)$$

exists (i.e., if  $\int_{\Omega} |x| \nu(dx) < \infty$ ), then  $E(X)$  is called the expectation of  $X$ .

Note: The expectation  $E(X)$  does not exist for every random variable. Counterexamples are, for example, Cauchy-distributed 꼬리가 두꺼운 Bell-shape 분포로 적분 결과가 발산함 random variables.

522

For discrete random variables  $X$  with values in  $\Omega \subseteq \mathbb{R}$  (e.g.,  $\Omega = \{0, 1\}$ ,  $\mathbb{N}$ , or  $\mathbb{Z}$ ), formula (4.8) simplifies to

$$E(X) := \sum_{x \in \Omega} x \operatorname{Prob}(X = x). \quad (4.9)$$

For any random variables  $X$  and  $Y$  regardless of independence,

$$\begin{aligned} E(X + Y) &= \sum_x \sum_y (x + y) P(X = x, Y = y) \\ &= \sum_x \sum_y x P(X = x, Y = y) + \sum_x \sum_y y P(X = x, Y = y) \\ &= \sum_x x \left( \sum_y P(X = x, Y = y) \right) + \sum_y y \left( \sum_x P(X = x, Y = y) \right) \\ &= \sum_x x P(X = x) + \sum_y y P(Y = y) \\ &= E(X) + E(Y) \end{aligned}$$

It can be generalized to  $E(\sum_{i=1}^n X_i) = \sum_{i=1}^n E(X_i)$ .

If  $X$  and  $Y$  are independent, then  $P(X = x, Y = y) = P(X = x)P(Y = y)$ . Thus,

$$\begin{aligned} E(XY) &= \sum_x \sum_y xy P(X = x, Y = y) \\ &= \sum_x \sum_y xy P(X = x) P(Y = y) \\ &= \left( \sum_x x P(X = x) \right) \left( \sum_y y P(Y = y) \right) \\ &= E(X)E(Y) \end{aligned}$$

If  $X$  assumes values in  $\mathbb{R}^m$  and has Lebesgue density  $f$  then (4.8) reads

$$E(X) := \int_{\mathbb{R}^m} x f(x) dx. \quad (4.10)$$

In the context of **PTRNG** evaluations, we are usually faced with these two special cases.

523

Note: For random variables with values in  $\{0, 1\}^n$ , no meaningful definition for the mean is evident.

524

The variance of a real-valued random variable  $X$  is defined by

$$\text{Var}(X) := E(E(X) - X)^2. \quad (4.11)$$

provided that both expectations exist. This is not always the case. Cauchy-dist.는 EX가 없으므로 Var도 없음

$$\begin{aligned} \text{Var} \left( \sum_{i=1}^n X_i \right) &= E \left( E \sum_{i=1}^n X_i - \sum_{i=1}^n X_i \right)^2 \\ &= E \left( \mu - \sum_{i=1}^n X_i \right)^2 \\ &= E \left[ \mu^2 - 2\mu \sum_{i=1}^n X_i + \left( \sum_{i=1}^n X_i \right)^2 \right] \\ &= \mu^2 - 2\mu E \sum_{i=1}^n X_i + E \left( \sum_{i=1}^n X_i \right)^2 \\ &= (EX)^2 - 2(EX)^2 + E \left( \sum_{i=1}^n X_i \right)^2 \\ &= -(EX)^2 + EX^2 \end{aligned}$$

525

Assume that  $\text{Var}(X)$  exists. Then

$$\sigma_X := \sqrt{\text{Var}(X)} \quad (4.12)$$

is the standard deviation of  $X$ .



[sum of normal distributions] If  $X_1$  and  $X_2$  denote **independent** normally distributed random variables with expectations  $\mu_1, \mu_2$  and variances  $\sigma_1^2, \sigma_2^2$ , then  $X_1 + X_2$  is normally distributed with expectation  $\mu_1 + \mu_2$  and variance  $\sigma_1^2 + \sigma_2^2$ . More generally, if the random variables  $X_1, \dots, X_n$  are iid  $N(\mu, \sigma^2)$ -distributed then the sum  $X_1 + \dots + X_n$  is  $N(n\mu, n\sigma^2)$ -distributed.

$$\begin{aligned}
 E\left(\sum X_i\right) &= \sum E(X_i) = n\mu \\
 E\left(\sum_{i=1}^n X_i\right)^2 &= E\left(\sum_{i=1}^n X_i^2 + \sum_{i \neq j} X_i X_j\right) = \sum_{i=1}^n E X_i^2 + \sum_{i \neq j} E(X_i X_j) \\
 \text{Var}\left(\sum_{i=1}^n X_i\right) &= E\left(\sum_{i=1}^n X_i - E \sum_{i=1}^n X_i\right)^2 \\
 &= E\left(\sum_{i=1}^n (X_i - EX_i)\right)^2 \\
 &= E\left(\sum_{i=1}^n (X_i - \mu_i)\right)^2 \\
 &= E\left(\sum_{i=1}^n (X_i - \mu_i)^2 + \sum_{i \neq j} (X_i - \mu_i)(X_j - \mu_j)\right) \\
 &= \sum_{i=1}^n E(X_i - \mu_i)^2 + \sum_{i \neq j} E[(X_i - \mu_i)(X_j - \mu_j)] \\
 &= \sum_{i=1}^n \text{Var}(X_i) + \sum_{i \neq j} \text{Cov}(X_i, X_j)
 \end{aligned}$$

If  $X_1, \dots, X_n$  are independent,

$$\text{Var}\left(\sum_{i=1}^n X_i\right) = \sum_{i=1}^n \text{Var}(X_i) = \sum_{i=1}^n \sigma^2 = n\sigma^2$$

because  $\text{Cov}(X_i, X_j) = 0$  for all  $i, j$  s.t.  $i \neq j$ .

527

[Gamma distribution] The Gamma distribution with the shape parameter  $\alpha > 0$  and rate parameter  $\beta > 0$  has the density  $\gamma_{\alpha,\beta}(\cdot)$ , cf. par. 516. A random variable that is Gamma distributed with parameters  $\alpha$  and  $\beta$  has mean  $\mu = \alpha/\beta$  and variance  $\sigma^2 = \alpha/\beta^2$ .

$$\Gamma(\alpha) = \int_0^\infty x^{\alpha-1} e^{-x} dx \quad \Gamma(\alpha + 1) = \alpha \Gamma(\alpha) \quad f(x; \alpha, \beta) = \frac{\beta^\alpha x^{\alpha-1} e^{-\beta x}}{\Gamma(\alpha)}, \quad x > 0$$

$$\begin{aligned} \mu = E(X) &= \frac{\alpha}{\beta} \int_0^\infty x f(x; \alpha, \beta) dx \\ &= \int_0^\infty x \frac{\beta^\alpha x^{\alpha-1} e^{-\beta x}}{\Gamma(\alpha)} dx \\ &= \int_0^\infty \frac{\beta^\alpha x^\alpha e^{-\beta x}}{\Gamma(\alpha)} dx \\ &= \frac{\alpha}{\beta} \int_0^\infty \frac{\beta^{\alpha+1} x^\alpha e^{-\beta x}}{\Gamma(\alpha + 1)} dx \quad (\because \Gamma(\alpha + 1) = \alpha \Gamma(\alpha)) \\ &= \frac{\alpha}{\beta} \int_0^\infty f(x; \alpha + 1, \beta) dx \\ &= \frac{\alpha}{\beta} \end{aligned}$$

$$\begin{aligned} E(X^2) &= \int_0^\infty x^2 \frac{\beta^\alpha x^{\alpha-1} e^{-\beta x}}{\Gamma(\alpha)} dx \\ &= \frac{\alpha(\alpha + 1)}{\beta^2} \int_0^\infty \frac{\beta^{\alpha+2} x^{\alpha+1} e^{-\beta x}}{\Gamma(\alpha + 2)} dx \\ &= \frac{\alpha(\alpha + 1)}{\beta^2} \int_0^\infty f(x; \alpha + 2, \beta) dx \\ &= \frac{\alpha(\alpha + 1)}{\beta^2} \\ &= \frac{\alpha^2}{\beta^2} + \frac{\alpha}{\beta^2} \end{aligned}$$

$$\sigma^2 = \text{Var}(X) = EX^2 - (EX)^2 = \left( \frac{\alpha^2}{\beta^2} + \frac{\alpha}{\beta^2} \right) - \left( \frac{\alpha}{\beta} \right)^2 = \frac{\alpha}{\beta^2}$$

528

[sum of Gamma distributions] If  $X$  and  $Y$  are **independent** random variables with densities  $\gamma_{\alpha_1,\beta}(\cdot)$  and  $\gamma_{\alpha_2,\beta}(\cdot)$ , respectively, then  $X + Y$  is Gamma-distributed with density  $\gamma_{\alpha_1+\alpha_2,\beta}(\cdot)$ . Consequently, if the random variables  $X_1, \dots, X_n$  are iid Gamma distributed with parameters  $\alpha$  and  $\beta$  then the sum  $X_1 + \dots + X_n$  is Gamma distributed with parameters  $n\alpha$  and  $\beta$ .

529

The random variables  $X_1, X_2, \dots, X_k$  are said to **be independent** if for each  $k$ -tuple  $(A_1, \dots, A_k)$  of measurable sets **the equality**

$$\text{Prob}(X_1 \in A_1, \dots, X_k \in A_k) = \prod_{j=1}^k \text{Prob}(X_j \in A_j). \quad (4.13)$$

**holds.**

530

More generally, the (infinite) sequence  $X_1, X_2, \dots$  of random variables is said to be independent if for each integer  $k' \geq 1$  and for each  $k'$ -tuple  $(A_1, \dots, A_{k'})$  **not distinct** of measurable sets, condition (4.13) is valid (with  $k'$  in place of  $k$ ).

Note: Independence can be generalized to uncountable index sets.

531

For discrete random variables  $X_1, X_2, \dots$  with values in  $\Omega$ , condition (4.13) simplifies to

$$\text{Prob}(X_1 = x_1, \dots, X_k = x_k) = \prod_{j=1}^k \text{Prob}(X_j = x_j) \quad (4.14)$$

for each  $k$ -tuple  $(x_1, \dots, x_k) \in \Omega^k$ . **xi={xi} in Omega**

532

In the context of random variables  $X_1, X_2, \dots$ , the abbreviation **iid** stands for “**independent and identically distributed.**”

533

Mathematically, **a sequence of iid uniformly distributed random variables  $X_1, X_2, \dots$  on a finite set  $\Omega$  (e.g.,  $\Omega = \{0, 1\}$ ) describes an ideal RNG.**



534

Assume that the random variables  $X_1, X_2, \dots, X_n$ , resp.  $X_1, X_2, \dots$  are independent. If  $X_j \sim \nu_j$ , the joint distribution of  $(X_1, X_2, \dots, X_n)$ , resp. of the sequence  $X_1, X_2, \dots$  is given by the product measure  $\otimes_{j=1}^n \nu_j$ , resp. by  $\otimes_{j=1}^\infty \nu_j$ . These product measures are characterized by the conditions from pars. 529 and 530. If the random variables  $X_1, X_2, \dots$  are identically distributed, i.e., if  $\nu_1 = \nu_2 = \dots = \nu_n$ , we alternatively also use the notation  $\nu^n$  and  $\nu^\mathbb{N}$ .

For the joint distribution

$$1. \sum_x \sum_y \Pr(X = x, Y = y) = 1$$

$$\begin{aligned} \Pr[X = x \text{ and } Y = y] &= \Pr[X = x|Y = y] \Pr[Y = y] \\ &= \Pr[Y = y|X = x] \Pr[X = x] \end{aligned}$$

$$2. \int_x \int_y f_{x,y} \, dy dx = 1$$

$$\begin{aligned} f_{x,y}(X = x \text{ and } Y = y) &= f_{x,y}(X = x|Y = y) f_y(Y = y) \\ &= f_{x,y}(Y = y|X = x) f_x(X = x) \end{aligned}$$

535

Assume that for the real-valued random variables  $X$  and  $Y$ , expectations and variances exist. Then the right-hand sides of (4.15) and (4.16) exist

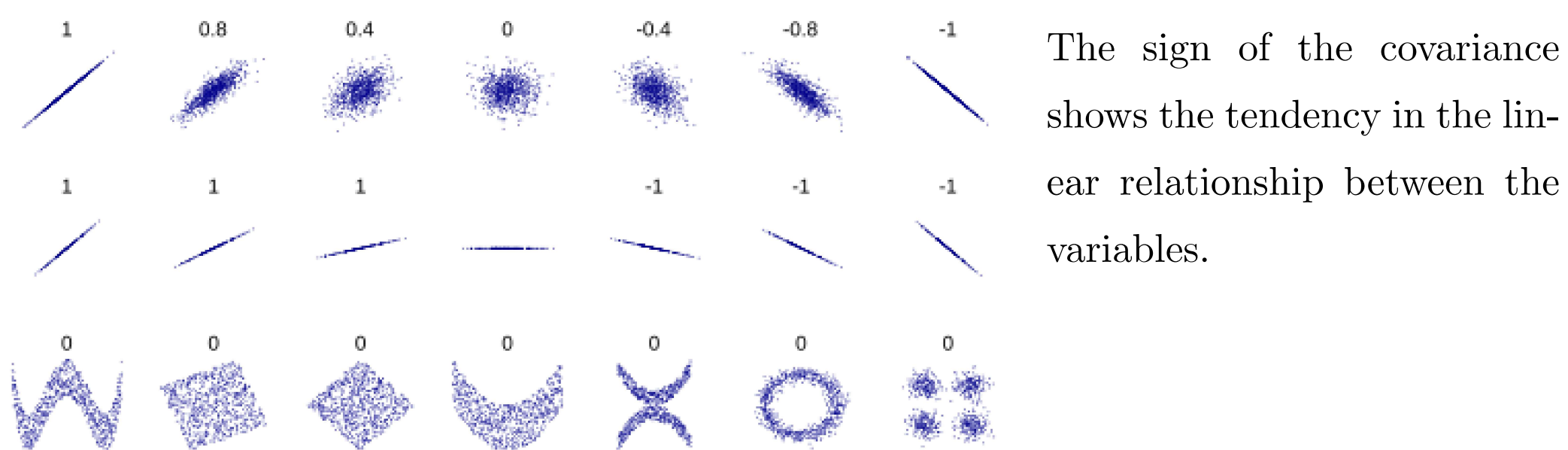
$$\text{Cov}(X, Y) := E(XY) - E(X)E(Y) \quad (\text{covariance}) \quad (4.15)$$

$$\begin{aligned} \text{Cov}(X, Y) &= E[(X - E(X))(Y - E(Y))] \\ &= E[(X - \mu_X)(Y - \mu_Y)] \\ &= E(XY - \mu_X Y - \mu_Y X + \mu_X \mu_Y) \\ &= E(XY) - \mu_X E(Y) - \mu_Y E(X) + \mu_X \mu_Y \\ &= E(XY) - \mu_X \mu_Y - \mu_X \mu_Y + \mu_X \mu_Y \\ &= E(XY) - \mu_X \mu_Y \\ &= E(XY) - E(X)E(Y) \end{aligned}$$

$$\text{corr}(X, Y) := \frac{\text{Cov}(X, Y)}{\sigma_X \cdot \sigma_Y} = \frac{E(XY) - E(X)E(Y)}{\sqrt{\text{Var}(X)}\sqrt{\text{Var}(Y)}} \quad (\text{correlation coefficient}) \quad (4.16)$$

Cov=0일 때가 있어서 일반적으로 sqrt(Cov)는 정의하지 않음

If  $\text{Cov}(X, Y) = 0$ , we say that  $X$  and  $Y$  are uncorrelated.



536

Independence implies uncorrelatedness but, in general, the converse is not true (cf. pars. 537 and 543).

If  $X, Y$  are independent,  $\text{Cov}(X, Y) = 0$  because  $E(XY) = E(X)E(Y)$ .

537

Counterexample ([Geor15], Beispiel (4.26)): Assume that  $X$  and  $Y$  are random variables that assume values in  $\Omega_1 = \{-1, 0, 1\}$  and in  $\Omega_2 = \{0, 1\}$ , respectively. Assume further that  $\text{Prob}(X = 1, Y = 0) = \text{Prob}(X = 0, Y = 1) = \text{Prob}(X = -1, Y = 0) = 1/3$ . Hence,  $\text{Prob}(X = 0) = \text{Prob}(X = 1) = \text{Prob}(X = -1) = 1/3$  and thus  $E(X) = 0$ . Similarly,  $\text{Prob}(Y = 0) = 2/3$ ,  $\text{Prob}(Y = 1) = 1/3$  and thus  $E(Y) = 1/3$ . Finally,

$$\begin{aligned} \text{Cov}(X, Y) &= E(XY) - 0 \cdot \frac{1}{3} = \sum_{x \in \Omega_1, y \in \Omega_2} xy \text{Prob}(X = x, Y = y) - 0 \\ &= \left(1 \cdot 0 \cdot \frac{1}{3} + 0 \cdot 1 \cdot \frac{1}{3} - 1 \cdot 0 \cdot \frac{1}{3}\right) = 0. \end{aligned}$$

Thus,  $X$  and  $Y$  are uncorrelated, but since  $\text{Prob}(X = 1, Y = 1) = 0 \neq 1/9 = \text{Prob}(X = 1) \cdot \text{Prob}(Y = 1)$ , the random variables  $X$  and  $Y$  are not independent.

538

**[Strong law of large numbers]** Assume that the random variables  $X_1, X_2, \dots$  have expectations  $E(X_1), E(X_2), \dots$ . We say that the sequence  $X_1, X_2, \dots$  satisfies the strong law of large numbers if

$$\text{Prob} \left( \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{j=1}^N (X_j - E(X_j)) = 0 \right) = 1. \quad (4.17)$$

If the random variables  $X_1, X_2, \dots$  are identically distributed with expectation  $E(X_j) = \mu$ , (4.17) simplifies to

$$\text{Prob} \left( \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{j=1}^N X_j = \mu \right) = 1. \quad (4.18)$$

**[Chebyshev inequality]**

Assume there exists  $E(|X|^p)$  for r.v.  $X$ . If  $X \geq 0$  and  $a > 0$ ,

$$\Pr(|X| \geq a) \leq \frac{1}{a^p} E(|X|^p)$$

Proof. Let  $X$  be a random variable and  $a \in \mathbb{R}^+$ . We assume  $X$  has a density function  $f_X$ .

Then

$$\begin{aligned} E(X^2) &= \int_{\mathbb{R}} x^2 f_X(x) dx \geq \int_{|x| \geq a} x^2 f_X(x) dx \\ &\geq a^2 \int_{|x| \geq a} f_X(x) dx = a^2 \Pr(|X| \geq a). \end{aligned}$$



That is,

$$\Pr(|X| \geq a) \leq \frac{1}{a^2} E(X^2).$$

We can generalize this to any moment  $p > 0$ :

$$\begin{aligned} E(|X|^p) &= \int_{\mathbb{R}} |x|^p f_X(x) dx \geq \int_{|x| \geq a} |x|^p f_X(x) dx \\ &\geq a^p \int_{|x| \geq a} f_X(x) dx = a^p \Pr(|X| \geq a). \end{aligned}$$

That is, we have proved

$$\Pr(|X| \geq a) \leq \frac{1}{a^p} E(|X|^p)$$

for any  $p = 1, 2, \dots$  (Of course, this assumes that  $E(|X|^p) < \infty$  for otherwise the inequality would not be saying much.)  $\square$

#### [Proof of the Weak Law of Large Numbers]

Let  $X_1, X_2, X_3, \dots$  be a sequence of independent random variables with common distribution function.

Set  $\mu = E(X_j)$  and  $\sigma^2 = \text{Var}(X_j)$ .

As usual we define  $S_n = X_1 + X_2 + \dots + X_n$  and let

$$S_n^* = \frac{S_n}{n} - \mu.$$

We apply Chebyshev's inequality to the random variable  $S_n^*$ .

Since  $E(S_n^*) = 0$  and  $\text{Var}(S_n^*) = \frac{\sigma^2}{n}$ ,

$$\Pr(|S_n^*| \geq \epsilon) \leq \frac{1}{\epsilon^2} \text{Var}(S_n^*).$$

for every  $\epsilon > 0$  by the Chebyshev inequality.

Writing this out explicitly:

$$\Pr\left(\left|\frac{X_1 + X_2 + \dots + X_n}{n} - \mu\right| \geq \epsilon\right) \leq \frac{1}{\epsilon^2} \frac{\sigma^2}{n}.$$

Thus for every  $\epsilon > 0$ ,

$$\lim_{n \rightarrow \infty} \Pr\left(\left|\frac{X_1 + X_2 + \dots + X_n}{n} - \mu\right| \geq \epsilon\right) = 0.$$

$\square$

#### [Borel-Cantelli Lemma]

Let  $A_1, A_2, \dots$  be an infinite sequence of events in  $\Omega$ . Consider the sequence of events

$$\bigcup_{n=1}^{\infty} A_n, \bigcup_{n=2}^{\infty} A_n, \bigcup_{n=3}^{\infty} A_n, \dots$$

Observe that this is a decreasing sequence in the sense that

$$\bigcup_{n=m+1}^{\infty} A_n \subseteq \bigcup_{n=m}^{\infty} A_n$$

for all  $m = 1, 2, \dots$

We are interested in those events that lie in infinitely many  $A_n$ . Such  $\omega$  would lie in  $\bigcup_{n=m}^{\infty} A_n$  for every  $m$ . Thus we define

$$\limsup A_n = \lim_{m \rightarrow \infty} \bigcup_{n=m}^{\infty} A_n = \{\omega \text{ that are in infinitely many } A_n\}.$$

We write this event as

$$\limsup A_n = \{\omega \in A_n \text{ i.o.}\}$$

where "i.o." is read as "infinitely often."

We can now state the **Borel-Cantelli Lemma**:

$$\sum_{n=1}^{\infty} \Pr(A_n) < \infty \implies \Pr(\omega \in A_n \text{ i.o.}) = 0.$$

Proof: First, observe that

$$0 \leq \limsup A_n \subseteq \bigcup_{n=m}^{\infty} A_n$$

for every  $m$  since the sequence is a decreasing sequence of events. Thus

$$0 \leq \Pr(\limsup A_n) \leq \sum_{n=m}^{\infty} \Pr(A_n)$$

for every  $m$ .

But we are assuming that the series  $\sum_{n=1}^{\infty} \Pr(A_n)$  converges. This means that

$$\lim_{m \rightarrow \infty} \sum_{n=m}^{\infty} \Pr(A_n) = 0.$$

Consequently,

$$0 \leq \Pr(\limsup A_n) \leq \sum_{n=m}^{\infty} \Pr(A_n) = 0. \therefore \Pr(\limsup A_n) = 0.$$

□

**[Proof of the Strong Law of Large Numbers]**

As above, let  $X_1, X_2, X_3, \dots$  denote an infinite sequence of independent random variables with a common distribution. Set

$$S_n = X_1 + \dots + X_n.$$

Let  $\mu = E(X_j)$  and  $\sigma^2 = \text{Var}(X_j)$ . The weak law of large numbers says that for every sufficiently large fixed  $n$  the average  $S_n/n$  is likely to be near  $\mu$ . The strong law of large numbers asks the question in what sense can we say

$$\lim_{n \rightarrow \infty} \frac{S_n(\omega)}{n} = \mu.$$

Clearly, it cannot be true for all  $\omega \in \Omega$ . (Take, for instance, in coin tossing the elementary event  $\omega = HHHH \dots$  for which  $S_n(\omega) = 1$  for every  $n$  and hence  $\lim_{n \rightarrow \infty} S_n(\omega)/n = 1$ .)

Thus, we want to look at the event

$$\mathcal{E} = \left\{ \omega \in \Omega : \lim_{n \rightarrow \infty} \frac{S_n(\omega)}{n} = \mu \right\}.$$

The Strong Law of Large Numbers says that

$$P(\mathcal{E}) = 1.$$

We will prove this under the additional restriction that  $\sigma^2 = E(X_j^2) < \infty$  and  $E(X_j^4) < \infty$ .

For the only  $X^2$ , we cannot demonstrate converges because infinite sum of  $1/n$  diverges generated from the  $X^2$ .

It is no loss of generality to assume  $\mu = 0$ . (Simply look at the new random variables  $Y_j = X_j - \mu$ .)

Now if

$$\lim_{n \rightarrow \infty} \frac{S_n(\omega)}{n} \neq 0,$$

then there exist  $\epsilon > 0$  such that for infinitely many  $n$

$$\left| \frac{S_n(\omega)}{n} \right| > \epsilon.$$

Thus, to prove the theorem, we prove that for every  $\epsilon > 0$

$$\Pr \left( \left| \frac{S_n(\omega)}{n} \right| > \epsilon \text{ i.o.} \right) = 0.$$



This then shows (by looking at the complement of this event) that

$$\Pr(\mathcal{E}) = \Pr\left(\frac{S_n}{n} = 0\right) = 1.$$

We use the Borel-Cantelli lemma applied to the events

$$A_n = \left\{ \omega \in \Omega : \left| \frac{S_n(\omega)}{n} \right| \geq \epsilon \right\}.$$

To estimate  $P(A_n)$ , we use the generalized Chebyshev inequality with  $p = 4$ . Thus we must compute  $E(S_n^4)$  or  $E(S^4/n^4)$  which equals

$$E\left(\sum_{1 \leq i, j, k, l \leq n} X_i X_j X_k X_l\right).$$

When the sums are multiplied out, there will be terms of the form  $E(X_i^3 X_j)$ ,  $E(X_i^2 X_j X_k)$ ,  $E(X_i X_j X_k X_l)$  with  $i, j, k, l$  all distinct. These terms are all equal to zero since  $E(X_i) = 0$  and the random variables are independent (and the subscripts are distinct). (Recall  $E(XY) = E(X)E(Y)$  when  $X$  and  $Y$  are independent.) If we select  $p = 2$ ,  $X^2$  is formed by  $XY$  from  $X^2$ . But  $X^2$  can't produce a convergent sum.

Thus the nonzero terms in the above sum are

$$E(X_i^4) \quad \text{and} \quad E(X_i^2 X_j^2) = (E(X_i^2))^2.$$

There are  $n$  terms of the form  $E(X_i^4)$ . The number of terms of the form  $E(X_i^2 X_j^2)$  is  $\binom{4}{2} \binom{n}{2} = 3n(n-1)$ . Thus we have shown

$$E(S_n^4) = nE(X_i^4) + 3n(n-1)\sigma^4.$$

For  $n$  sufficiently large there exists a constant  $C$  such that

$$\frac{1}{n^4} E(S_n^4) = \frac{nE(X_i^4) + 3n(n-1)\sigma^4}{n^4} \leq \frac{C}{n^2}.$$

(For  $n$  sufficiently large,  $C$  can be chosen to be  $3\sigma^4 + 1$ .) That is,

$$E\left(\frac{S_n^4}{n^4}\right) = \frac{1}{n^4} E(S_n^4) \leq \frac{C}{n^2}.$$

Then the Chebyshev inequality (with  $p = 4$ ) together with it gives

$$\Pr\left(\left|\frac{S_n^4}{n^4}\right| \geq \epsilon\right) \leq \frac{1}{\epsilon^4} E\left(\frac{S_n^4}{n^4}\right) \leq \frac{C}{\epsilon^4 n^2}.$$



Thus

$$\sum_{n \geq n_0} \Pr \left( \left| \frac{S_n}{n} \right| \geq \epsilon \right) = \sum_{n \geq n_0} \Pr \left( \left| \frac{S_n}{n} \right|^4 \geq \epsilon^4 \right) \leq \sum_{n \geq n_0} \frac{C}{\epsilon^4 n^2} < \infty.$$

(Here  $n_0$  is the first  $n$  so that the inequality  $E(S_n^4) \leq n^2 C$  holds. Since we are neglecting a finite set of terms in the sum, this cannot affect the convergence or divergence of the infinite series.)

Thus by the Borel-Cantelli lemma

$$\Pr \left( \left| \frac{S_n}{n} \right| \geq \epsilon \text{ i.o.} \right) = 0.$$

Since this holds for every  $\epsilon > 0$ , we have proved the strong law of large numbers.  $\square$

Remarks: Our proof assumed that the moments  $E(X_i^4)$  and  $E(X_i^2)$  are finite. It can be shown that the strong law of large numbers holds only under the assumption  $E(|X_i|) < \infty$ . Of course, we are still taking  $X_i$  to be independent with common distribution.

The proving process of the Weak Law of Large Numbers

[Theorem. Markov Inequality] If  $X \geq 0$  is a random variable and  $a > 0$ , then

$$P(X \geq a) \leq \frac{1}{a} EX.$$

Proof. Let the indicator function  $I_{X \geq a}$  be defined as follows.

$$I := \begin{cases} 1 & \text{if } X \geq a, \\ 0 & \text{if } X < a. \end{cases}$$

Then, here is the crucial observation:

$$I_{\{X \geq a\}} \leq \frac{1}{a} X.$$

Indeed, if  $X < a$ , the left-hand side is 0 and the right-hand side is nonnegative; if  $X \geq a$ , the left-hand side is 1, and the right-hand side is at least 1. Taking the expectation of both sides, we get

$$P(X \geq a) = E(I_{\{X \geq a\}}) \leq \frac{1}{a} EX. \quad \square$$

[Theorem. Chebyshev inequality] If  $EX = \mu$  and  $\text{Var}(X) = \sigma^2$  are both finite and  $k > 0$ , then

$$P(|X - \mu| \geq k) \leq \frac{\sigma^2}{k^2}.$$

Proof. By the Markov inequality,

$$\begin{aligned} P(|X - \mu| \geq k) &= P((X - \mu)^2 \geq k^2) \\ &\leq \frac{1}{k^2} E(X - \mu)^2 \\ &= \frac{1}{k^2} \text{Var}(X). \quad \square \end{aligned}$$

The Chebyshev inequality is useful if either  $\sigma$  is small or  $k$  is large.

[Theorem. Weak Law of Large Numbers] If  $X, X_1, X_2, \dots$  are iid R.V.s with finite expectation and variance, then  $\frac{X_1 + \dots + X_n}{n}$  converges to  $EX$  in the sense that, for any fixed  $\epsilon > 0$ ,

$$P\left(\left|\frac{X_1 + \dots + X_n}{n} - EX\right| \geq \epsilon\right) \rightarrow 0,$$

as  $n \rightarrow \infty$ .

Proof. Denote  $\mu = EX$ ,  $\sigma^2 = \text{Var}(X)$ , and let  $S_n = X_1 + \dots + X_n$ . Then,

$$ES_n = EX_1 + \dots + EX_n = n\mu \quad \text{and} \quad \text{Var}(S_n) = n\sigma^2.$$

Therefore, by the Chebyshev inequality,

$$\begin{aligned} P\left(\left|\frac{S_n}{n} - EX\right| > \epsilon\right) &= P(|S_n - n\mu| \geq n\epsilon) \\ &\leq \frac{n\sigma^2}{n^2\epsilon^2} \\ &= \frac{\sigma^2}{n\epsilon^2} \rightarrow 0, \quad \text{as } n \rightarrow \infty. \quad \square \end{aligned}$$

In particular, if  $S_n$  is the number of successes in  $n$  independent trials, each of which is a success with probability  $p$ , then, as we have observed before,  $S_n = I_1 + \dots + I_n$ , where  $I_i = I_{\{\text{success at trial } i\}}$ . So, for every  $\epsilon > 0$ ,

$$P\left(\left|\frac{S_n}{n} - p\right| \geq \epsilon\right) \rightarrow 0, \quad \text{as } n \rightarrow \infty.$$

Thus, the proportion of successes converges to  $p$  in this sense.

1.  $X_n \rightarrow X$  in **probability**, if for all  $\epsilon > 0$ ,  $\mathbb{P}(|X_n - X| > \epsilon) \rightarrow 0$  as  $n \rightarrow \infty$ .
2.  $X_n \rightarrow X$  in distribution, if  $\mathbb{P}(X_n \leq x) \rightarrow \mathbb{P}(X \leq x)$  as  $n \rightarrow \infty$ , for all  $x \in \mathbb{R}$  at which  $x \mapsto \mathbb{P}(X \leq x)$  is continuous.

3.  $X_n \rightarrow X$  in  $L^1$ , if  $\mathbb{E}(|X_n|) < \infty$  for all  $n \geq 1$  and  $\mathbb{E}(|X_n - X|) \rightarrow 0$  as  $n \rightarrow \infty$ .

4.  $X_n \rightarrow X$  **almost surely** (a.s.), if  $\mathbb{P}(X_n \rightarrow X \text{ as } n \rightarrow \infty) = 1$ .

$X_n \rightarrow X$  almost surely (Strong)

$\Downarrow$

$X_n \rightarrow X$  in probability (Weak)  $\Rightarrow X_n \rightarrow X$  in distribution

$\Uparrow$

$X_n \rightarrow X$  in  $L^1 \quad \Rightarrow \quad \mathbb{E}(X_n) \rightarrow \mathbb{E}(X)$

539

[Strong law of large numbers] If the random variables  $X_1, X_2, \dots$  fulfill one of the following conditions, they satisfy the strong law of large numbers.

- i The random variables  $X_1, X_2, \dots$  are iid with expectation  $E(X_j) = \mu$ .
- ii The random variables  $X_1, X_2, \dots$  are identically distributed with expectation  $E(X_j) = \mu$  and mutually independent.

Note 1: Condition (ii) generalizes condition (i).

Note 2: Under suitable conditions, the strong law of large numbers also applies to dependent random variables.



540

[Convolution of probability measures] Let  $X$  and  $Y$  denote independent real-valued random variables that are  $\eta$ -distributed and  $\tau$ -distributed, respectively. The distribution of the sum  $X + Y$  is the **convolution** product of  $\eta$  and  $\tau$ , denoted by  $\eta * \tau$ . If  $X$  has density  $g(\cdot)$  and  $Y$  has density  $h(\cdot)$ , then  $X + Y$  has density

$$f(x) := \int_{-\infty}^{\infty} g(x-y)h(y) dy. \quad (4.19)$$

$Z=X+Y=z$ 가 되는 모든  $(x,y)$ 에 대한 joint distribution density의 누적합

The CDF of  $Z = X + Y$  is

$$\begin{aligned} \Pr(X + Y \leq z) &= \int f * g(z) dz = \int \int_{x+y \leq z} f(x)g(y) dx dy \\ &= \int_{-\infty}^{\infty} g(y) \int_{-\infty}^{z-y} f(x) dx dy \\ \therefore f * g(z) dz &= \int_{-\infty}^{\infty} g(y) \left( \frac{d}{dz} \int_{-\infty}^{z-y} f(x) dx \right) dy \\ &= \int_{-\infty}^{\infty} g(y) f(z-y) dy \quad \because f(-\infty) \rightarrow 0 \end{aligned}$$

Replace  $z$  with  $x$  and denote the convolution  $f * g$  as  $h$  then, 본문은 각 분포에 대한 density가  $h=f * g$ 로 표기됨

$$h(x) dx = \int_{-\infty}^{\infty} f(x-y)g(y) dy$$

Note 1: The convolution of special distributions is discussed in pars. 1202 to 1206.

$$\begin{aligned} \Pr(Z = z) &= \Pr[(X, Y) \in A] && \text{Let } A := \{(x, y) : x + y = z\} \\ &= \sum_y P(X = z - y | Y = y) P(Y = y) \\ &= \sum_y P(X = z - y) P(Y = y) && \because X, Y \text{ are independent} \\ &= \sum_y f(z - y) g(y) \end{aligned}$$

Note 2: The term  $\tau^{*k}$  denotes the  $k$ -fold convolution product  $\tau * \cdots * \tau$ .

541

[Convolution of probability measures on groups] The convolution of probability measures is not only defined on  $\mathbb{R}$  but also on groups. In our context, finite groups are relevant. Assume that the independent random variables  $X$  and  $Y$  take on values on a finite (not necessarily commutative) group  $G$  with group operation “ $\circ$ ”, and that  $X$  and  $Y$  are  $\tau$ -distributed and  $\eta$ -distributed, respectively. In particular,

$$\tau * \eta(g) = \sum_{h \in G} \eta(g \circ h^{-1}) \tau(h) \quad \text{for each } g \in G. \quad (4.20)$$

$$\begin{aligned} \tau * \eta(z) &= \Pr(Z = z) \\ &= \Pr((X, Y) \in A) \quad \text{Let } A := \{(x, y) \in G \times G \mid y \circ x = z\} \\ &= \sum_{(x, y) \in A} \Pr(X = x, Y = y) \\ &= \sum_{x \in G} \Pr(X = x, Y = z \circ x^{-1}) \\ &= \sum_{x \in G} P(Y = z \circ x^{-1}) P(X = x) \quad \because X, Y \text{ are independent} \\ &= \sum_{x \in G} \eta(z \circ x^{-1}) \tau(x) \end{aligned}$$

Replace  $z$  and  $x$  with  $g$  and  $h$ , respectively, then,

$$\tau * \eta(g) = \sum_{h \in G} \eta(g \circ h^{-1}) \tau(h)$$

Example:

1.  $(G, \circ) = (\mathbb{Z}_n, +(\bmod n))$ .
2.  $(G, \circ) = (\{0, 1\}^\ell, \oplus)$ , where “ $\oplus$ ” denotes the bitwise addition mod 2.
3.  $(G, \circ) = \text{GL}(n, \text{GF}(2))$ , the group of invertible  $(n \times n)$ -matrices over  $\text{GF}(2)$ . Here, the group operation is the matrix multiplication. The group  $\text{GL}(n, \text{GF}(2))$  is not commutative for  $n \geq 2$ .

Note 1: In an additive group,  $X \circ Y$  and  $g \circ h^{-1}$  read  $X + Y$  and  $g - h$ , respectively.

Note 2: Pars. 596 to 603 provide useful theorems.

542

다변량 RNG의 Stochastic model 구성 시

Definition: Let  $C$  is a positive definite  $(k \times k)$ -matrix. The term  $N(\vec{\mu}, C)$  denotes the  $k$ -dimensional normal (Gaussian) distribution with expectation  $\vec{\mu}$  and covariance matrix  $C$ . It has the  $k$ -dimensional density

$$f_C(x) := \frac{1}{\sqrt{(2\pi)^k \det C}} e^{-0.5(\vec{x}-\vec{\mu})^T C^{-1}(\vec{x}-\vec{\mu})}. \quad (4.21)$$

Let  $\mathbf{z} \sim N(\mathbf{0}, I)$ . (It means  $E[\mathbf{z}] = \vec{0}$ ,  $\text{Cov}(\mathbf{z}) = I$ .)

Note: Covariance of a 1-dim Normal distribution is  $\text{Cov}(z_1, z_1) = \sigma^2$

The joint PDF of  $\mathbf{z}$  is

$$\begin{aligned} f_{\mathbf{z}}(\mathbf{z}) &= \prod_{i=1}^k f(z_i) = \prod_{i=1}^k \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}z_i^2} \\ &= \frac{1}{(2\pi)^{k/2}} e^{-\frac{1}{2}\sum z_i^2} \\ &= \frac{1}{(2\pi)^{k/2}} e^{-\frac{1}{2}\mathbf{z}^T \mathbf{z}} \end{aligned}$$

Now we want to replace  $\mathbf{x}$  with  $\mathbf{z}$ . The  $\mathbf{x}$  has two conditions in this proof.

Let  $A$  be the  $k \times k$  invertible matrix, and the covariance matrix of  $\mathbf{x}$  is  $C := AA^T$  (Cholesky decomposition).

1.  $\text{Cov}(\mathbf{x}) = C$ .

$$\begin{aligned} \text{Cov}(\mathbf{x}) &= C \\ &= AA^T \\ &= AIA^T \\ &= A\text{Cov}(\mathbf{z})A^T \quad \because \text{Cov}(\mathbf{z}) = I \\ &= \text{Cov}(A\mathbf{z}) \\ &= \text{Cov}(A\mathbf{z} + \mathbf{c}) \quad \text{for some vector } \mathbf{c} \end{aligned}$$

2.  $E[\mathbf{x}] = \boldsymbol{\mu}$ .

$$\begin{aligned} E[\mathbf{x}] &= \boldsymbol{\mu} \\ &= B \cdot \mathbf{0} + \boldsymbol{\mu} \quad \text{for some matrix } B \\ &= B \cdot E[\mathbf{z}] + \boldsymbol{\mu} \quad \because E[\mathbf{z}] = \mathbf{0} \\ &= E[B\mathbf{z} + \boldsymbol{\mu}] \end{aligned}$$



Since the  $\mathbf{x}$  need to satisfy both 1. and 2.,

$$\mathbf{x} = A\mathbf{z} + \boldsymbol{\mu}$$

and consequently,

$$\therefore \mathbf{z} = A^{-1}(\mathbf{x} - \boldsymbol{\mu})$$

Next, we find the PDF of  $\mathbf{x}$ .

$$\begin{aligned} \int_{\Omega_1} f_{\mathbf{x}}(\mathbf{x}) d\mathbf{x} &= \int_{\Omega_2} f_{\mathbf{z}}(\mathbf{z}) d\mathbf{z} && \because P(\mathbf{x} \in \Omega_1) = P(\mathbf{z} \in \Omega_2) \\ &= \int_{\Omega_1} f_{\mathbf{z}}(\mathbf{z}) \left| \det \frac{d\mathbf{z}}{d\mathbf{x}} \right| d\mathbf{x} && \because \frac{d\mathbf{z}}{d\mathbf{x}} = \text{Vol} \left( \frac{d\mathbf{z}}{d\mathbf{x}} \right) = \left| \det \frac{d\mathbf{z}}{d\mathbf{x}} \right| \\ \therefore f_{\mathbf{x}}(\mathbf{x}) &= f_{\mathbf{z}}(\mathbf{z}) \left| \det \frac{d\mathbf{z}}{d\mathbf{x}} \right| \end{aligned}$$

where

$$\begin{aligned} \left| \det \left( \frac{d\mathbf{z}}{d\mathbf{x}} \right) \right| &= |\det A^{-1}| \quad \because \frac{d\mathbf{z}}{d\mathbf{x}} = \frac{d}{d\mathbf{x}}(A^{-1}(\mathbf{x} - \boldsymbol{\mu})) = A^{-1} \\ &= \frac{1}{|\det A|} \\ &= \frac{1}{\sqrt{|\det C|}} \quad \because \det C = \det(AA^T) = (\det A)^2 \end{aligned}$$

Therefore,

$$\begin{aligned} f_{\mathbf{x}}(\mathbf{x}) &= f_{\mathbf{z}}(\mathbf{z}) \left| \det \left( \frac{d\mathbf{z}}{d\mathbf{x}} \right) \right| \\ &= \frac{1}{(2\pi)^{k/2}} e^{-\frac{1}{2}\mathbf{z}^T \mathbf{z}} \cdot \frac{1}{\sqrt{|\det C|}} \\ &= \frac{1}{\sqrt{(2\pi)^k |\det C|}} e^{-\frac{1}{2}(A^{-1}(\mathbf{x} - \boldsymbol{\mu}))^T (A^{-1}(\mathbf{x} - \boldsymbol{\mu}))} \\ &= \frac{1}{\sqrt{(2\pi)^k |\det C|}} e^{-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^T (A^{-1})^T A^{-1}(\mathbf{x} - \boldsymbol{\mu})} \\ &= \frac{1}{\sqrt{(2\pi)^k |\det C|}} e^{-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^T C^{-1}(\mathbf{x} - \boldsymbol{\mu})} \quad \because (A^{-1})^T = (A^T)^{-1} \end{aligned}$$

Note 1: If  $\vec{X} = (X_1, \dots, X_k)$  is  $N(\vec{\mu}, C)$ -distributed, its expectation  $E(\vec{X}) = \vec{\mu} \in \mathbb{R}^k$ , and  $C = (c_{ij})_{1 \leq i, j \leq k} = (\text{Cov}(X_i, X_j))_{1 \leq i, j \leq k}$  is the covariance matrix of  $\vec{X}$ .

Note 2: Definition (4.21) generalizes (4.4). In the one-dimensional case,  $\vec{\mu} = \mu$ , and the covariance matrix  $C$  equals the variance  $\sigma^2$ .

Multivariate normal distribution  $\Rightarrow$  each is normally distributed (the converse is not true)



Note 3: If the random variables  $X_1, \dots, X_k$  are iid  $N(0, 1)$ -distributed, the covariance matrix is given by the  $k$ -dimensional identity mapping.

Note 4: Above, we assume that the covariance matrix  $C$  is regular. It should be noted that normal distributions with singular covariance matrices 역행렬이 없는 정사각 행렬  $C$  also exist. These distributions do not have  $k$ -dimensional densities.

543

Assume that the random variables  $X$  and  $Y$  are bivariate normally distributed. If  $X$  and  $Y$  are uncorrelated,  $X$  and  $Y$  are independent.

Note: In general, this conclusion is not true; cf. pars. 536 and 537.

544

Let  $(\Omega, \mathcal{A}, P)$  be a probability space. Formally, a **stochastic process** process에 time의 의미가 내포되는 경우가 많음  $(X_t)_{t \in T}$  with state space  $\Omega$  is a collection of real-valued random variables  $\{X_t \mid t \in T\}$ , where the index  $t$  is usually interpreted as “time.”

545

If  $T \subseteq \mathbb{R}$  is an interval (e.g.,  $T = (a, b)$ ,  $T = [0, \infty)$  or  $T = \mathbb{R}$ ), we speak of (time-)continuous stochastic processes. If  $T \subseteq \Delta\mathbb{Z}$  for some  $\Delta > 0$ , e.g.,  $T = \mathbb{Z}$ ,  $T = \mathbb{N}$  or  $T = \mathbb{N}_0$ , the stochastic process is called (time-)discrete.

546

Example: Markov chains (time-discrete stochastic process; cf. par. 578), Gaussian process (time-continuous stochastic process; cf. par. 547), Wiener process (time-continuous stochastic process).

[Wiener process] For the R.V.s  $W_{t_1}, W_{t_2}, W_{t_3}, W_{t_4}$ , a Markov chain is established by  $(W_{t_2} - W_{t_1}), (W_{t_4} - W_{t_3})$  with  $\text{Cov} = 0$ . And  $W_{t_j} - W_{t_i} \sim N(0, t_j - t_i)$

547

A time-continuous real-valued stochastic process  $(X_t)_{t \in T}$  is called a Gaussian process if the random vector  $(X_{t_1}, \dots, X_{t_k})$  is (multivariate) normally (Gaussian) distributed for each finite index set  $\{t_1, \dots, t_k\} \subseteq T$ ; cf. par. 542.

For any  $T' \subseteq T$ ,  $\{X_t \mid t \in T'\}$  consist multivariate normal distribution.  
Each  $X_i$  satisfy  $X_i \sim N(\mu, \sigma^2)$ .

548

A stochastic process  $(X_t)_{t \in T}$  is called **stationary** (or: stationary in a strict sense) if

$$\text{Prob}(X_{t_1} \in A_1, X_{t_2} \in A_2, \dots, X_{t_k} \in A_k) = \text{Prob}(X_{t_1+\tau} \in A_1, X_{t_2+\tau} \in A_2, \dots, X_{t_k+\tau} \in A_k) \quad (4.22)$$

for each  $k \in \mathbb{N}$ ,  $\tau > 0$ , all  $t_1 < \dots < t_k$  with  $t_j, t_j + \tau \in T$  ( $j \leq k$ ), and all measurable sets  $A_1, \dots, A_k$ .

If the random variables  $X_j$  are discrete, (4.22) simplifies to

$$\text{Prob}(X_{t_1} = x_1, X_{t_2} = x_2, \dots, X_{t_k} = x_k) = \text{Prob}(X_{t_1+\tau} = x_1, X_{t_2+\tau} = x_2, \dots, X_{t_k+\tau} = x_k) \quad (4.23)$$

for each  $k \in \mathbb{N}$ ,  $\tau > 0$ , all  $t_1 < \dots < t_k$  with  $t_j, t_j + \tau \in T$  ( $j \leq k$ ), and all  $x_1, \dots, x_k \in \Omega$ .

$(t_1, \dots, t_k)$  and  $(t_{\tau+1}, \dots, t_{\tau+k})$  can overlap. This means the distributions at  $t_i$  and  $t_j = \tau + t_i$  ( $t_i < t_j$ ) are identical for any  $\tau \in T$ . For example,  $\text{Pr}[(1, 0, 1) \text{ in } t_1] = \text{Pr}[(1, 0, 1) \text{ in } t_3 = t_1 + \tau]$ . This property is called time-invariance.

549

Stationarity means that the distribution of the stochastic process is time-invariant. In other words: For admissible shifts  $\tau$  (that is,  $T + \tau \subseteq T$ ), the stochastic processes  $(X_t)_{t \in T}$  and  $(X_{t+\tau})_{t \in T}$  are identically distributed. If  $T = \mathbb{R}$  or  $T = [0, \infty)$ , for example, any  $\tau > 0$  is admissible. For  $T = \mathbb{Z}$  or  $T = \mathbb{N}$  (time-discrete stochastic processes), the shift parameter  $\tau$  must be a positive integer.

550

A stochastic process  $(X_t)_{t \in T}$  is stationary in a weak sense (or: stationary in a wide sense) if

expectation

$$E(X_t) = E(X_{t+\tau}) \quad (4.24)$$

covariance

$$E((X_{t_1} - \mu)(X_{t_2} - \mu)) = E((X_{t_1+\tau} - \mu)(X_{t_2+\tau} - \mu)) \quad (4.25)$$

for all  $t, t + \tau \in T, \tau > 0$ .

551

Assume that  $(X_t)_{t \in T}$  is a stochastic process on the state space  $\mathbb{R}$  for which  $E(X_t)$  and  $\text{Var}(X_t)$  exist for all  $t \in T$ . Then

$$\overline{K}_X(t_1, t_2) := E((X_{t_1} - E(X_{t_1}))(X_{t_2} - E(X_{t_2}))) \quad (t_1, t_2 \in T) \quad (4.26)$$

is the autocovariance  $\text{하나의 } X \text{로부터 } X_{t1}, X_{t2} \text{를 관측함}$  function of  $(X_t)_{t \in T}$ . Furthermore,

$$\overline{\gamma}_X(t_1, t_2) := \frac{E((X_{t_1} - E(X_{t_1}))(X_{t_2} - E(X_{t_2})))}{\sqrt{\text{Var}(X_{t_1})\text{Var}(X_{t_2})}} \quad (t_1, t_2 \in T) \quad (4.27)$$

defines the autocorrelation function of  $(X_t)_{t \in T}$ .

552

Assume that  $(X_t)_{t \in T}$  is a stochastic process on the state space  $\mathbb{R}$  for which  $E(X_t)$  and  $\text{Var}(X_t)$  exist for all  $t \in T$ . If  $(X_t)_{t \in T}$  is stationary or stationary in a weak sense, (4.26) and (4.27) only depend on  $|t_1 - t_2|$ .

Analogously to par. 551, we define the autocovariance function

$$K_X(h) := E((X_t - E(X_t))(X_{t+h} - E(X_{t+h}))) \quad (h \geq 0; t, t+h \in T) \quad (4.28)$$

and the autocorrelation function (ACF)

$$\gamma_X(h) := \frac{E((X_t - E(X_t))(X_{t+h} - E(X_{t+h})))}{\sqrt{\text{Var}(X_t)\text{Var}(X_{t+h})}} \quad (h \geq 0; t, t+h \in T). \quad (4.29)$$

Note: If  $T = [0, \infty)$ , for example,  $h \in [0, \infty)$  while  $h \in \mathbb{N}_0$  if  $T \in \{\mathbb{Z}, \mathbb{N}_0, \mathbb{N}\}$ .



553

Pars. 585 and 587 to 590 collect useful facts on stationary stochastic processes. Stationarity plays an important role in stochastic models for PTRNGs; cf. Section 4.5. It captures the desired feature that if a PTRNG is analyzed in a certain period in time, its stochastic behavior should be the same at different times.

Note: For stochastic models of physical noise sources, the requirement is relaxed to time-local stationarity; cf. pars. 788 to 790.

554

If a (time-continuous or time-discrete) stationary stochastic process is ergodic, statistical properties of this stochastic process can be deduced from a single, sufficiently long realization of this stochastic process with probability 1.

[Ergodic] The process  $X(t)$  is said to be (mean-)ergodic if the time average estimate converges in squared mean to the ensemble calculated over all possible sample functions  $X$ , not time averages average as time goes to infinity.

$$\text{time-dependent} \quad \lim_{T \rightarrow \infty} \left( \hat{\mu}_X = \frac{1}{T} \int_0^T X(t) dt \right) = \mu_X \quad \text{time-independent}$$

or for the discret-time r.v.

$$\lim_{N \rightarrow \infty} \left( \hat{\mu}_X = \frac{1}{N} \sum_{n=1}^N X[n] \right) = EX$$

Ergodicity allows a single RNG to function as multiple RNGs. For example, we can generate a random bit sequence with a fair coin by tossing it 100 times instead of using 100 separate fair coins. An RNG can demonstrate ergodicity by restarting itself.

Note 1: In the context of the evaluation of PTRNGs, this feature is exploited for the estimation of parameters, by online tests and by statistical tests applied by the evaluator, for example.

Note 2: There exist several equivalent formal definitions for ergodicity, e.g., that the invariant events are attained with probability 0 or 1. We refer the interested reader to the relevant literature, e.g., to [KaTa75], Chapter 9.

Note 3: Pars. 555 and 556 provide an example of an ergodic process and a counterexample. Loosely speaking, to ensure ergodicity, long-term dependencies of the stochastic process need to decrease sufficiently fast.

555

Example: Assume that the random variables  $X_1, X_2, \dots$  are iid  $B(1, p)$ -distributed. If we observe a realization sequence  $x_1, x_2, \dots$ , the empirical mean  $n^{-1} \sum_{j=1}^n x_j$  converges to  $p$  with probability 1 (Strong law of large numbers; cf. pars. 538 and 539). If the random variables model the repeated tossing of a particular coin (cf. Subsec. 4.5.2), a sequence of realizations can easily be obtained by tossing this coin several times, which allows the estimation of the (unknown) parameter  $p$ . The random variables  $X_1, X_2, \dots$  define a stationary ergodic process (cf. par. 556).

556

Counterexample: Assume that the random variables  $X_1, X_2, \dots$  are identically  $B(1, p)$ . Unlike in par. 555, these random variables are not independent but fully dependent, namely  $X_1 = X_2 = \dots$ . Then the realization of  $X_1$  determines the whole realization sequence. In this case, one can only observe the realization sequences  $1, 1, \dots$  (with probability  $p$ ) or  $0, 0, \dots$  (with probability  $1 - p$ ). Hence, it is not possible to estimate  $p$  on the basis of a single realization sequence. This stochastic process is stationary but not ergodic.

557

[empirical mean and empirical variance] Assume that  $x_1, x_2, \dots, x_m$  are realizations of the iid random variables  $X_1, X_2, \dots, X_m$ . Assume further that the expectation  $\mu = E(X_j)$  and the variance  $\sigma^2 = \text{Var}(X_j)$  exist. The arithmetic mean  $\bar{x}$  and the empirical variance  $\bar{s}^2$  of  $x_1, x_2, \dots, x_m$  are given by

$$\bar{x} := \frac{x_1 + x_2 + \dots + x_m}{m}, \quad (4.30)$$

$$\bar{s}^2 := \frac{1}{m-1} \sum_{j=1}^m (x_j - \bar{x})^2. \quad (4.31)$$

$\bar{x}$  and  $\bar{s}^2$  are \*unbiased\* estimators of  $\mu$  and  $\sigma^2$ . In this context, unbiased means, that if the sample values  $x_j$  in the right-hand sides of (4.30) and (4.31) are replaced by random variables  $X_j$ , the expectation of these terms is  $\mu$  and  $\sigma^2$ , respectively. **Unbiased Estimators**

$$\begin{aligned} E[\bar{x}] &= E \left[ \frac{x_1 + \dots + x_m}{m} \right] \\ &= \frac{1}{m} [Ex_1 + \dots + Ex_m] \\ &= \frac{1}{m} [\mu + \dots + \mu] = \frac{1}{m} m\mu \\ &= \mu = EX \end{aligned}$$



$$\begin{aligned}
E[\bar{s}^2] &= E\left[\frac{1}{m-1} \sum_{j=1}^m (x_j - \bar{x})^2\right] \\
&= \frac{1}{m-1} \sum_{j=1}^m E(x_j - \bar{x})^2 \\
&= \frac{1}{m-1} \sum_{j=1}^m \left(\text{Var}(x_j - \bar{x}) + [E(x_j - \bar{x})]^2\right) \\
&= \frac{1}{m-1} \sum_{j=1}^m \text{Var}(x_j - \bar{x}) \quad \because E[\bar{x}] = \mu \text{ and } Ex_j = \mu \implies E(x_j - \bar{x}) = 0 \\
&= \frac{1}{m-1} \sum_{j=1}^m \text{Var}\left(\frac{m-1}{m}x_j - \frac{1}{m} \sum_{i \neq j} x_i\right) \quad \because \bar{x} = \frac{x_1 + \cdots + x_m}{m} \\
&= \frac{1}{m-1} \sum_{j=1}^m \left[\left(\frac{m-1}{m}\right)^2 \text{Var}(x_j) + \left(-\frac{1}{m}\right)^2 \sum_{i \neq j} \text{Var}(x_i)\right] \\
&= \frac{1}{m-1} \sum_{j=1}^m \left[\left(\frac{m-1}{m}\right)^2 \sigma^2 + \left(\frac{1}{m}\right)^2 \sum_{i \neq j} \sigma^2\right] \\
&= \frac{m}{m-1} \left[\frac{(m-1)^2}{m^2} \sigma^2 + \frac{m-1}{m^2} \sigma^2\right] \\
&= \frac{m-1}{m} \sigma^2 + \frac{1}{m} \sigma^2 \\
&= \sigma^2 \\
&= \text{Var}(X)
\end{aligned}$$

Note: Occasionally, formula (4.31) is used with factor  $1/m$  in place of  $1/(m-1)$ . In this case, the estimator is biased (but asymptotically unbiased).

558

[empirical mean and empirical variance] Assume that the random variables  $X_1, X_2, \dots, X_m$  are iid  $N(\mu, \sigma^2)$ -distributed. Then

$$\frac{X_1 + X_2 + \dots + X_m}{m} \sim N\left(\mu, \frac{\sigma^2}{m}\right) \quad (4.32)$$

and

$$\frac{m-1}{\sigma^2} \cdot \frac{1}{m-1} \sum_{j=1}^m (X_j - \bar{X})^2 \sim \chi_{m-1}^2 \quad (4.33)$$

where  $\chi_{m-1}^2$  denotes the  $\chi^2$ -distribution with  $m-1$  degrees of freedom. Formula (4.33) is a well-known corollary from Cochran's Theorem. 정규모집단에서 표본평균과 표본분산의 분포

$$\begin{aligned} E\left[\frac{X_1 + X_2 + \dots + X_m}{m}\right] &= \frac{1}{m}E[X_1] + \dots + \frac{1}{m}E[X_m] = \mu \\ \text{Var}\left[\frac{X_1 + X_2 + \dots + X_m}{m}\right] &= \frac{1}{m^2}\text{Var}[X_1] + \dots + \frac{1}{m^2}\text{Var}[X_m] = \frac{1}{m}\sigma^2 \end{aligned}$$

For the iid R.V.s s.t.  $(X_i - \mu)/\sigma \sim N(0, 1)$  ( $i = 1, 2, \dots, m$ ),

$$\sum_{i=1}^m \left(\frac{X_i - \mu}{\sigma}\right)^2 \sim \chi^2(m)$$

Since,  $n\bar{X} = \sum_{i=1}^m X_i$ , and  $S^2 = \sum_{i=1}^m (X_i - \bar{X})^2 / (m-1)$ ,

$$\begin{aligned} \sum_{i=1}^m \left(\frac{X_i - \mu}{\sigma}\right)^2 &= \sum_{i=1}^m \left(\frac{X_i - \bar{X}}{\sigma}\right)^2 + m \left(\frac{\bar{X} - \mu}{\sigma}\right)^2 + \frac{2}{\sigma^2}(\bar{X} - \mu) \sum_{i=1}^m (X_i - \bar{X}) \\ &= \sum_{i=1}^m \left(\frac{X_i - \bar{X}}{\sigma}\right)^2 + m \left(\frac{\bar{X} - \mu}{\sigma}\right)^2 \quad \because \sum_{i=1}^m (X_i - \bar{X}) = \sum_{i=1}^m X_i - m\bar{X} = 0 \\ &= \left(\frac{(m-1)S^2}{\sigma^2}\right) + \left(\frac{\bar{X} - \mu}{\sigma/\sqrt{m}}\right)^2 \end{aligned}$$

[Cochran's Theorem] Let  $U_1, \dots, U_N$  be i.i.d. standard normally distributed random variables, and  $U = [U_1, \dots, U_N]^T$ . Let  $B^{(1)}, B^{(2)}, \dots, B^{(k)}$  be symmetric matrices. Define  $r_i$  to be the rank of  $B^{(i)}$ . Define  $Q_i = U^T B^{(i)} U$ , so that the  $Q_i$  are quadratic forms. Further assume  $\sum_i Q_i = U^T U$ .

Cochran's theorem states that the following are equivalent:

- $r_1 + \dots + r_k = N$
- the  $Q_i$  are independent
- each  $Q_i$  has a chi-squared distribution with  $r_i$  degrees of freedom.

The rank  $\sum_{i=1}^m \left(\frac{X_i - \mu}{\sigma}\right)^2 = m$  is trivial. Informally, the rank of  $\sum_{i=1}^m \left(\frac{X_i - \bar{X}}{\sigma}\right)^2$  is  $m - 1$  because the r.v.s are independent except only one r.v. (e.g.  $X_m$ ), and the rank of  $\left(\frac{\bar{X} - \mu}{\sigma/\sqrt{m}}\right)^2$  is 1 because  $\bar{X} \sim N(\mu, \sigma^2/m)$  is the single normal standard dist. So,  $\text{rank} \sum_{i=1}^m \left(\frac{X_i - \mu}{\sigma}\right)^2 = \text{rank} \sum_{i=1}^m \left(\frac{X_i - \bar{X}}{\sigma}\right)^2 + \text{rank} \left(\frac{\bar{X} - \mu}{\sigma/\sqrt{m}}\right)^2$ , and it means  $\sum_{i=1}^m \left(\frac{X_i - \mu}{\sigma}\right)^2$  and  $\left(\frac{\bar{X} - \mu}{\sigma/\sqrt{m}}\right)^2$  are independent by the Cochran's thm.

Furthermore,

$$\sum_{i=1}^m \left(\frac{X_i - \mu}{\sigma}\right)^2 \sim \chi^2(m), \quad \left(\frac{\bar{X} - \mu}{\sigma/\sqrt{m}}\right)^2 \sim \chi^2(1).$$

Therefore, by the additivity of the Chi-square dist. for the independent terms  $\sum_{i=1}^m \left(\frac{X_i - \mu}{\sigma}\right)^2, \left(\frac{\bar{X} - \mu}{\sigma/\sqrt{m}}\right)^2$

$$\frac{(m-1)S^2}{\sigma^2} \sim \chi^2(m-1).$$

559

[empirical mean and empirical variance] If the random variables  $X_1, X_2, \dots, X_m$  are iid (but not necessarily normally distributed), then

$$E \left( \frac{1}{m-1} \sum_{j=1}^m (X_j - \bar{X})^2 \right) = \sigma^2 \quad (4.34)$$

$$\text{Var} \left( \frac{1}{m-1} \sum_{j=1}^m (X_j - \bar{X})^2 \right) = \frac{1}{m} \left( E((X - \mu)^4) - \frac{m-3}{m-1} \sigma^4 \right) \quad (4.35)$$

$$\bar{s}^2 = \frac{1}{m-1} \sum (X_i - \bar{X})^2$$

Let  $Y_i = X_i - \bar{X}$  be a r.v. for all  $i$ . Then, we can see the

$$\bar{Y} := EY_i = \mu = 0$$

and

$$\begin{aligned} \text{Var}(Y_i) &= E(Y_i^2) - (EY_i)^2 \\ &= E(Y_i^2) \\ &= E(X_i^2 - 2X_i\bar{X} + \bar{X}^2) \\ &= EX_i^2 - 2\bar{X}EX_i + \bar{X}^2 \\ &= EX_i^2 - (EX_i)^2 \\ &= \sigma^2 \end{aligned}$$



That is,  $\text{Var}(Y_i) = E(Y_i^2) = \sigma^2$ .

Thus,

$$\bar{s}^2 = \frac{1}{m-1} \sum_{i=1}^m (Y_i - \bar{Y})^2 = \frac{1}{m-1} \left( \sum_{i=1}^m Y_i^2 - m\bar{Y}^2 \right)$$

Next, we want to take the variance of  $\bar{s}^2$ , i.e.,

$$\text{Var}(\bar{s}^2) = \frac{1}{(m-1)^2} \text{Var} \left( \sum_{i=1}^m Y_i^2 - m\bar{Y}^2 \right)$$

So, we need to find the  $\text{Var} \left( \sum Y_i^2 - m\bar{Y}^2 \right)$ .

$$\text{Var} \left( \sum_{i=1}^m Y_i^2 - m\bar{Y}^2 \right) = \text{Var} \left( \sum_{i=1}^m Y_i^2 \right) + m^2 \text{Var}(\bar{Y}^2) - 2 \text{Cov} \left( \sum_{i=1}^m Y_i^2, m\bar{Y}^2 \right)$$

First,

$$\begin{aligned} \text{Var} \left( \sum_{i=1}^m Y_i^2 \right) &= \sum_{i=1}^m \text{Var}(Y_i^2) \\ &= \sum_{i=1}^m \left[ EY^4 - (EY^2)^2 \right] \\ &= m \left[ \mu_4 - \sigma^4 \right] \quad \because E(Y_i^2) = \sigma^2 \text{ and define } EY^4 = \mu_4 \end{aligned}$$

and

$$\begin{aligned} m^2 \text{Var}(\bar{Y}^2) &= m^2 \left[ E\bar{Y}^4 - (E\bar{Y}^2)^2 \right] \\ &= m^2 \left[ E\bar{Y}^4 - \left( \frac{\sigma^2}{m} \right)^2 \right] \quad \because E(\bar{Y}^2) = \text{Var}(\bar{Y}) + (E\bar{Y})^2 = \frac{\sigma^2}{m} - 0 = \frac{\sigma^2}{m} \\ &= m^2 \left[ \frac{\mu_4}{m^3} + \frac{3(m-1)\sigma^4}{m^3} - \frac{\sigma^4}{m^2} \right] \\ &= m^2 \left[ \frac{\mu_4}{m^3} + \frac{2m-3}{m^3} \sigma^4 \right] \\ &= \frac{\mu_4}{m} + \frac{2m-3}{m} \sigma^4 \end{aligned}$$

where

$$\begin{aligned} E \left( \sum Y_i \right)^4 &= m\mu_4 + 3m(m-1)\sigma^4 \quad \because E[Y_i Y_j Y_k Y_l] = E[Y_i Y_j^3] = 0 \\ \implies E(\bar{Y}^4) &= \frac{1}{m^4} \{ m\mu_4 + 3m(m-1)\sigma^4 \} \quad \because \bar{Y} = \frac{\sum Y_i}{m} \implies \bar{Y}^4 = \frac{(\sum Y_i)^4}{m^4} \\ &= \frac{\mu_4}{m^3} + \frac{3(m-1)\sigma^4}{m^3} \end{aligned}$$

Last, because

$$2 \operatorname{Cov} \left( \sum_{i=1}^m Y_i^2, m\bar{Y}^2 \right) = 2m \left( E \left[ \sum_{i=1}^m Y_i^2 \cdot \bar{Y}^2 \right] - E \sum_{i=1}^m Y_i^2 \cdot E\bar{Y}^2 \right),$$

$$\begin{aligned} E \left[ \sum_{i=1}^m Y_i^2 \cdot \bar{Y}^2 \right] &= E \left[ \left( \sum Y_i^2 \right) \cdot \left( \frac{1}{m} \sum Y_j \right)^2 \right] \\ &= \frac{1}{m^2} E \left[ \sum_i Y_i^2 \cdot \sum_j \sum_k Y_j Y_k \right] \\ &= \frac{1}{m^2} [m\mu_4 + m(m-1)\sigma^4] \quad \because E[Y_i Y_j Y_k Y_l] = E[Y_i Y_j^3] = 0 \\ &= \frac{\mu_4}{m} + \frac{m-1}{m} \sigma^4 \end{aligned}$$

and

$$E \sum_{i=1}^m Y_i^2 \cdot E\bar{Y}^2 = (m\sigma^2) \left( \frac{\sigma^2}{m} \right) = \sigma^4$$

Thus,

$$2 \operatorname{Cov} \left( \sum_{i=1}^m Y_i^2, m\bar{Y}^2 \right) = 2m \left[ \left( \frac{\mu_4}{m} + \frac{m-1}{m} \sigma^4 \right) - \sigma^4 \right] = 2(\mu_4 - \sigma^4)$$

Consequently,

$$\begin{aligned} \operatorname{Var} \left( \sum_{i=1}^m Y_i^2 - m\bar{Y}^2 \right) &= \operatorname{Var} \left( \sum_{i=1}^m Y_i^2 \right) + m^2 \operatorname{Var} (\bar{Y}^2) - 2 \operatorname{Cov} \left( \sum_{i=1}^m Y_i^2, m\bar{Y}^2 \right) \\ &= m [\mu_4 - \sigma^4] + \left( \frac{\mu_4}{m} + \frac{2m-3}{m} \sigma^4 \right) - 2(\mu_4 - \sigma^4) \\ &= \frac{(m-1)^2}{m} \mu_4 - \frac{(m-1)(m-3)}{m} \sigma^4 \end{aligned}$$

Finally,

$$\begin{aligned} \operatorname{Var} \left( \frac{1}{m-1} \sum (X_i - \bar{X})^2 \right) &= \operatorname{Var}(\bar{s}^2) \\ &= \frac{1}{(m-1)^2} \operatorname{Var} \left( \sum_{i=1}^m Y_i^2 - m\bar{Y}^2 \right) \\ &= \frac{1}{(m-1)^2} \left( \frac{(m-1)^2}{m} \mu_4 - \frac{(m-1)(m-3)}{m} \sigma^4 \right) \\ &= \frac{1}{m} \mu_4 - \frac{m-3}{m(m-1)} \sigma^4 \\ &= \frac{1}{m} E(X - \mu)^4 - \frac{m-3}{m(m-1)} \sigma^4 \quad \because EY^4 = \mu_4 \text{ and } Y_i = X_i - \bar{X} \quad \square \end{aligned}$$

560

[Allan variance] When estimating the jitter of digital clock signals, for example, the empirical variance may overestimate the jitter if low frequency noise such as flicker noise is present. In such scenarios, often the (empirical) Allan variance is used instead; cf., e.g., [ASPB+18]. Assume that the measurement values  $x_1, x_2, \dots, x_m$  are taken at times  $\tau, 2\tau, \dots, m\tau$  (in practice, the  $x_j$  often are fractional frequencies that have been averaged over an interval of length  $\tau$ ). The (empirical) Allan variance of  $x_1, x_2, \dots, x_m$  is defined by

$$\overline{\text{AVar}} = \frac{1}{2(m-1)} \sum_{j=1}^{m-1} (x_j - x_{j+1})^2. \quad (4.36)$$

Allan variance analyzes the difference between consecutive measurements rather than comparing each measurement to the overall mean. Typically, implementations use this approach because measured values tend to change monotonically over time in the real world. For example, in the case of thermal noise, the noise level can differ during the day, when temperatures rise, and at night, when temperatures fall. Calculating the variance with this in mind helps prevent the variance from increasing unnecessarily.

Note 1: By construction, the Allan variance is only a little sensitive to slow drifts of the distributions of the corresponding random variables  $X_1, X_2, \dots, X_m$ .

Note 2: The definition of the Allan variance is not uniform in the literature.

561

[Allan variance] Assume that the measurement values  $x_1, x_2, \dots$  are realizations of the random variables  $X_1, X_2, \dots$ . If the random variables are stationarily distributed and uncorrelated (i.e.,  $\text{Cov}(X_i, X_j) = 0$  for  $i \neq j$ ), the Allan variance coincides with the “usual” variance [ASPB+18], Theorem 1.

The stationary distribution, such as IID, also conforms to the Allan variance.



### 4.2.2 Useful Theorems and Facts

562

Subsection 4.2.2 provides facts and theorems that can be useful in the context of this document.

563

[Stirling's approximation]

$$\sqrt{2\pi n} \left(\frac{n}{e}\right)^n e^{\frac{1}{12n+1}} < n! < \sqrt{2\pi n} \left(\frac{n}{e}\right)^n e^{\frac{1}{12n}} \quad (\text{Stirling's approximation}) \quad (4.37)$$

Stirling's approx. (general)

$$n! \approx \sqrt{2\pi n} \left(\frac{n}{e}\right)^n \quad \text{for large } n$$

proof. We use Laplace's method

$$\int_a^b e^{Mf(x)} dx \approx \sqrt{\frac{2\pi}{M|f''(x_0)|}} e^{Mf(x_0)}$$

as  $M \rightarrow \infty$  for  $x_0$  is a maximum point. Then,

$$\begin{aligned} n! &= \Gamma(n+1) \\ &= \int_0^\infty x^n e^{-x} dx = \int_0^\infty e^{n \ln x} e^{-x} dx = \int_0^\infty e^{n \ln x - x} dx \\ &= \int_0^\infty e^{n \ln ny - ny} n dy \quad (\text{changing variables to } x = ny \text{ for applying the Laplace's method}) \\ &= \int_0^\infty n e^{n \ln n} e^{n(\ln y - y)} dy = n e^{n \ln n} \int_0^\infty e^{n(\ln y - y)} dy = n \cdot n^n \int_0^\infty e^{n(\ln y - y)} dy \\ &\approx n \cdot n^n \sqrt{\frac{2\pi}{n}} e^{-n} \\ &= \sqrt{2\pi n} \left(\frac{n}{e}\right)^n \end{aligned}$$

[Euler-Maclaurin formula] for  $B_{2k}$ : Bernoulli numbers

$$\sum_{i=m}^n f(i) \approx \int_m^n f(x) dx + \frac{f(m) + f(n)}{2} + \sum_{k=1}^p \frac{B_{2k}}{(2k)!} \left( f^{(2k-1)}(n) - f^{(2k-1)}(m) \right)$$

[Robbins Inequality]

$$\ln n! = \sum_{k=1}^n \ln k = \left(n + \frac{1}{2}\right) \ln n - n + \ln \sqrt{2\pi} + \mu(n)$$

For  $\mu(n)$ ,

$$\mu(n) - \mu(n+1) = \frac{1}{3(2n+1)^2} + \frac{1}{5(2n+1)^4} + \frac{1}{7(2n+1)^6} + \dots$$

where  $x = \frac{1}{2n+1}$ , because

$$\mu(n) - \mu(n+1) = \left(n + \frac{1}{2}\right) \ln\left(1 + \frac{1}{n}\right) - 1.$$

Then, the upper bound is

$$\mu(n) - \mu(n+1) > \frac{1}{3(2n+1)^2} > \frac{1}{12n(n+1)} = \frac{1}{12n} - \frac{1}{12(n+1)} \implies \mu(n) < \frac{1}{12n}$$

and the lower bound is

$$\mu(n) - \mu(n+1) < \frac{1}{12n+1} - \frac{1}{12(n+1)+1} \implies \mu(n) > \frac{1}{12n+1}$$

Therefore,

$$\ln \sqrt{2\pi n}^{(n+1/2)} / e^n + \ln e^{\frac{1}{12n+1}} < \ln n! < \ln \sqrt{2\pi n}^{(n+1/2)} / e^n + \ln e^{\frac{1}{12n}}$$

$$\therefore \sqrt{2\pi n} \left(\frac{n}{e}\right)^n e^{\frac{1}{12n+1}} < n! < \sqrt{2\pi n} \left(\frac{n}{e}\right)^n e^{\frac{1}{12n}} \quad \square$$

564

[Stirling's approximation] If  $n$ ,  $k$ , and  $n-k$  are large, applying the lower bound in Stirling's formula (4.37) to the factorials of  $\binom{n}{k}$  yields the approximation:

$$\binom{n}{k} \approx \sqrt{\frac{n}{2\pi k(n-k)}} \cdot \frac{n^n}{k^k (n-k)^{n-k}} \quad (4.38)$$

$$\begin{aligned} \binom{n}{k} &= \frac{n!}{k!(n-k)!} \\ &\approx \frac{\sqrt{2\pi n} \cdot \frac{n^n}{e^n}}{\left(\sqrt{2\pi k} \cdot \frac{k^k}{e^k}\right) \left(\sqrt{2\pi(n-k)} \cdot \frac{(n-k)^{n-k}}{e^{n-k}}\right)} \quad (\text{by Stirling's approx.}) \\ &= \sqrt{\frac{n}{2\pi k(n-k)}} \cdot \frac{n^n}{k^k (n-k)^{n-k}} \end{aligned}$$

565

[Expectation: computation rules] Assume that for the (not necessarily independent nor identically distributed) random variables  $X_1, \dots, X_k$ , the expectations  $E(X_j)$  exist. Let  $Y = a_1X_1 + \dots + a_kX_k$  with  $a_1, \dots, a_k \in \mathbb{R}$ . Then the expectation of  $Y$  exists. More precisely,

$$E(Y) = E(a_1X_1 + \dots + a_kX_k) = \sum_{j=1}^k a_j E(X_j). \quad (4.39)$$

If the random variables are iid and  $a_j = 1/k$  for each  $j \leq k$ , then  $E(Y) = E(X_1) = \dots = E(X_k)$ .

$$\begin{aligned} E(aX) &= \sum_i (ax_i)P(X = x_i) = a \sum_i x_i P(X = x_i) = aE(X) \\ \therefore E(ax) &= aE(X) \end{aligned}$$

Next, for  $P(X_1 = x_i, X_2 = y_j)$ ,

$$\begin{aligned} E(X_1 + X_2) &= \sum_i \sum_j (x_i + y_j)P(X_1 = x_i, X_2 = y_j) \\ &= \sum_i \sum_j x_i P(X_1 = x_i, X_2 = y_j) \\ &\quad + \sum_i \sum_j y_j P(X_1 = x_i, X_2 = y_j) \\ &= \sum_i x_i \left[ \sum_j P(X_1 = x_i, X_2 = y_j) \right] \\ &\quad + \sum_j y_j \left[ \sum_i P(X_1 = x_i, X_2 = y_j) \right] \\ &\quad \because \sum_j P(X_1 = x_i, X_2 = y_j) = P(X_1 = x_i) \\ &= \sum_i x_i P(X_1 = x_i) + \sum_j y_j P(X_2 = y_j) \\ &= E(X_1) + E(X_2) \end{aligned}$$

$$\therefore E(X_1 + X_2) = E(X_1) + E(X_2)$$

Moreover,

$$E(Y) = \sum_{j=1}^k \frac{1}{k} E(X_j) = \frac{1}{k} \sum_{j=1}^k \mu = \frac{1}{k} \cdot k\mu = \mu$$



566

[Variance: computation rules] Assume that for the independent (but not necessarily identically distributed) random variables  $X_1, \dots, X_k$ , the variances  $\text{Var}(X_j)$  exist. Let  $Y = a_1 X_1 + \dots + a_k X_k$  for  $a_1, \dots, a_k \in \mathbb{R}$ . Then the expectation of  $Y$  exists. More precisely,

$$\text{Var}(Y) = \text{Var}(a_1 X_1 + \dots + a_k X_k) = \sum_{j=1}^k a_j^2 \text{Var}(X_j). \quad (4.40)$$

If we drop the assumption that the random variables  $X_1, \dots, X_k$  are independent, then (4.40) becomes more complicated:

$$\begin{aligned} \text{Var}(Y) &= \text{Var}(a_1 X_1 + \dots + a_k X_k) \\ &= \sum_{j=1}^k a_j^2 \text{Var}(X_j) + \sum_{i \neq j} a_i a_j \text{Cov}(X_i, X_j). \end{aligned} \quad (4.41)$$

$$\begin{aligned} \text{Var}(Y) &= E \left[ \left( \sum_{j=1}^k a_j X_j - \sum_{j=1}^k a_j \mu_j \right)^2 \right] \quad \text{by (4.11)} \\ &= E \left[ \left( \sum_{j=1}^k a_j (X_j - \mu_j) \right)^2 \right] \\ &= E \left[ \sum_{j=1}^k a_j^2 (X_j - \mu_j)^2 + \sum_{i \neq j} a_i a_j (X_i - \mu_i)(X_j - \mu_j) \right] \\ &= \sum_{j=1}^k a_j^2 E[(X_j - \mu_j)^2] + \sum_{i \neq j} a_i a_j E[(X_i - \mu_i)(X_j - \mu_j)] \\ \therefore \text{Var}(Y) &= \sum_{j=1}^k a_j^2 \text{Var}(X_j) + \sum_{i \neq j} a_i a_j \text{Cov}(X_i, X_j) \end{aligned}$$

If  $X_1, \dots, X_k$  are Independent,  $\text{Cov}(X_i, X_j) = 0$  for all  $i \neq j$ . Thus

$$\text{Var}(Y) = \sum_{j=1}^k a_j^2 \text{Var}(X_j)$$

567

Example: Expectation and variance of  $B(n, p)$ -distributed random variables.

The random variable  $Y = Y_1 + \dots + Y_n \sim B(n, p)$  if  $Y_1, \dots, Y_n$  are iid  $B(1, p)$ -distributed. By (4.39) and (4.40) we conclude that  $E(Y) = E(Y_1) + \dots + E(Y_n) = np$  and  $\text{Var}(Y) = \text{Var}(Y_1) + \dots + \text{Var}(Y_n) = np(1 - p)$ .



568

[Central Limit Theorem (CLT)] Assume that the **real-valued** random variables  $X_1, X_2, \dots$  are **iid** with expectation  $\mu$  and variance  $\sigma^2$ . For  $n = 1, 2, \dots$ ,

$$S_n^* := \frac{X_1 + \dots + X_n - n\mu}{\sqrt{n}\sigma} \quad (4.42)$$

define normalized partial sums. The Central Limit Theorem (CLT) applies to the sequence  $X_1, X_2, \dots$ . More precisely,

$$\lim_{n \rightarrow \infty} \text{Prob}(S_n^* \leq x) = \Phi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x e^{-\frac{t^2}{2}} dt \quad \text{for all } x \in \mathbb{R}. \quad (4.43)$$

확률변수의 합 = convolution; 이항정리에서  $\binom{2n}{n}$ 이 가장 큰 값인 것처럼 평균이 될 수 있는 확률변수 tuple의 조합이 가장 많고 꼬리 값이 나오는 조합이 가장 적으므로, convolution을 하면 끝 값의 convolution estimate value는 점점 낮아지고 중앙값의 convolution estimate value는 점점 높아지는 형태가 된다. 결과적으로 확률 변수를 많이 더할수록 분포는 평균 값에 대하여 가장 높은 확률을 가지고 평균과 멀어질수록 낮은 확률을 가지는 완만한 형태를 보이게 된다. 한편, convolution은 주변 값과 평균을 맞춰주므로 그래프가 완만한 형태를 가지게 된다.

First, we need a few facts about moment generating functions(mgf). Recall that the mgf of a r.v.  $X$  is  $M_X(t) = E(e^{tX})$ . We'll write this as

$$\int_{-\infty}^{\infty} e^{tx} f(x) dx$$

and think of  $f$  as a continuous random variable, although the result holds for discrete r.v.s as well. If we differentiate the mgf  $r$  times, we get

$$M^{(r)}(t) = \frac{d^r}{dt^r} \int_{-\infty}^{\infty} e^{tx} f(x) dx = \int_{-\infty}^{\infty} \frac{d^r}{dt^r} [e^{tx} f(x)] dx = \int_{-\infty}^{\infty} x^r e^{tx} f(x) dx.$$

The second equality requires some justification, in order to interchange differentiation and integration, but this should seem believable.

Therefore, we get

$$M^{(r)}(0) = \int_{-\infty}^{\infty} x^r e^{0 \cdot x} f(x) dx = E(X^r).$$

**Thm.1** Let  $F_n$  be a sequence of cumulative distribution functions(cdf) with the correspond mgfs  $M_n$ . Let  $F$  be a cdf with the mgf  $M$ . If  $M_n(t) \rightarrow M(t)$  for all  $t$  in an open interval containing zero, then  $F_n(x) \rightarrow F(x)$  for all  $x$  at which  $F$  is continuous.

This says that if the moment generating functions of a sequence of distributions approach some limit, then that limiting mgf is the mgf of the limit of that sequence of distributions. So in order to prove the CLT, it will be enough to show that the mgf of a standardized sum of  $n$  iid r.v.s approaches the mgf of a standard normal as  $n \rightarrow \infty$ .

**Thm.2 (CLT)** Let  $X_1, X_2, \dots$  be a sequence of independent r.v.s having mean  $\mu$  and variance  $\sigma^2$ . Let each  $X_i$  have the cdf  $P(X_i \leq x) = F(x)$  and the mgf  $M(t) = E(e^{tX_i})$ . Let  $S_n = \sum_{i=1}^n X_i$ . Then

$$\lim_{n \rightarrow \infty} P\left(\frac{S_n - \mu n}{\sigma\sqrt{n}} \leq x\right) = \Phi(x)$$

for  $-\infty < x < \infty$

*Proof.* It suffices to do the proof in the case  $\mu = 0$ . If  $\mu \neq 0$ , let  $Y_i = X_i - \mu$  for each  $i$ . Let  $T_n = Y_1 + \dots + Y_n$ . Then we have

$$\lim_{n \rightarrow \infty} P\left(\frac{S_n - \mu n}{\sigma\sqrt{n}} \leq x\right) = \lim_{n \rightarrow \infty} P\left(\frac{T_n}{\sigma\sqrt{n}} \leq x\right)$$

and so it suffices to prove the central limit thm. in the case  $\mu = 0$ . Let  $Z_n = S_n/\sigma\sqrt{n}$ . We'll show that the mgf of  $Z_n$  tends to the mgf  $M_Z(t) = \exp(t^2/2)$  of the standard normal dist. Since  $S_n$  is a sum of independent r.v.s,

$$M_{S_n}(t) = E[e^{tS_n}] = E[e^{t(X_1 + \dots + X_n)}] \stackrel{\text{indep}}{=} \prod_{i=1}^n E[e^{tX_i}] \stackrel{\text{i.d.}}{=} E[e^{tX}]^n = [M_X(t)]^n$$

and we have

$$M_{Z_n}(t) = \left[M\left(\frac{t}{\sigma\sqrt{n}}\right)\right]^n.$$

It will suffice to show that  $\lim_{n \rightarrow \infty} n \log M(t/\sigma\sqrt{n}) = t^2/2$  because of the moment function of the standard normal dist.; then we can take exponentials and use Thm.1 to get the result. Call this limit  $L$ . Let  $x = 1/\sqrt{n}$  in this limit; then we need to find

$$L = \lim_{n \rightarrow \infty} \frac{\ln M(t/\sigma\sqrt{n})}{(1/\sqrt{n})^2} = \lim_{x \rightarrow 0} \frac{\ln M(tx/\sigma)}{x^2}.$$

This is of indeterminate form  $0/0$ , since  $M(0) = 1$ . Differentiating (L'Hopital's rule) gives

$$L = \lim_{x \rightarrow 0} \frac{\frac{M'(tx/\sigma)}{M(tx/\sigma)} \cdot \frac{t}{\sigma}}{2x}$$

and we can pull out a constant to get

$$L = \frac{t}{2\sigma} \lim_{x \rightarrow 0} \frac{M'(tx/\sigma)}{xM(tx/\sigma)}.$$

This is again indeterminate of form  $0$  over  $0$ . Differentiating again gives

$$L = \frac{t}{2\sigma} \lim_{x \rightarrow 0} \frac{M''(tx/\sigma) \cdot \frac{t}{\sigma}}{M(tx/\sigma) + xM'(tx/\sigma) \cdot \frac{t}{\sigma}}.$$

Pulling out a constant and rearranging limits gives

$$L = \frac{t^2}{2\sigma^2} \frac{\lim_{x \rightarrow 0} M''(tx/\sigma)}{\lim_{x \rightarrow 0} M(tx/\sigma) + \frac{t}{\sigma} \lim_{x \rightarrow 0} xM'(tx/\sigma)} = \frac{t^2}{2\sigma^2} \frac{M''(0)}{M(0)}.$$

Now we recall  $M^{(r)}(0) = E(X^r)$ ; thus  $M(0) = E(1)$ ,  $M'(0) = E(X) = 0$ ,  $M''(0) = E(X^2) = E(X)^2 + \text{Var}(X) = 0 + \sigma^2 = \sigma^2$ . This finally gives

$$L = \frac{t^2}{2\sigma^2} \frac{\sigma^2}{1} = \frac{t^2}{2},$$

which is what we wanted. □

569

[tail of the standard normal distribution] ([GaSt77], Lemma 1.19.2) For  $x > 0$ , it is

$$\left(\frac{1}{x} - \frac{1}{x^3}\right) \frac{1}{\sqrt{2\pi}} e^{-x^2/2} \leq 1 - \Phi(x) = \Phi(-x) \leq \frac{1}{x} \frac{1}{\sqrt{2\pi}} e^{-x^2/2}. \quad (4.44)$$

For the  $\Phi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x e^{-\frac{t^2}{2}} dt$ ,

$$1 - \Phi(x) = \frac{1}{\sqrt{2\pi}} \int_x^\infty \frac{1}{u} \cdot (ue^{-u^2/2}) du \quad \text{for } x > 0.$$

and then we can calculate

$$\int_x^\infty \frac{1}{u} \cdot (ue^{-u^2/2}) du = \left[ -\frac{1}{u} e^{-u^2/2} \right]_x^\infty - \int_x^\infty \frac{1}{u^2} e^{-u^2/2} du = \frac{1}{x} e^{-x^2/2} - \int_x^\infty \frac{1}{u^2} e^{-u^2/2} du$$

using the integration by parts. Since the term  $\int_x^\infty \frac{1}{u^2} e^{-u^2/2} du$  is positive,

$$1 - \Phi(x) < \frac{1}{\sqrt{2\pi} x} e^{-x^2/2}.$$

Similarly, for the remain term  $\int_x^\infty \frac{1}{u^2} e^{-u^2/2} du$ ,

$$\int_x^\infty \frac{1}{u^2} e^{-u^2/2} du = \left[ -\frac{1}{u^3} e^{-u^2/2} \right]_x^\infty - \int_x^\infty \frac{3}{u^4} e^{-u^2/2} du = \frac{1}{x^3} e^{-x^2/2} - \int_x^\infty \frac{3}{u^4} e^{-u^2/2} du.$$

Consequently,

$$\begin{aligned} 1 - \Phi(x) &= \frac{1}{\sqrt{2\pi}} \left( \frac{1}{x} e^{-x^2/2} - \left( \frac{1}{x^3} e^{-x^2/2} - \int_x^\infty \frac{3}{u^4} e^{-u^2/2} du \right) \right) \\ &= \frac{1}{\sqrt{2\pi}} \left( \frac{1}{x} - \frac{1}{x^3} \right) e^{-x^2/2} + \frac{3}{\sqrt{2\pi}} \int_x^\infty \frac{1}{u^4} e^{-u^2/2} du. \end{aligned}$$

Since the term  $\frac{3}{\sqrt{2\pi}} \int_x^\infty \frac{1}{u^4} e^{-u^2/2} du$  is positive,

$$1 - \Phi(x) \geq \left( \frac{1}{x} - \frac{1}{x^3} \right) \frac{1}{\sqrt{2\pi}} e^{-x^2/2} \quad \square$$



570

[CLT, parameter estimation] Assume that  $X_1, X_2, \dots$  are iid  $B(1, p)$ -distributed. Then the CLT implies

$$\begin{aligned}
 \text{Prob} \left( \left| \frac{1}{N} \sum_{j=1}^N X_j - p \right| > \epsilon \right) &= \text{Prob} \left( \left| \frac{\sum_{j=1}^N X_j - Np}{N} \right| > \epsilon \right) \\
 &= \text{Prob} \left( \left| \frac{\sum_{j=1}^N X_j - Np}{\sqrt{N} \sqrt{p(1-p)}} \right| > \frac{\epsilon \sqrt{N}}{\sqrt{p(1-p)}} \right) \\
 &= 2\Phi \left( \frac{-\epsilon \sqrt{N}}{\sqrt{p(1-p)}} \right) \\
 &\leq 2\Phi(-2\epsilon \sqrt{N}). \quad (4.45)
 \end{aligned}$$

By the assumption,  $E[X_j] = p$ ,  $\text{Var}(X_j) = p(1-p)$  for all  $j$  and  $E\bar{X} = E \left[ \frac{1}{N} \sum_{j=1}^N X_j \right] = p$ ,  $\text{Var}(\bar{X}) = \frac{p(1-p)}{N}$  for  $N$  independent samples.

$$\begin{aligned}
 \text{Prob}(|\bar{X} - p| > \epsilon) &= \text{Prob} \left( \left| \frac{\sum_{j=1}^N X_j}{N} - p \right| > \epsilon \right) \\
 &= \text{Prob} \left( \left| \sum_{j=1}^N X_j - Np \right| > N\epsilon \right) \\
 &= \text{Prob} \left( \frac{|\sum X_j - Np|}{\sqrt{Np(1-p)}} > \frac{\epsilon \sqrt{N}}{\sqrt{p(1-p)}} \right) \\
 &= 2 \cdot \text{Prob} \left( \frac{\sum X_j - Np}{\sqrt{Np(1-p)}} < -\frac{\epsilon \sqrt{N}}{\sqrt{p(1-p)}} \right) \\
 &\quad (\text{take the negative part to modify into the form } \Phi)
 \end{aligned}$$

Moreover, the function  $f(p) = p(1-p)$  has maximum  $1/4$  where  $p = 1/2$ . Therefore,

$$\frac{-1}{\sqrt{p(1-p)}} \leq -2 \text{ and } \epsilon \sqrt{N} > 0$$

$$\begin{aligned}
 \text{Prob}(|\bar{X} - p| > \epsilon) &= 2 \text{ Prob} \left( \frac{\sum X_j - Np}{\sqrt{Np(1-p)}} < -\frac{\epsilon \sqrt{N}}{\sqrt{p(1-p)}} \right) \\
 &= 2\Phi \left( \frac{-\epsilon \sqrt{N}}{\sqrt{p(1-p)}} \right) \leq 2\Phi(-2\epsilon \sqrt{N})
 \end{aligned}$$

571

[CLT] Par. 568 formulates the Central Limit Theorem (CLT) for **iid** random variables. **The CLT is very robust and holds under weak conditions.** Under suitable conditions, the iid assumption and even the independence property can be dropped. Some special cases are covered in paragraphs 572, 583, and 584.

Background information: **If the CLT applies, the random variables  $S_1^*, S_2^*, \dots$  converge to  $N(0, 1)$  in distribution.** We do not go deeper but refer the interested reader to ([Geor15], Subsection 5.3).

572

[CLT] Assume that the real-valued random variables  $X_1, X_2, \dots$  are independent (but not necessarily iid) with expectations  $E(X_j) = \mu_j$  and variances  $\text{Var}(X_j) = \sigma_j^2$  for  $j \in \mathbb{N}$ . For  $n = 1, 2, \dots$ ,

$$S_n^* := \frac{\sum_{j=1}^n (X_j - \mu_j)}{\sqrt{s_n^2}} \quad \text{with } s_n^2 := \sum_{j=1}^n \sigma_j^2 \quad (4.46)$$

defines normalized partial sums.

$$\begin{aligned} E[S_n^*] &= E \left[ \frac{\sum_{j=1}^n (X_j - \mu_j)}{s_n} \right] = \frac{1}{s_n} \sum_{j=1}^n (E[X_j] - \mu_j) = \frac{1}{s_n} \sum_{j=1}^n (\mu_j - \mu_j) = 0 \\ \text{Var}(S_n^*) &= \text{Var} \left[ \frac{\sum_{j=1}^n (X_j - \mu_j)}{s_n} \right] = \frac{1}{s_n^2} \sum_{j=1}^n \text{Var}(X_j - \mu_j) = \frac{1}{s_n^2} \sum_{j=1}^n \sigma_j^2 = \frac{s_n^2}{s_n^2} = 1 \end{aligned}$$

Assume further that the Lindeberg condition holds

$$\begin{aligned} \lim_{n \rightarrow \infty} L_n(\delta) &= 0 \quad \text{for all } \delta > 0 \\ \text{where } L_n(\delta) &:= \frac{1}{s_n^2} \sum_{j=1}^n E[(X_j - \mu_j)^2 1_{\{|X_j - \mu_j| \geq \delta s_n\}}]. \end{aligned} \quad (4.47)$$

The equation

$$L_n(\delta) := \frac{1}{s_n^2} \sum_{j=1}^n E[(X_j - \mu_j)^2 \cdot 1_{\{|X_j - \mu_j| \geq \delta s_n\}}] \rightarrow 0$$

implies that the sum of the  $(X_j - \mu_j)^2$  exceeding a certain value  $\delta s_n$  converges to 0 as  $n \rightarrow \infty$ . In other words, there is no r.v. indicating overwhelming deviation.

Then the Central Limit Theorem (CLT) applies to the sequence  $X_1, X_2, \dots$ . In particular,

$$\lim_{n \rightarrow \infty} \text{Prob}(S_n^* \leq x) = \Phi(x) \quad \text{for all } x \in \mathbb{R}. \quad (4.48)$$

다음 쪽에서는 대략적인 식의 성립 과정을 보임. 구체적인 증명은 [link] 참고



Let  $Y_j = \frac{X_j - \mu_j}{s_n}$  is a characteristic function. Then,  $S_n^* = \sum Y_j$ . The characteristic function of  $S_n^*$  is

$$\varphi_{S_n^*}(t) = \prod_{j=1}^n \varphi_{Y_j}(t) = \prod_{j=1}^n E[e^{itY_j}] \quad (\varphi_X(t) := E[e^{itX}])$$

with the independent property.

Applying Taylor series on  $E[e^{itY_j}]$  at  $t = 0$ ,

$$\begin{aligned} e^{itY_j} &= 1 + itY_j + \frac{(itY_j)^2}{2} + R(itY_j) \\ &= 1 + itY_j - \frac{t^2 Y_j^2}{2} + R(itY_j) \\ \varphi_{Y_j}(t) &= E[e^{itY_j}] = E[1] + itE[Y_j] - \frac{t^2 E[Y_j^2]}{2} + E[R(itY_j)] \\ &= 1 - \frac{t^2 \sigma_j^2}{2s_n^2} + E[R(itY_j)] \quad (\because E[Y_j] = 0, E[Y_j^2] = \text{Var}(Y_j) = \frac{\sigma_j^2}{s_n^2}) \end{aligned}$$

where  $R_{nj}(t)$  is remainder.

Next, use the property  $\ln(1+x) \approx x$ .

$$\begin{aligned} \ln \varphi_{S_n^*}(t) &= \sum_{j=1}^n \ln \left( 1 - \frac{t^2 \sigma_j^2}{2s_n^2} + R_{nj}(t) \right) \\ &\approx \sum_{j=1}^n \left( -\frac{t^2 \sigma_j^2}{2s_n^2} + R_{nj}(t) \right) \\ &= -\frac{t^2}{2s_n^2} \sum \sigma_j^2 + \sum R_{nj}(t) \\ &= -\frac{t^2}{2} + \sum R_{nj}(t). \end{aligned}$$

If the condition of (4.47) satisfies,  $\sum R_{nj}(t) \rightarrow 0$  Higher-order terms converge faster than the first and second-order terms ( $\because$  Lindeberg condition holds) as  $n \rightarrow \infty$ . Consequently

$$\lim_{n \rightarrow \infty} \ln \varphi_{S_n^*}(t) = -\frac{t^2}{2} \implies \lim_{n \rightarrow \infty} \varphi_{S_n^*}(t) = e^{-t^2/2}$$

By the Levy's Continuity Theorem, the characteristic function converges to that of normal distribution. Thus, r.v.  $S_n^*$  converges in normal distribution.

$$\lim_{n \rightarrow \infty} \text{Prob}(S_n^* \leq x) = \Phi(x)$$

573

[CLT] Assume that the random variables  $X_1, X_2, \dots$  are **iid** and that besides  $E(X_1)$  and  $E(X_1^2)$ , also the third moment  $E(X_1^3)$  exist. Then the well-known Berry-Esséen-Theorem provides an upper bound for the maximal difference between the exact cumulative distribution function of  $S_n^*$  and  $\Phi(\cdot)$ .  $n \rightarrow \infty$ 를 이야기하지 않음. 즉, standard normal dist.를 만들기 위해 필요한 확률 변수의 수  $n$ 을 구하기 위한 식으로 사용 가능함 It is

$$(\text{Berry-Esséen-Theorem}) \quad |\text{Prob}(S_n^* \leq x) - \Phi(x)| \leq C \frac{E(|X_1 - E(X_1)|^3)}{(\text{Var}(X_1))^{1.5}} \frac{1}{\sqrt{n}} \quad \text{for each } x \in \mathbb{R} \quad (4.49)$$

for a suitable constant  $C$  (cf. [Geor15], Bemerkung (5.31), with  $C = 0.8$ ). In [Shev11], it is proved that  $C < 0.4748$ . In particular, (4.49) says that the rate of convergence is  $O(n^{-0.5})$ .

First, we assume that  $X_1, X_2, \dots, X_n$  are iid and satisfy the follow conditions.

$$E(X_1) = 0, \quad \text{Var}(X_1) = \sigma^2, \quad E|X_1|^3 = \rho < \infty, \quad S_n^* = \frac{\sum_{j=1}^n X_j}{\sigma\sqrt{n}}$$

The most powerful tool for proving Esseen's Smoothing Lemma is the inequality below, which transforms a gap between CDFs into a gap between characteristic functions. The following conditions hold for all  $T > 0$ .

$$\sup_{x \in \mathbb{R}} |F_n(x) - \Phi(x)| \leq \frac{k}{T} + \frac{1}{\pi} \int_{-T}^T \left| \frac{\varphi_n(t) - e^{-t^2/2}}{t} \right| dt$$

where  $\varphi_n(t) = E[e^{itS_n^*}]$  is the characteristic function of  $S_n^*$  and  $k$  is a constant.

We expand the characteristic function  $\varphi(t)$  of  $X_j$  to third-order at  $t = 0$ .

$$\varphi(t) = 1 - \frac{\sigma^2 t^2}{2} + \theta \frac{\rho |t|^3}{6}, \quad (|\theta| \leq 1)$$

The characteristic function  $\varphi_n(t)$  of the standardized sum  $S_n^*$  is shown as follows due to its independence.

$$\varphi_n(t) = \left[ \varphi\left(\frac{t}{\sigma\sqrt{n}}\right) \right]^n = \left[ 1 - \frac{t^2}{2n} + \theta \frac{\rho |t|^3}{6\sigma^3 n^{3/2}} \right]^n$$

Next, we estimate the error using the log approximation  $\ln(1+z) \approx z$  as  $n \rightarrow \infty$ .

$$\ln \varphi_n(t) \approx n \left( -\frac{t^2}{2n} + \frac{\rho |t|^3}{6\sigma^3 n^{3/2}} \right) = -\frac{t^2}{2} + \frac{\rho |t|^3}{6\sigma^3 \sqrt{n}}$$

Let  $T = \frac{\sigma^3 \sqrt{n}}{4\rho}$ . Then

$$|\varphi_n(t) - e^{-t^2/2}| \leq \frac{\rho |t|^3}{\sigma^3 \sqrt{n}} e^{-t^2/4}.$$

Now we substitute it into the integration of Esseen's Smoothing Lemma.

$$\int_{-T}^T \frac{|\varphi_n(t) - e^{-t^2/2}|}{|t|} dt \leq \frac{\rho}{\sigma^3 \sqrt{n}} \int_{-\infty}^{\infty} t^2 e^{-t^2/4} dt$$

Because the integral term on the right side converges to a constant, the full term is proportional to  $\frac{\rho}{\sigma^3 \sqrt{n}}$ .

The first term  $k/T$  is also proportional to  $T \propto \sqrt{n}$ , thus it is of the same order as  $1/\sqrt{n}$ . Eq.(4.49) is derived by combining two terms as follows.

$$|\Pr(S_n^* \leq x) - \Phi(x)| \leq C \cdot \frac{E|X_1 - E(X_1)|^3}{(\text{Var}(X_1))^{1.5}} \cdot \frac{1}{\sqrt{n}}.$$

As a result, the error size decreases at a rate of  $1/\sqrt{n}$ ;  $O(n^{-0.5})$ . Additionally, the case where  $C < 0.4748$  has been examined.

574

A sequence  $X_1, X_2, \dots$  of random variables is called  **$q$ -dependent** if the random vectors  $(X_1, \dots, X_u)$  and  $(X_v, \dots, X_n)$  are **independent** for all  $1 \leq u < v \leq n$  with  $v - u > q$ .

Note: The components of each vector need not be independent.

Simply, it means “if variables are separated by a distance greater than  $q$ , they have no relationship (they are independent).”

The formula  $v - u > q$  implies the following:

- First Group: The set from  $X_1$  to  $X_u$  (past data)
- Second Group: The set from  $X_v$  to  $X_n$  (future data)
- Condition: If the distance between the start of the second group ( $v$ ) and the end of the first group ( $u$ ) is greater than  $q$ , these two groups are mathematically independent.

The ‘Note’ at the bottom of the image is a crucial distinction. This means that while groups separated by a large distance are independent, items within the same group or close neighbors (like  $X_1$  and  $X_2$ ) do not need to be independent (i.e., they can have dependence).



575

CLT의 R.V.들이 IID라는 가정을 거리가  $q$ 보다 멀 때 independent라는 가정으로 완화함

[CLT for  $q$ -dependent random variables, [HoRo48]] Let  $X_1, X_2, \dots$  be a  $q$ -dependent (not necessarily stationary) sequence of random variables such that  $E[|X_i|^3]$  is uniformly bounded for all  $i \in \mathbb{N}$ .

$$A_i := \text{Var}(X_{i+q}) + 2 \sum_{j=1}^q \text{Cov}(X_{i+q-j}, X_{i+q}) \quad \text{for } i \in \mathbb{N}. \quad (4.50)$$

Since,

$$\text{Var}(S_n) = \sum_{j=1}^n \text{Var}(X_j) + 2 \sum_{1 \leq j < k \leq n} \text{Cov}(X_j, X_k)$$

for the sum of r.v.s  $S_n = \sum_{j=1}^n X_j$ .

By the definition of the  $q$ -dependent,  $\text{Cov}(X_i, X_k) = 0$  if  $|k - j| > q$ . Thus, the series of the covariance term can be truncated at the  $q$ -th term.

$$\text{Var}(S_n) = \sum_{j=1}^n \text{Var}(X_j) + 2 \sum_{j=q+1}^n \sum_{k=1}^q \text{Cov}(X_j, X_{j-k}) + C$$

where  $C$  indicates truncated terms.

For any (time) index  $i$ ,  $A_i$  is the sum of the variance at  $i + q$  and the covariance of  $q$  terms.

$$A_i = \text{Var}(X_{i+q}) + 2 \sum_{j=1}^q \text{Cov}(X_{i+q-j}, X_{i+q}).$$

Then,  $\sum A_i$  forms  $S_n$  excluding the boundaries of each  $A_i$ . As a result, each  $A_i$  contributes as ‘Incremental Variance’ to show the average rate at which variance increases as  $n$  increases.

If the limit  $A := \lim_{u \rightarrow \infty} u^{-1} \sum_{h=1}^u A_{i+h}$  exists uniformly for all  $i \in \mathbb{N}$ , then

$$\begin{aligned} \lim_{n \rightarrow \infty} \text{Prob} \left( \frac{\sum_{j=1}^n (X_j - E(X_j))}{\sqrt{An}} \leq x \right) &= \Phi(x) \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x e^{-\frac{t^2}{2}} dt \quad \text{for all } x \in \mathbb{R}. \end{aligned} \quad (4.51)$$

It shows that the convergence of the distribution of the total sum as  $n$  increases.

First, let  $A$ , the average rate at which variance increases, as follow.

$$A := \lim_{u \rightarrow \infty} u^{-1} \sum_{h=1}^u A_{i+h}$$

If it exists,  $\text{Var}(S_n)$  increases linearly with  $A_n$  as  $n \rightarrow \infty$ ;

$$\mathrm{Var}(S_n) \stackrel{q\text{-dep}}{=} \sum_{i=1}^n A_i \stackrel[\text{exist}]{\text{uniformly}}{\approx} An > 0.$$

For using Bernstein's Blocking Method, we split the number of r.v.s  $n$  into  $p$  and  $q$ ;  $n = k(p + q) + r$  ( $0 \leq r < p + q$ ). It serves block sizes;  $p$ -size large and  $q$ -size small blocks. Let

$$U_j = \sum_{i=(j-1)(p+q)+1}^{(j-1)(p+q)+p} (X_i - E[X_i]), \quad j = 1, \dots, k$$

$$V_j = \sum_{i=(j-1)(p+q)+p+1}^{j(p+q)} (X_i - E[X_i]), \quad j = 1, \dots, k$$

$$R_n = \sum_{i=k(p+q)+1}^n (X_i - E[X_i]).$$

$$\begin{array}{ccccccccc} U_1 & V_1 & U_2 & V_2 & U_3 & \dots & U_k & V_k & R_n \\ X_1 & X_2 & X_3 & X_4 & X_5 & X_6 & X_7 & X_8 & X_9 & X_{10} & X_{11} & X_{12} & X_{13} & X_{14} & X_{15} & X_{16} & X_{17} & X_{18} & X_{19} & X_{20} & X_{21} & \dots & X_{n-10} & X_{n-9} & X_{n-8} & X_{n-7} & X_{n-6} & X_{n-5} & X_{n-4} & X_{n-3} & X_{n-2} & X_{n-1} & X_n \end{array}$$

$U_j$  represents the sum of  $p$ -size blocks and  $V_j$  is the sum of  $q$ -size blocks ( $R_n$  is remainder). Then,

$$S_n = \sum_{j=1}^k U_j + \sum_{j=1}^k V_j + R_n.$$

According to the definition of the  $q$ -dependence,  $X_i$  and  $X_j$  are independent if  $|i - j| > q$ ,  $U_j$  and  $U_m$  are independent for any  $j \neq m$  because the last index of  $U_i$  is  $(j - 1)(p + q) + p$  and the first index of  $U_{j+1}$  is  $j(p + q) + 1$ ;  $\{j(p + q) + 1\} - \{(j - 1)(p + q) + p\} = q + 1 > q$ . Next, we adjust the parameters so that  $p \rightarrow \infty$  and  $\frac{q}{p} \rightarrow 0$  as  $n \rightarrow \infty$ . Based on the assumption,  $\text{Var}(S_n) \sim An$  and  $E[V_j^2] \approx Aq$ , where the block size of  $V_j$  is  $q$ . Since  $V_j$  are independent, and  $k = \frac{n-r}{p+q} \approx n/p \because p \gg q, p \gg r$ ,

$$\frac{\text{Var}(\sum V_j)}{\text{Var}(S_n)} \approx \frac{k \cdot Aq}{An} \approx \frac{(n/p)Aq}{An} = \frac{q}{p} \rightarrow 0.$$

By the Chebyshev's inequality,  $\sum V_j$  converges to 0 in probability as  $n \rightarrow \infty$ .

Let expectation of the r.v.  $X$  be  $E(X) = \mu$ , and the variance be the  $\text{Var}(X) = \sigma^2$ .

$$P(|X - \mu| \geq \epsilon) \leq \frac{\sigma^2}{\epsilon^2}$$

for given positive constant  $\epsilon > 0$ .

Similarly, the block size of  $R_n$  is less than  $p + q$ ,

$$\frac{\text{Var}(R_n)}{An} \leq \frac{k \cdot Aq}{An} \rightarrow 0 \quad \text{and} \quad R_n \rightarrow 0.$$

By the Slutsky's Theorem, the limiting distribution of  $S_n/\sqrt{An}$  depends only on the  $(\sum U_j)/\sqrt{An} \xrightarrow{d} \sum V_j \rightarrow 0, R_n \rightarrow 0$ , which are the sum of the independent terms.

Assume that the following conditions hold for the two r.v.s  $X_n$  and  $Y_n$ :

$$X_n \xrightarrow{d} X \quad \text{and} \quad Y_n \xrightarrow{p} c.$$

Then, these conditions apply.

1.  $X_n + Y_n \xrightarrow{d} X + c$
2.  $X_n Y_n \xrightarrow{d} cX$
3.  $X_n/Y_n \xrightarrow{d} X/c$  if  $c \neq 0$

Now we check Lyapunov's condition to show that the CLT holds for the independent  $U_j$ .

$$\lim_{n \rightarrow \infty} \frac{1}{s_n^3} \sum_{i=1}^n E[|X_i - \mu_i|^3] = 0$$

Since  $E|X_i|^3$  is uniformly bounded,

$$\frac{\sum_{i=1}^k E|U_j|^3}{s_k^3} = \frac{\sum_{j=1}^k E|U_j|^3}{(\text{Var}(\sum U_j))^{3/2}} \leq \frac{k \cdot C \cdot p^{3/2}}{(Akp)^{3/2}} = \frac{k \cdot C}{(Ak)^{3/2}} = \frac{C}{A^{3/2} \cdot \sqrt{k}} \approx \frac{1}{\sqrt{k}} \rightarrow 0$$

where  $C$  is a proportional constant of the third moment of the  $X_i$ . Therefore standardized  $\sum U_j$  converges to the standard normal distribution because this condition holds for the given  $U_i$ .

Finally, First condition of the Slutsky's Theorem

$$\frac{S_n}{\sqrt{An}} = \underbrace{\frac{\sum U_j}{\sqrt{An}}}_{X_n} + \underbrace{\frac{\sum V_j + R_n}{\sqrt{An}}}_{Y_n} \xrightarrow{d} N(0, 1) + 0$$

and then

$$\lim_{n \rightarrow \infty} \text{Prob} \left( \frac{\sum_{j=1}^n (X_j - E(X_j))}{\sqrt{An}} \leq x \right) = \Phi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x e^{-\frac{t^2}{2}} dt$$

for all  $x \in \mathbb{R}$ . □



576

[CLT for q-dependent random variables] If the random variables  $X_1, X_2, \dots$  in par. 575 are **stationary**, the necessary conditions simplify considerably: It suffices that  $E(|X_i|^3)$  and

$$A = \sigma^2 := \text{Var}(X_1) + 2 \sum_{j=1}^q \text{Cov}(X_1, X_{1+j}). \quad (4.52)$$

exist.

Regarding eq.(4.50)

$$A_i = \text{Var}(X_{i+q}) + 2 \sum_{j=1}^q \text{Cov}(X_{i+q-j}, X_{i+q}),$$

the stationary property provides the following conditions.

$$\text{Var}(X_{i+q}) = \text{Var}(X_1), \quad \text{Cov}(X_{i+q-j}, X_{i+q}) = \text{Cov}(X_1, X_{1+j})$$

Therefore,  $A_i$  is constant as follows for all  $i \in \mathbb{N}$ ,

$$A_i = \text{Var}(X_1) + 2 \sum_{j=1}^q \text{Cov}(X_1, X_{1+j})$$

and we then redefined it as  $A$  or  $\sigma^2$ , that is,

$$A = \sigma^2 := \text{Var}(X_1) + 2 \sum_{j=1}^q \text{Cov}(X_1, X_{1+j}).$$

In particular, for  $\mu := E(X_1)$ , we obtain the equivalent to (4.51)

$$\lim_{n \rightarrow \infty} \text{Prob} \left( \frac{\sum_{j=1}^n (X_j - \mu)}{\sigma \sqrt{n}} \leq x \right) = \Phi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x e^{-\frac{t^2}{2}} dt \quad \text{for all } x \in \mathbb{R}. \quad (4.53)$$

Eq.(4.51) is

$$\lim_{n \rightarrow \infty} \text{Prob} \left( \frac{\sum_{j=1}^n (X_j - E(X_j))}{\sqrt{An}} \leq x \right) = \Phi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x e^{-\frac{t^2}{2}} dt.$$

$E[X_j] = \mu$  is trivial for the stationary r.v.s and we obtain  $A = \sigma^2$  in eq.(4.52). Therefore,

$$\lim_{n \rightarrow \infty} \text{Prob} \left( \frac{\sum_{j=1}^n (X_j - \mu)}{\sigma \sqrt{n}} \leq x \right) = \Phi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x e^{-t^2/2} dt.$$

577

[CLT, dependent random variables] The CLT may even hold if  $X_1, X_2, \dots$  has no finite memory, provided that the dependencies decrease sufficiently fast. If the sequence  $X_1, X_2, \dots$  is stationary and, e.g., strongly mixing, the CLT holds if some further conditions are fulfilled. If needed, the reader is referred to [Jone04], Section 4, for details.

578

[Markov chains] Assume that the random variables  $X_0, X_1, \dots$  take on values in a countable set  $\Omega$ . Assume that

$$\text{Prob}(X_{n+1} = x_{n+1} | X_1 = x_1, \dots, X_n = x_n) = \text{Prob}(X_{n+1} = x_{n+1} | X_n = x_n) \quad (4.54)$$

for each  $n \in \mathbb{N}_0$  and all  $x_0, x_1, \dots, x_{n+1} \in \Omega$  provided that both conditional probabilities in (4.54) are well-defined (note: the latter is the case if  $\text{Prob}(X_1 = x_1, \dots, X_n = x_n) > 0$ ). We say that  $X_0, X_1, \dots$  is a (time-discrete) **Markov chain** on the state space  $\Omega$ . If the right-hand conditional probabilities in (4.54) do not depend on  $n$ , the Markov chain is **homogeneous**.  $\text{Pr}[X_{n+1} | X_n] = p_{i,j}$  for constant  $p_{i,j}$ ; 시간에 따라 전이 확률 행렬  $P$ 가 변하지 않음

579

[Markov chains] Condition (4.54) says that  $X_{n+1}$  may depend on  $X_n$  but any further information on the preceding random variables  $X_0, \dots, X_{n-1}$  does not provide additional information on the outcome of  $X_{n+1}$ .

Note 1: Equation (4.54) does not imply that  $X_{n+1}$  and  $X_{n-1}$  are independent.

Note 2: Condition (4.54) can be generalized to time-continuous stochastic processes on arbitrary state spaces (Markov processes).

580

[Markov chains] Assume that  $X_0, X_1, \dots$  is a homogeneous Markov chain on the finite state space  $\Omega = \{\omega_1, \dots, \omega_k\}$ . The **transition matrix**  $P = (p_{ij})_{1 \leq i, j \leq k}$  is defined by  $p_{ij} = \text{Prob}(X_{n+1} = \omega_j | X_n = \omega_i)$ . If the row vector  $\nu_j$  denotes the distribution of  $X_j$ , then  $\nu_{n+1} = \nu_n P$ . distribution  $\nu_j$  is  $1 \times k$  matrix with coordinate function that  $w_i$  into  $\nu_j(w_i)$ .

$$P = (p_{ij})_{1 \leq i, j \leq k}, \quad p_{ij} = \text{Pr}[X_{n+1} = \omega_j | X_n = \omega_i]$$

row vector  $\nu_l$ :  $X_l$ 의 분포 ;  $\nu_l = (\text{Prob}(X_l = \omega_1), \text{Prob}(X_l = \omega_2), \dots, \text{Prob}(X_l = \omega_k))$



$$\begin{aligned}
\Pr[X_{n+1} = \omega_j] &= \sum_{i=1}^k \Pr[X_{n+1} = \omega_j, X_n = \omega_i] \\
&= \sum_{i=1}^k \Pr[X_{n+1} = \omega_j \mid X_n = \omega_i] \Pr[X_n = \omega_i] \\
\nu_{n+1}(\omega_j) &= \sum_{i=1}^k p_{ij} \nu_n(\omega_i) = (\nu_n P)(\omega_j) \quad (\because \nu_{n+1} = \nu_n P)
\end{aligned}$$

Note: In the literature on Markov chains, traditionally row vectors are used instead of column vectors.

Generally, matrices are denoted as  $P_{k \times k}(\nu_n)_{k \times 1}$ , but since time progression is often thought of as moving from left to right,  $\nu_n$  is frequently represented as a row vector:  $(\nu_n)_{1 \times k} P_{k \times k}$ .  
(time)  $\longrightarrow$

581

[Markov chains] Assume that  $X_0, X_1, \dots$  is a homogeneous Markov chain  $\Pr[X_{n+1} = x_{n+1} \mid X_n = x_n]$  do not depend on  $n$  on the finite state space  $\Omega = \{\omega_1, \dots, \omega_k\}$  with transition matrix  $P$ . Assume further that there is an integer  $m \in \mathbb{N}$  for which all entries of  $P^m$  are positive Regular  $\implies$  irreducible, aperiodic; 충분한 시간인  $m$ 번의 step이 지나면, 시작 위치와 관계 없이 공간 내 모든 상태를 지날 확률이 0보다 큼. Then for each initial distribution  $\nu_0$  시작 위치와 관계 없음을 의미, the sequence of distributions  $\nu_0, \nu_1, \dots$  converges to a limit distribution  $\nu$   $\nu_n = \nu_0 P^n$ 은 유일한 극한 분포  $\nu$ 로 수렴; 즉,  $\lim_{n \rightarrow \infty} \nu_n = \nu$   $\lim_{i \rightarrow \infty} \Pr(X_i = \omega_j) = \nu(\omega_j)$  for all  $j = 1, \dots, k$  with  $\nu(\omega_j) > 0$  모든 상태가 장기적으로 유의미한 전이 확률을 가짐 for all  $j \leq k$ . The limit distribution  $\nu$  is the unique left eigenvector of  $P$   $\nu = \nu P$ , 즉, 극한 분포  $\nu$ 는  $P$ 로 한 스텝을 더 곱해도 변하지 않는 stationary dist. to the eigenvalue 1  $\sum \nu_j = 1$ 로 정규화  $\because$  eigenvalue = 1  $\implies P$  keeps the norm of the vector. The convergence rate is exponentially (e.g., [Geor15], Subsection 6.3.1).

[Perron-Frobenius Theorem] If  $P$  is a non-negative matrix and there exists an integer  $m$  such that all entries of  $P^m$  are positive, then  $P$  has a unique largest positive eigenvalue  $\lambda_1$  and the corresponding left and right eigenvectors have strictly positive components.

The transition matrix  $P$  is a stochastic matrix, which means that each row of  $P$  sums to 1  $\sum_i p_{ij} = \sum_i \Pr[X_{n+1} = \omega_j \mid X_n = \omega_i]$ , so it has the eigenvalue  $\lambda_1 = 1$ . The condition that all entries of  $P^m$  are positive means that the other eigenvalues  $\lambda_2, \dots, \lambda_k$  satisfy  $|\lambda_i| < 1$  for  $i = 2, \dots, k$ . Then, by the squaring of the transition matrix  $P^n$  as  $n \rightarrow \infty$ ,  $|\lambda_i|^n$  are exponentially decreasing to 0 except for  $\lambda_1$ . Therefore, the limit distribution  $\nu$  is the unique left eigenvector of  $P$  to the eigenvalue 1, and the convergence rate is exponentially fast.

582

If the Markov chain from par. 581 already starts in the equilibrium state, namely if  $\nu_0 = \nu$   $\nu_0$ 가 처음부터 극한 분포  $\nu$ 인 상태, then  $\nu_0 = \nu_1 = \dots = \nu$ . Then the Markov chain  $X_0, X_1, \dots$  is stationary 분포가 항상  $\nu$ 로 일정하게 유지됨. and ergodic stationary  $\implies$  ergodic (see, e.g., [Geor 15], Subsection 6.3.1).

If the Markov chain starts in the equilibrium state  $\nu$ , then  $\nu_0 = \nu$  and

$$\nu_1 = \nu_0 P = \nu P = \nu.$$

By induction,  $\nu_n = \nu$  for all  $n \in \mathbb{N}$ . It means the marginal probability distribution is constant. Moreover, the Markov chain is stationary. It needs to be shown that the joint distribution of  $(X_{t_1}, \dots, X_{t_k})$  is the same regardless of the time shift. Since the Markov chain is homogeneous, the transition probabilities do not depend on time, and thus the joint distribution is invariant under time shifts.

$$P(X_t = i, X_{t+1} = j) = P(X_t = i)P(X_{t+1} = j \mid X_t = i)$$

Moreover,  $P(X_t = i) = \nu(i)$  because the initial dist. is equilibrium, and the transition probabilities are  $P_{ij}$  such as

$$P(X_t = i, X_{t+1} = j) = \nu(i)P_{ij}$$

that does not depend on  $t$ . Therefore, the Markov chain is stationary holding the joint distribution of  $(X_{t_1}, \dots, X_{t_k})$  invariant under time shifts. Finally, the Markov chain is ergodic because it converges to the unique stationary distribution  $\nu$ .

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n f(X_k) \xrightarrow{a.s.} E_\nu[f(X)] = \sum_{j=1}^k f(\omega_j) \nu(\omega_j).$$



583

[CLT, Markov chain] Assume that  $X_0, X_1, \dots$  is a homogeneous Markov chain on the finite state space  $\Omega = \{\omega_1, \dots, \omega_k\}$  with transition matrix  $P$ . Assume further that the Markov chain converges to a limit distribution  $\nu$  regardless of  $\nu_0$  다른 논문들에는 stationary라고 가정함 (as in par. 581). Let  $g : \Omega \rightarrow \mathbb{R}$  any mapping. Then the Central Limit Theorem applies to  $g(X_1), g(X_2), \dots$ . The normalized partial sums (see par. 568) are given by

$$S_n^* := \frac{g(X_1) + \dots + g(X_n) - n\mu}{\sqrt{n}\sigma} \quad \text{with} \\ \mu := E[g(X_1)] \quad \text{and} \quad \sigma^2 := \text{Var}(g(X_1)) + \sum_{k=1}^{\infty} \text{Cov}(g(X_1), g(X_{1+k})) \quad (4.55).$$

The Markov chain defined over the finite state space  $\Omega$ , and each state  $\omega_i$  has not to be the number. Therefore, we need to apply the function  $g$  to transform the state  $\omega_i$  into a real number. For this way, we get the sequence of the real-valued random variables  $Y_i = g(X_i)$ , and we can apply the CLT to the sequence  $Y_1, Y_2, \dots$ .

First,  $\mu = \sum_{j=1}^n E[g(X_j)]/n = E[g(X_1)]$  is trivial because of the par. 581 and 582 ???  $\nu_0 \rightarrow \nu \implies \nu_i \rightarrow \nu$  for all  $i > 0$  and then  $X_i \rightarrow X$  regardless of the  $\nu_0$  (and  $X_0$ ). The expectation of the limit dist.  $\mu$  controls the mean of the distribution of the sum asymptotically. Then, the Markov chain is stationary and ergodic.

Next, the limit variance  $\sigma^2$  concludes the sum with  $k \rightarrow \infty$  but it is well-defined because the Markov chain converges to a limit distribution  $\nu$  as in par. 581, and thus the covariance term  $\text{Cov}(g(X_1), g(X_{1+k}))$  decreases as  $k$  increases under the assumption that the limit distribution  $\nu$  converges regardless of the initial distribution.

$$|\text{Cov}(g(X_1), g(X_{1+k}))| \leq C\rho^k \quad \text{for some constant } C > 0 \text{ and } 0 < \rho < 1.$$

On this basis, the following is satisfied with the CLT by the par. 572.

$$S_n^* := \frac{g(X_1) + \dots + g(X_n) - n\mu}{\sqrt{n}\sigma}$$

“On the Markov chain Central Limit Theorem” by Galin.

584

The version of the CLT from par. 583 also applies under more general conditions. infinite, continuous

Note: If the random variables  $X_1, X_2, \dots$  are independent (special case of a Markov chain), (4.55) coincides with (4.42), applied to the random variables  $g(X_1), g(X_2), \dots$

If  $X_1, X_2, \dots$  are independent, then  $\text{Cov}(g(X_1), g(X_{1+k})) = 0$  and therefore, the variance  $\sigma^2$  simplifies to  $\sigma^2 = \text{Var}(g(X_1))$ . It gives equation (4.42) that is the general form of the CLT for independent random variables, that is,

$$S_n^* := \frac{X_1 + \dots + X_n - n\mu}{\sqrt{n}\sigma}.$$

585

[Stationarity, stationarity in a weak sense]

- a Assume that a stochastic process  $(X_t)_{t \in T}$  is **strict** stationary and that  $E(X_t)$  and  $\text{Var}(X_t)$  exist. Then  $(X_t)_{t \in T}$  is also stationary in a weak sense.

It is trivial in general but there is a counterexample that the expectation and variance do not exist with strict stationarity: the Cauchy distribution. The conditions that expectation and variance exist are needed to ensure the mathematical well-definedness of the weak stationarity.

- b Assume that a Gaussian stochastic process  $(X_t)_{t \in T}$  (cf. par. 542) is stationary in a weak sense. Then  $(X_t)_{t \in T}$  is stationary.

In general, the weak stationarity does not imply the strict stationarity. However, for the Gaussian stochastic process, the weak stationarity implies the strict stationarity because the p.d.f. of the Gaussian dist. is fully determined by the mean and the covariance.

$$f(\mathbf{x}) = \frac{1}{\sqrt{(2\pi)^k |\boldsymbol{\Sigma}|}} \exp \left( -\frac{1}{2} (\mathbf{x} - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu}) \right)$$

That is, if the Gaussian stochastic process is weakly stationary, the mean and the covariance are constant over time, and thus the joint dist. that determine the process is strictly stationary.

Note 1: In the context of PTRNGs, the condition that expectation and variance exist should usually be fulfilled.

현실의 온도나 전압, 전류는 무한대로 발산할 수 없는 유계(Bounded) 현상이므로, 코시 분포처럼 분산이 발산하는 현상은 일어나지 않음. 따라서 현실에서 기댓값과 분산이 존재한다는 가정은 항상 충족됨.

Note 2: Assertion b) can be very useful in the evaluation of PTRNGs.

암호학에서 물리적 난수 생성기를 평가할 때, 그 출력값이 'strictly stationary'인지 검증하는 것은 무한 차원의 결합 분포를 모두 추정해야 하므로 통계적으로 불가능함. 그러나 열 잡음과 같은 많은 물리적 잡음원은 본질적으로 CLT에 의해 Gaussian process로 모델링할 수 있음. 따라서 평가자는 weak sense에서 stationarity만 평가해도 됨(표본 평균과 표본 자기상관함수(ACF)가 시간에 따라 일정한지만 확인).

586

[Stationarity] The pars. 587 to 590 provide useful facts on stationary stochastic processes for which stationarity is “inherited”. These results can be very useful for the analysis of PTRNGs.



587

[Stationarity] Proposition: Assume that the stochastic process  $(X_t)_{t \in T}$  is stationary on the state space  $\Omega = \mathbb{R}^m$  for some integer  $m \geq 1$ . Furthermore, let  $f : \Omega \rightarrow \Omega' := \mathbb{R}^s$  denote a measurable mapping (as usual, we consider the Borel- $\sigma$ -algebras on  $\mathbb{R}^m$  and  $\mathbb{R}^s$ ).

a) The stochastic process  $(f(X_t))_{t \in T}$  is stationary, too.  $\nu \neq \nu'$ 이지만,  $\nu' \circ \tau$ 에 대하여 일정하게 유지됨

*Proof:* Since  $f$  is measurable,  $f^{-1}(B') \in \mathcal{B}(\mathbb{R}^m)$  for all  $t \in T$  and  $B' \in \mathcal{B}(\mathbb{R}^s)$ , and thus the stationarity of  $(X_t)_{t \in T}$  implies that  $(f(X_t))_{t \in T}$  is stationary, too.  $\square$

b) Let  $\Delta t > 0$ . The stochastic process  $(Y_t)_{t \in T}$  given by  $Y_t := X_{t+\Delta t} - X_t$  defines a stationary process on  $\mathbb{R}^m$  (we tacitly assume that  $T + \Delta t \subseteq T$ ).

*Proof:* For given  $t_1, \dots, t_k, \tau$  we temporarily define the sets

$$S := \{t_1, t_1 + \Delta t, t_2, t_2 + \Delta t, \dots, t_k, t_k + \Delta t\} \quad \text{and} \\ S_\tau := \{t_1 + \tau, t_1 + \Delta t + \tau, t_2 + \tau, t_2 + \Delta t + \tau, \dots, t_k + \tau, t_k + \Delta t + \tau\}.$$

Assume for the moment Moments of a function in mathematics are certain quantitative measures related to the shape of the function's graph. 모든 모멘트의 집합이 확률 분포를 유일하게 결정함 that  $|S| = 2k$  (and thus also  $|S_\tau| = 2k$ ). The stationarity of  $(X_t)_{t \in T}$  implies

$$\begin{aligned} & \text{Prob}(X_{t_1+\Delta t} \in B_1, X_{t_1} \in C_1, \dots, X_{t_k+\Delta t} \in B_k, X_{t_k} \in C_k) \\ &= \text{Prob}(X_{t_1+\Delta t+\tau} \in B_1, X_{t_1+\tau} \in C_1, \dots, X_{t_k+\Delta t+\tau} \in B_k, X_{t_k+\tau} \in C_k) \\ & \quad \text{for all } B_1, C_1, \dots, B_k, C_k \in \mathcal{B}(\mathbb{R}^m). \end{aligned} \quad (4.56)$$

Since the set  $\{B'_1 \times C'_1 \times B'_2 \times \dots \times C'_k \mid B'_j, C'_j \in \mathcal{B}(\mathbb{R}^m)\}$  is stable under intersections and generates the product  $\sigma$ -algebra  $\mathcal{B}(\mathbb{R}^m) \otimes \mathcal{B}(\mathbb{R}^m) \otimes \dots \otimes \mathcal{B}(\mathbb{R}^m) = \mathcal{B}(\mathbb{R}^{2km})$  on  $\mathbb{R}^{2km}$ , the random vectors  $Z' := (X_{t_1+\Delta t}, X_{t_1}, \dots, X_{t_k+\Delta t}, X_{t_k})$  and  $Z'_\tau := (X_{t_1+\Delta t+\tau}, X_{t_1+\tau}, \dots, X_{t_k+\Delta t+\tau}, X_{t_k+\tau})$  are identically distributed on  $\mathbb{R}^{2km}$ .  $\text{dist.}(X_{t_i+\Delta t}) = \text{dist.}(X_{t_i+\Delta t+\tau}), \text{dist.}(X_{t_i}) = \text{dist.}(X_{t_i+\tau}) \implies Y_t = X_{t+\Delta t} - X_t = X_{t+\Delta t+\tau} - X_{t+\tau} = Y_{t+\tau}$  Consequently,  $\vec{Y} := (Y_{t_1}, \dots, Y_{t_k})$  and  $\vec{Y}_\tau := (Y_{t_1+\tau}, \dots, Y_{t_k+\tau})$  are identically distributed on  $\mathbb{R}^{mk}$ . In particular,

$$\begin{aligned} & \text{Prob}(Y_{t_1} \in A_1, \dots, Y_{t_k} \in A_k) = \text{Prob}(Y_{t_1+\tau} \in A_1, \dots, Y_{t_k+\tau} \in A_k) \\ & \quad \text{for all } A_1, \dots, A_k \in \mathcal{B}(\mathbb{R}^m). \end{aligned} \quad (4.57)$$

For the case  $|S| < 2k$ , an analogous procedure (considering random variables  $\vec{Z}'$  and  $\vec{Z}'_\tau$  on  $\mathbb{R}^{|S|m}$ ) also verifies (4.57). Since  $t_1, \dots, t_k$  were arbitrary, this completes the proof of b).  $\square$

c) Let  $(Z_t)_{t \in T'}$  with  $T' \subseteq T$  and  $Z_t = X_t$  for each  $t \in T'$ . Then  $(Z_t)_{t \in T'}$  is stationary, too.

*Proof:*  $\tau$ 가 존재하지 않는  $T'$ 은 고려하지 않음 Compared to  $(X_t)_{t \in T}$  the “restricted” process  $(Z_t)_{t \in T'}$  only has to satisfy a subset of conditions of type (4.22) to be stationary.  $\square$



588

[Stationarity] Proposition: Assume that the stochastic process  $(X_n)_{n \in \mathbb{N}}$  is stationary on a state space  $\Omega = \mathbb{R}^m$  for some integer  $m \geq 1$ .

- a) The stochastic process  $\vec{Y}_1, \vec{Y}_2, \dots$  with  $\vec{Y}_n := (X_{(n-1)t+1}, \dots, X_{nt})$  is stationary on  $\mathbb{R}^{mt}$ . (The random vectors  $\vec{Y}_1, \vec{Y}_2, \dots$  are non-overlapping.)

Proof: For given  $j_1, \dots, j_k, \tau$ , we temporarily define the set  $M = \{j_1, j_1 + 1, \dots, j_1 + t - 1, j_2, \dots, j_k + t - 1\}$ . The stationarity of  $(X_n)_{n \in \mathbb{N}}$  implies

$$\begin{aligned} & \text{Prob}(\vec{Y}_{j_i} \in (B_{(j_i-1)t+1} \times \dots \times B_{j_it}) \text{ for } i \leq k) \\ &= \text{Prob}(X_i \in B_i \text{ for } i \in M) \\ &= \text{Prob}(X_{i+\tau} \in B_i \text{ for } i \in M) \\ &= \text{Prob}(\vec{Y}_{j_i+\tau} \in (B_{(j_i-1)t+1} \times \dots \times B_{j_it}) \text{ for } i \leq k) \\ & \quad \text{for each } \tau \in \mathbb{N} \text{ and all } B_i \in \mathcal{B}(\mathbb{R}^m), i \in M. \end{aligned} \quad (4.58)$$

Since the set  $\{B'_1 \times B'_2 \times \dots \times B'_{kt} \mid B'_i \in \mathcal{B}(\mathbb{R}^m)\}$  is stable under intersections and generates  $\mathcal{B}(\mathbb{R}^{kmt})$  the vectors  $(\vec{Y}_1, \dots, \vec{Y}_k)$  and  $(\vec{Y}_{1+\tau}, \dots, \vec{Y}_{k+\tau})$  are identically distributed on  $\mathbb{R}^{kmt}$ . Therefore, (4.58) implies

$$\begin{aligned} & \text{Prob}(\vec{Y}_{j_i} \in A_{j_i} \text{ for } i \leq k) \\ &= \text{Prob}(\vec{Y}_{j_i+\tau} \in A_{j_i} \text{ for } i \leq k) \text{ for each } \tau \in \mathbb{N} \text{ and } A_{j_i} \in \mathcal{B}(\mathbb{R}^{mt}). \end{aligned} \quad (4.59)$$

□

- b) The stochastic process  $\vec{Z}_1, \vec{Z}_2, \dots$  with  $\vec{Z}_n := (X_n, \dots, X_{n+t-1})$  is stationary on  $\mathbb{R}^{mt}$ . (The random vectors  $\vec{Z}_1, \vec{Z}_2, \dots$  are overlapping.) stationary는 모든  $\tau$ 에 대하여 성립해야 하므로

Proof: The proof of b) is analogous to the proof of a). In the second and the third term of (4.58), components  $X_i$  can appear several times. This is not a problem. It is, for example,  $(X_i \in A', X_i \in A'')$  equivalent to  $X_i \in A' \cap A''$ . □

589

[Stationarity] Proposition: Given a (not necessarily stationary) stochastic process  $(X_n)_{n \in \mathbb{N}}$  with state space  $\Omega = \mathbb{R}^m$  for some integer  $m \geq 1$ . Furthermore, assume that the stochastic processes  $(\vec{Y}_n)_{n \in \mathbb{N}}$  (non-overlapping random vectors with  $t$  components) and  $(\vec{Z}_n)_{n \in \mathbb{N}}$  (overlapping random vectors with  $t$  components) are defined as in par. 588.

- a) Assume that the stochastic process  $(\vec{Y}_n)_{n \in \mathbb{N}}$  is stationary. Then the stochastic process  $(X_n)_{n \in \mathbb{N}}$  need not be stationary.

Proof: Counterexample ( $m = 1$ ): Let  $X_1, X_2, \dots$  denote independent binary-valued random variables. Furthermore,  $X_i \sim B(1, 0.5)$  for  $i \in \{t, 2t, \dots\}$  and  $X_i \sim B(1, 0)$  else. Then  $(\vec{Y}_n)_{n \in \mathbb{N}}$  is stationary but not  $(X_n)_{n \in \mathbb{N}}$ .  $\square$

- b) If the stochastic process  $(\vec{Z}_n)_{n \in \mathbb{N}}$  is stationary, the underlying stochastic process  $(X_n)_{n \in \mathbb{N}}$  is stationary, too.

Proof: The second equality of (4.60) follows from the stationarity of  $(\vec{Y}_n)_{n \in \mathbb{N}}$ .

$$\begin{aligned} \text{Prob}(X_{j_1} \in A_1, \dots, X_{j_k} \in A_k) &= \text{Prob}(\vec{Y}_{j_1} \in A_1 \times \Omega^{t-1}, \dots, \vec{Y}_{j_k} \in A_k \times \Omega^{t-1}) \\ &= \text{Prob}(\vec{Y}_{j_1+\tau} \in A_1 \times \Omega^{t-1}, \dots, \vec{Y}_{j_k+\tau} \in A_k \times \Omega^{t-1}) \\ &= \text{Prob}(X_{j_1+\tau} \in A_1, \dots, X_{j_k+\tau} \in A_k) \end{aligned} \quad (4.60)$$

 $\square$

590

[Stationarity] Proposition: Assume that the stochastic processes  $(X_t)_{t \in T}$  and  $(Y_t)_{t \in T}$  is stationary on the state space  $\Omega = \mathbb{R}^m$  for some integer  $m \geq 1$ .

- a) If the stochastic processes  $(X_t)_{t \in T}$  and  $(Y_t)_{t \in T}$  are independent and  $\odot \in \{+, -\}$ , the stochastic process  $(Z_t)_{t \in T}$  with  $Z_t = X_t \odot Y_t$  is stationary, too.

Proof: For given  $t_1, \dots, t_k$ , stationarity and independence imply

$$\begin{aligned} & \text{Prob}(X_{t_1} \in B_1, Y_{t_1} \in C_1, \dots, X_{t_k} \in B_k, Y_{t_k} \in C_k) \\ &= \text{Prob}(X_{t_1} \in B_1, \dots, X_{t_k} \in B_k) \cdot \text{Prob}(Y_{t_1} \in C_1, \dots, Y_{t_k} \in C_k) \\ &= \text{Prob}(X_{t_1+\tau} \in B_1, \dots, X_{t_k+\tau} \in B_k) \cdot \text{Prob}(Y_{t_1+\tau} \in C_1, \dots, Y_{t_k+\tau} \in C_k) \\ &= \text{Prob}(X_{t_1+\tau} \in B_1, Y_{t_1+\tau} \in C_1, \dots, X_{t_k+\tau} \in B_k, Y_{t_k+\tau} \in C_k) \\ & \quad \text{for all } B_1, C_1, \dots, B_k, C_k \in \mathcal{B}(\mathbb{R}^m). \end{aligned} \quad (4.61)$$

Since the set  $\{B_1 \times C_1 \times \dots \times B_k \times C_k \mid B_j, C_j \in \mathcal{B}(\mathbb{R}^m)\}$  is stable under intersection and generates the  $\sigma$ -algebra  $\mathcal{B}(\mathbb{R}^{2km})$ , and since the componentwise addition or subtraction of random vectors is measurable, as in par. 587, proof of assertion b), this implies

$$\begin{aligned} & \text{Prob}(Z_{t_1} \in A_1, \dots, Z_{t_k} \in A_k) = \text{Prob}(Z_{t_1+\tau} \in A_1, \dots, Z_{t_k+\tau} \in A_k) \\ & \quad \text{for all } A_1, \dots, A_k \in \mathcal{B}(\mathbb{R}^m). \end{aligned} \quad (4.62)$$

Since  $t_1, \dots, t_k$  were arbitrary, this completes the proof of a).  $\square$

- b) Let  $m = 1$   $m \geq 2$ 이면 곱셈, 나눗셈 연산에 대하여 차원이 달라지거나 연산의 정의를 새롭게 해야 할 수 있어 stationary를 보장할 수 없음. If the stochastic processes  $(X_t)_{t \in T}$  and  $(Y_t)_{t \in T}$  are independent and  $\odot \in \{+, -, \cdot, /\}$ , the stochastic process  $(Z_t)_{t \in T}$  with  $Z_t = X_t \odot Y_t$  is stationary, too. For  $\odot = /$  the additional condition  $\text{Prob}(X_t = 0) = 0$  is required.

Proof: Assertion b) can be proved as a).  $\square$

- c) If the stochastic processes  $(X_t)_{t \in T}$  and  $(Y_t)_{t \in T}$  are not independent, in general a) and b) are not true.

Proof: Counterexample: Let  $V_0, V_1, \dots$  denote iid  $B(1, 0.5)$ -distributed random variables,  $Z_n = X_n + Y_n$  for  $n \in \mathbb{N}$ ,  $T = \mathbb{N}$ ,  $\odot = +$ , and  $m = 1$ . Let further  $X_n = V_n$  for  $n \geq 1$ ,  $Y_1 = V_0$ , and  $Y_n = V_n$  for  $n \geq 2$ . Then

$$\begin{aligned} & \text{Prob}(X_1 + Y_1 = 1) = \text{Prob}(V_1 = 1, V_0 = 0) + \text{Prob}(V_1 = 0, V_0 = 1) = 0.5 \quad \text{but} \\ & \text{Prob}(X_n + Y_n = 1) = \text{Prob}(V_n + V_n = 1) = 0 \quad \text{for } n \geq 2. \end{aligned} \quad (4.63)$$

Hence, the stochastic process  $(Z_n)_{n \in \mathbb{N}}$  is not stationary. Analogously, the stochastic process defined by  $Z'_n = X_n - Y_n$  is not stationary. Similarly, let  $X'_n = e^{V_n}$  for  $n \geq 1$ ,  $Y'_1 = e^{V_0}$ , and  $Y'_n = e^{V_n}$  for  $n \geq 2$ . Then the stochastic processes defined by  $Z''_n := X'_n \cdot Y'_n$  and  $Z'''_n := X'_n / Y'_n$  respectively, are not stationary.  $\square$



591

[Renewal process] Assume that  $T_1, T_2, \dots$  denote iid non-negative random variables, and that the expectation  $E(T_j) > 0$  exists. Furthermore, unless otherwise stated,  $\text{Prob}(T_j = 0) = 0$   $j-1$ 번째 사건과  $j$ 번째 사건이 동시에 발생할 수 없음 and there is no  $\Delta > 0$  such that  $\text{Prob}(T_j \in \{j\Delta \mid j \in \mathbb{N}\}) = 1$  ????. Then  $Z(t) := \inf\{k \mid T_1 + \dots + T_k > t\}$  defines a **renewal process**, where  $t$  ranges in  $[0, \infty)$ . The random variables  $T_j$  are often interpreted as lifetimes of machines and called the  $j^{\text{th}}$  holding time. In the context of physical noise sources, the random variable  $T_j$  often quantifies the intermediate time between the  $(j-1)^{\text{th}}$  and the  $j^{\text{th}}$  event; see Subsections 5.4.2, 5.4.3, and 5.4.4.

592

[Renewal process] A delayed renewal process considers independent, non-negative random variables  $T_0, T_1, T_2, \dots$ . Again, the random variables  $T_1, T_2, \dots$  are iid, while  $T_0$  can have a different distribution. Furthermore, it is assumed that  $E(T_0) \geq 0$  and  $E(T_j) > 0$  exist. Also, let

$$J_n := T_0 + T_1 + \dots + T_n. \quad (4.64)$$

The delayed renewal process is given by

$$Z(t) := \inf\{k \mid T_0 + T_1 + \dots + T_k > t\} \quad (4.65)$$

Note: The renewal process from par. 591 can be interpreted as a special case of a delayed renewal process with  $T_0 \equiv 0$ . It is also called a non-delayed renewal process.

Suppose a random number generator produces random numbers at intervals  $T_1, T_2, \dots$ , which are iid. When the generator is turned on, an additional time  $T_0$  is required before 'random number generation' begins, due to boot-up and filling the entropy pool. The process that accounts for this delay is called a delayed renewal process. A situation where  $T_0 = 0$  represents the ideal case, where random numbers can be obtained at intervals  $T_i$  immediately after powering on the generator.

## 4.3 Entropy and Guesswork

### 4.3.1 Entropy

605

For the evaluation of RNGs, Shannon entropy (4.88) and min-entropy (4.89) play an outstanding role. Both can be viewed as special cases of Renyi entropy, a more general definition of entropy. Collision entropy (4.90) also has some relevance.

Given  $X$  is a random variable, it implies that the probability distribution is specified. Let  $f(X)$  be a random variable. Then, the expectation of  $f(X)$  is defined by

$$\mathbb{E}[f(X)] = \sum_x f(X = x) \Pr(X = x).$$

Information indicates the level of uncertainty in a random variable. The uncertainty measures the likelihood(probability) of correctly guessing the value of that variable. That is, a random variable with high information has a low probability of being guessed correctly.

The information of an event  $X = x$  is defined using  $-\log_2 \Pr(X = x)$ , which is useful for the following reasons:

- The function  $-\log$  is monotonically decreasing for  $0 < p(x) \leq 1$ , which means that as the probability of an event increases, the information content decreases.
- $-\log \Pr(X = x) \geq 0$  for all  $x$  because  $0 \leq \Pr(X = x) \leq 1$ .
- It satisfies the fundamental axioms of probability and inheritance information theory, such as:
  - \*  $-\log \Pr(X = x) \leq 0$ .
  - \*  $-\log \Pr(X = x) = 0$  if  $\Pr(X = x) = 1$ .
  - \*  $-\log \Pr(X = x, Y = y) = -\log \Pr(X = x) + -\log \Pr(Y = y)$  if  $X$  and  $Y$  are independent.
- $-\log \Pr(X = x)$  is continuous.

The expectation of the information is

$$\mathbb{E}[I(X)] = \mathbb{E}[-\log \Pr(X)] = \sum_x \Pr[X = x] \cdot I(X = x) = - \sum_x \Pr(X = x) \log \Pr(X = x)$$

and we call it the Shannon entropy  $H(X)$ .

606

Let  $X$  be a random variable that assumes values in the finite set  $\Omega = \{\omega_1, \omega_2, \dots, \omega_k\}$ . The most general notion of entropy is the **Renyi entropy** of order  $\alpha$  ( $\alpha = 0, 1, \infty$ 일 때는  $\lim$ 를 이용하여 정의함), denoted by  $H_\alpha$ , where

$$H_\alpha(X) = \frac{1}{1-\alpha} \log_2 \sum_{i=1}^k (\text{Prob}(X = \omega_i))^\alpha, \quad 0 \leq \alpha < \infty. \quad (4.87)$$

Shannon entropy 등에서  $-\log_2$ 의  $-$ 는  $1/(1-\alpha)$ 에 의한 부호임 Formula (4.87) comprises infinitely many different definitions of entropy. Its most important representatives are Shannon entropy  $\alpha = 1$ , min-entropy  $\alpha \rightarrow \infty$ , and collision entropy  $\alpha = 2$ . For a given random variable  $X$ , the entropy values  $H_\alpha(X)$  are monotonically decreasing in  $\alpha$ .

[Proof the monotonically decreasing property of Renyi Entropy]

Let  $p_i = \text{Pr}(X = \omega_i)$  and  $f(\alpha) = \ln \left( \sum_{i=1}^k p_i^\alpha \right)$ . Then,

$$H_\alpha = \frac{f(\alpha)}{\ln 2 \cdot (1-\alpha)}.$$

To show  $H_\alpha$  is monotonically decreasing, we must prove

$$\frac{d}{d\alpha} H_\alpha = \frac{(1-\alpha)f'(\alpha) + f(\alpha)}{\ln 2 \cdot (1-\alpha)^2} \leq 0.$$

Since the denominator is always positive, we only need to show the numerator

$$N(\alpha) := (1-\alpha)f'(\alpha) + f(\alpha) \leq 0$$

where

$$f'(\alpha) = \frac{\sum_{i=1}^k p_i^\alpha \ln p_i}{\sum_{i=1}^k p_i^\alpha}.$$

Let

$$q_i = \frac{p_i^\alpha}{\sum_{j=1}^k p_j^\alpha},$$

so that  $\sum_{i=1}^k q_i = 1$ . Then,

$$f'(\alpha) = \sum_{i=1}^k q_i \ln p_i.$$

On the other hand, by the definition of  $f(\alpha)$ ,

$$\ln q_i = \alpha \ln p_i - f(\alpha)$$

$$f(\alpha) = \alpha \ln p_i - \ln q_i.$$



Therefore,

$$\begin{aligned}\sum_{i=1}^k q_i f(\alpha) &= \sum_{i=1}^k q_i (\alpha \ln p_i - \ln q_i) \\ f(\alpha) &= \alpha \sum_{i=1}^k q_i \ln p_i - \sum_{i=1}^k q_i \ln q_i \quad (\because f(\alpha) \sum_i q_i = f(\alpha) \cdot 1)\end{aligned}$$

Then,

$$\begin{aligned}N(\alpha) &= (1 - \alpha) \left[ \sum_{i=1}^k q_i \ln p_i \right] + \left[ \alpha \sum_{i=1}^k q_i \ln p_i - \sum_{i=1}^k q_i \ln q_i \right] \\ &= \sum_{i=1}^k q_i \ln p_i - \sum_{i=1}^k q_i \ln q_i \\ &= \sum_{i=1}^k q_i \ln \left( \frac{p_i}{q_i} \right).\end{aligned}$$

Next, we applying the inequality  $\ln x \leq x - 1$  for all  $x > 0$ .

$$N(\alpha) \leq \sum_{i=1}^k q_i \left( \frac{p_i}{q_i} - 1 \right) = \sum_{i=1}^k (p_i - q_i).$$

Since  $\sum_{i=1}^k p_i = 1$  and  $\sum_{i=1}^k q_i = 1$ ,

$$N(\alpha) \leq 1 - 1 = 0.$$

Consequently, it follows that  $\frac{dH_\alpha}{d\alpha} \leq 0$ . Thus, Renyi entropy  $H_\alpha(X)$  is monotonically decreasing with respect to  $\alpha$ .

607

The entropy  $H_\alpha(X)$  only depends on the distribution  $\mu$  of  $X$ . Thus, we synonymously use the notation  $H_\alpha(\mu)$ .

608

The special case  $\alpha = 1$  yields the well-known **Shannon entropy**. In particular, L'Hopital's rule then leads to the usual definition of Shannon entropy

$$H_1(X) = H(X) = - \sum_{i=1}^k \text{Prob}(X = \omega_i) \log_2(\text{Prob}(X = \omega_i)). \quad (4.88)$$

If  $\text{Prob}(X = \omega_i) = 0$ , by convention  $\text{Prob}(X = \omega_i) \log_2(\text{Prob}(X = \omega_i)) = 0$ . As usual, we use  $H$  in place of  $H_1$  to indicate the Shannon entropy.

Since  $\log_2 \sum_{i=1}^k (\text{Prob}(X = \omega_i))^1 = \log_2 1 = 0$ ,

$$\begin{aligned} \lim_{\alpha \rightarrow 1} H_\alpha(X) &= \lim_{\alpha \rightarrow 1} H_\alpha(X) \\ &= \lim_{\alpha \rightarrow 1} \frac{\log \sum_{i=1}^k (\text{Pr}(X = \omega_i))^\alpha}{(1 - \alpha)} \\ &= \lim_{\alpha \rightarrow 1} \frac{\frac{\sum_{i=1}^k \log \text{Pr}(X = \omega_i) \cdot (\text{Pr}(X = \omega_i))^\alpha}{\sum_{i=1}^k (\text{Pr}(X = \omega_i))^\alpha}}{-1} \quad (\text{by the L'Hopital's rule}) \\ &= \frac{\sum_{i=1}^k \log \text{Pr}(X = \omega_i) \cdot \text{Pr}(X = \omega_i)}{-\sum_{i=1}^k \text{Pr}(X = \omega_i)} \\ &= - \sum_{i=1}^k \log \text{Pr}(X = \omega_i) \cdot \text{Pr}(X = \omega_i) \end{aligned}$$

609

Shannon entropy  $H = H_1$  is sometimes also called average entropy or simply entropy due to its prevalence in information theory.

610

The parameter  $\alpha = \infty$  yields the **min-entropy**.

$$\lim_{\alpha \rightarrow \infty} H_\alpha(X) = -\log_2 \left( \max_{1 \leq i \leq k} \{\text{Prob}(X = \omega_i)\} \right) = H_{\min}(X). \quad (4.89)$$

Besides  $H_{\min}(\cdot)$ , the notation  $H_\infty(\cdot)$  is also common.

Let  $p_{\max} = \max_{1 \leq i \leq k} \text{Pr}(X = \omega_i)$ . Then,

$$\sum_{i=1}^k p_i^\alpha = p_{\max}^\alpha \sum_{i=1}^k \left( \frac{p_i}{p_{\max}} \right)^\alpha.$$

and

$$\begin{aligned} H_\alpha(X) &= \frac{1}{1-\alpha} \log_2 \left[ p_{\max}^\alpha \cdot \sum_{i=1}^k \left( \frac{p_i}{p_{\max}} \right)^\alpha \right] \\ &= \frac{\alpha}{1-\alpha} \log_2 p_{\max} + \frac{1}{1-\alpha} \log_2 \sum_{i=1}^k \left( \frac{p_i}{p_{\max}} \right)^\alpha. \end{aligned}$$

Therefore,

$$\begin{aligned} \lim_{\alpha \rightarrow \infty} H_\alpha(X) &= \lim_{\alpha \rightarrow \infty} \left[ \frac{\alpha}{1-\alpha} \log_2 p_{\max} + \frac{1}{1-\alpha} \log_2 \sum_{i=1}^k \left( \frac{p_i}{p_{\max}} \right)^\alpha \right] \\ &= -\log_2 p_{\max} + 0 \quad \left( \because \frac{p_i}{p_{\max}} \leq 1 \text{ for all } i \right) \\ &= -\log_2 p_{\max}. \end{aligned}$$

611

Finally,  $H_2$  defines collision entropy. Let  $X$  and  $X'$  be two **independent and identically-distributed** random variables with values in a finite set  $\Omega$ . The collision probability is  $\text{Prob}(X = X') = \sum_{x \in \Omega} (\text{Prob}(X = x))^2$ , and the collision entropy equals

$$H_2(X) = -\log_2 \left( \sum_{\omega \in \Omega} (\text{Prob}(X = \omega))^2 \right). \quad (4.90)$$



612

The inequality (4.91) explains the relation between Shannon entropy, min-entropy, and collision entropy.

$$H_{\min} \leq H_2 \leq H_1, \quad H_{\min} \leq H_2 \leq 2H_{\min}. \quad (4.91)$$

By par. 606, the min-entropy is the most conservative보수적인 entropy measure. Figure 16 plots  $H_1$ ,  $H_2$ , and  $H_{\infty}$  for binary-valued random variables.

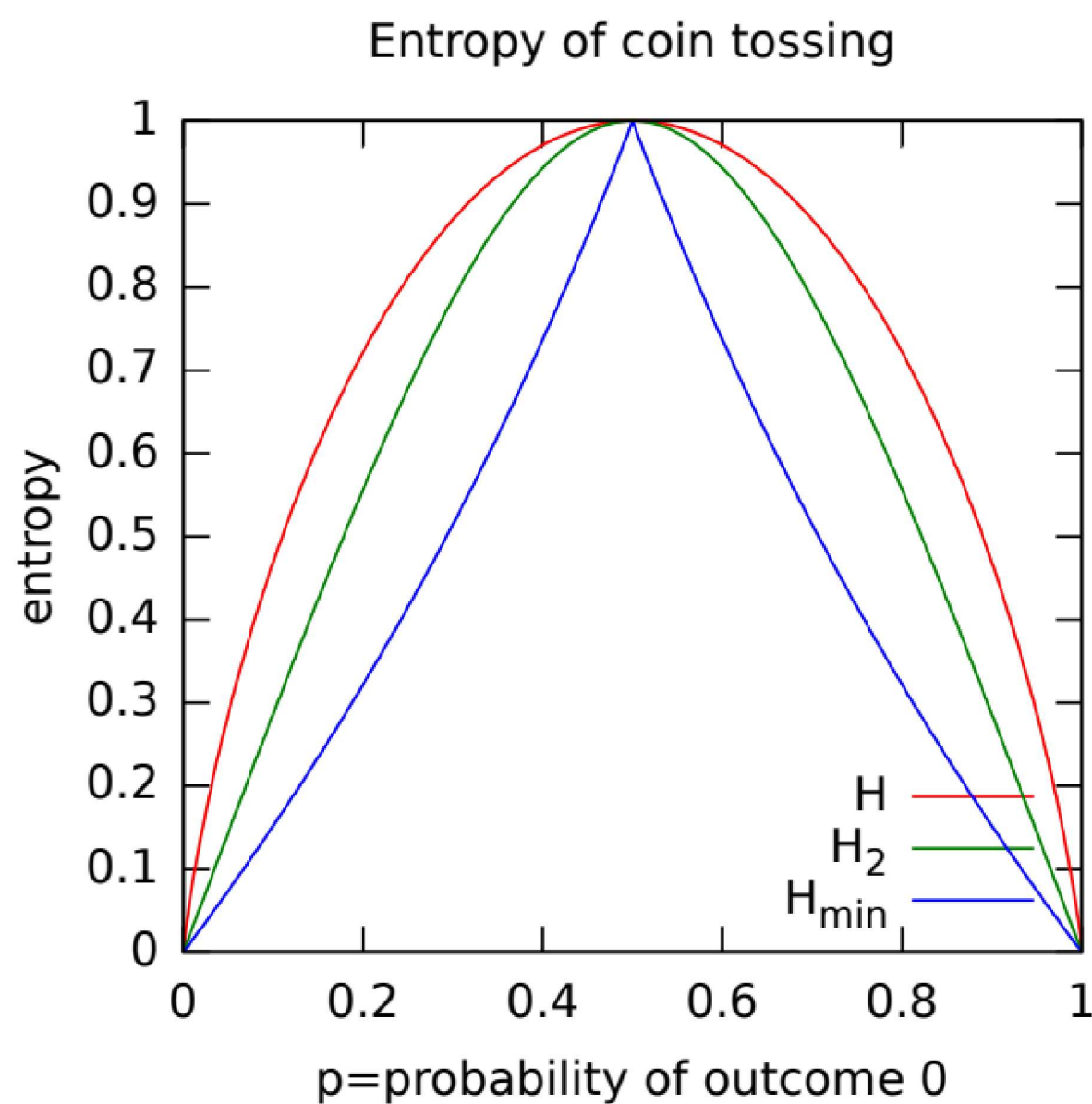


Figure 16: min-entropy  $H_{\min}$  (blue), collision entropy  $H_2$  (green), and Shannon entropy  $H$  (red) for  $B(1, p)$ -distributed random variables

- $H_{\min} \leq H_2$

Since  $0 \leq p_i \leq p_{\max} \leq 1$ ,

$$\begin{aligned} p_i^2 &\leq p_i \cdot p_{\max} \\ \sum_i p_i^2 &\leq \sum_i (p_i \cdot p_{\max}) \\ \sum_i p_i^2 &\leq p_{\max} \sum_i p_i \\ \sum_i p_i^2 &\leq p_{\max} \\ -\log_2 \sum_i p_i^2 &\geq -\log_2 p_{\max} \\ \therefore H_2 &\geq H_{\min}. \end{aligned}$$



- $H_2 \leq 2H_{\min}$

Let  $p_{\max} = p_j$  for  $j$  is the smallest index such that  $p_j = \max_i p_i$ . Then,

$$\begin{aligned} \sum_i p_i^2 &= p_{\max}^2 + \sum_{p_i \neq p_j} p_i^2 \\ \sum_i p_i^2 &\geq p_{\max}^2 \\ \log_2 \sum_i p_i^2 &\geq \log_2 p_{\max}^2 \\ &= 2 \log_2 p_{\max} \\ -\log_2 \sum_i p_i^2 &\leq -2 \log_2 p_{\max} \\ \therefore H_2 &\leq 2H_{\min} \end{aligned}$$

- $H_2 \leq H_1$

We use the property that  $f(x) = -\log_2 x$  is the convex function. So the following inequality holds for any random variable  $Y$ .

$$\begin{aligned} \mathbb{E}[-\log_2 Y] &\geq -\log_2(\mathbb{E}[Y]) \\ \sum_i p_i(-\log_2 p_i) &\geq -\log_2 \sum_i p_i \cdot p_i \\ -\sum_i p_i \log_2 p_i &\geq -\log_2 \sum_i p_i^2 \\ \therefore H_1 &\geq H_2 \end{aligned}$$

Consequently, we have  $H_{\min} \leq H_2 \leq H_1$  and  $H_{\min} \leq H_2 \leq 2H_{\min}$ .

613

The variation distance (a.k.a. **statistical distance**) between two probability distributions  $\tau = (\tau(\omega_1), \dots, \tau(\omega_k))$  and  $\eta = (\eta(\omega_1), \dots, \eta(\omega_k))$  on the finite set  $\Omega$  is defined by

$$\delta(\tau, \eta) := \max_{A \subseteq \Omega} \{|\tau(A) - \eta(A)|\} = \sum_{\omega \in \Omega \mid \tau(\omega) > \eta(\omega)} (\tau(\omega) - \eta(\omega)). \quad (4.92)$$

Note that  $\delta(\tau, \tau) = 0$  and

$$\delta(\tau, \eta) = 0.5 \|\tau - \eta\|_1 \quad (4.93)$$

where  $\|\cdot\|_1$  denotes the  $L^1$ -norm. For  $\nu = (\frac{1}{k}, \dots, \frac{1}{k})$  (uniform distribution on  $\Omega$ ), the variation distance simplifies to

$$\delta(\tau, \nu) = \tau(A_+) - \frac{|A_+|}{k}, \text{ with } A_+ = \left\{a \mid \tau(a) > \frac{1}{k}, a \in \Omega\right\}. \quad (4.94)$$

First, we divide the entire set  $\Omega$  into two subsets based on the magnitude relationship between the two distributions:

$$S_+ = \{\omega \in \Omega \mid \tau(\omega) > \eta(\omega)\},$$

$$S_- = \{\omega \in \Omega \mid \tau(\omega) \leq \eta(\omega)\}.$$

By the fundamental property of probability distributions, the sum of all probabilities is 1. That is,  $\sum_{\omega \in \Omega} \tau(\omega) = 1$  and  $\sum_{\omega \in \Omega} \eta(\omega) = 1$ . Therefore, the sum of their differences over the entire space is 0:

$$\sum_{\omega \in \Omega} (\tau(\omega) - \eta(\omega)) = 1 - 1 = 0.$$

If we split this summation into  $S_+$  and  $S_-$ , it becomes:

$$\sum_{\omega \in S_+} (\tau(\omega) - \eta(\omega)) + \sum_{\omega \in S_-} (\tau(\omega) - \eta(\omega)) = 0$$

and then

$$\sum_{\omega \in S_+} (\tau(\omega) - \eta(\omega)) = - \sum_{\omega \in S_-} (\tau(\omega) - \eta(\omega)) = \sum_{\omega \in S_-} (\eta(\omega) - \tau(\omega)).$$

In other words, the sum of the differences where  $\tau$  is greater than  $\eta$  is exactly equal to the sum of the differences where  $\eta$  is greater than  $\tau$ .

Now, let's look at the  $\max_{A \subseteq \Omega} \{|\tau(A) - \eta(A)|\}$  part. When choosing a subset  $A$ , we must only collect elements where  $\tau(\omega) - \eta(\omega)$  is positive (or negative for  $\eta(\omega) - \tau(\omega)$ ) to maximize the total sum. If any negative (or positive) elements (elements belonging to  $S_-$ ) are included, the total value will decrease. Therefore, the subset that yields the maximum value is

exactly when  $A = S_+$ . As a result,

$$\max_{A \subseteq \Omega} \{|\tau(A) - \eta(A)|\} = \sum_{\omega \in S_+} (\tau(\omega) - \eta(\omega))$$

holds true, which perfectly matches the right-hand side of Equation (4.92).

Equation (4.93) shows that the variation distance is equal to half of the  $L^1$ -norm. The definition of the  $L^1$ -norm is the sum of absolute errors for all elements.

$$\|\tau - \eta\|_1 = \sum_{\omega \in \Omega} |\tau(\omega) - \eta(\omega)|$$

To remove the absolute value signs, we use the  $S_+$  and  $S_-$  sets defined earlier.

$$\|\tau - \eta\|_1 = \sum_{\omega \in S_+} (\tau(\omega) - \eta(\omega)) + \sum_{\omega \in S_-} (\eta(\omega) - \tau(\omega))$$

In the above, we demonstrated that the values of these two summation terms are equal. Both terms are equal to the value of  $\delta(\tau, \eta)$ .

$$\|\tau - \eta\|_1 = \delta(\tau, \eta) + \delta(\tau, \eta) = 2\delta(\tau, \eta)$$

Dividing both sides by 2 yields Equation (4.93).

$$\delta(\tau, \eta) = 0.5\|\tau - \eta\|_1$$

이러한 이유로 variation distance는 분포 간 겹치지 않는 면적의 절반을 의미하기도 함

When the number of elements is  $k$ , the uniform distribution has a probability of  $\nu(\omega) = \frac{1}{k}$  for all elements. Let's apply the definition from Equation (4.92) when  $\eta = \nu$ .

$$\delta(\tau, \nu) = \sum_{\omega \in \Omega | \tau(\omega) > \nu(\omega)} (\tau(\omega) - \nu(\omega))$$

Here, we will define the set of elements satisfying the condition  $\tau(\omega) > \frac{1}{k}$  as  $A_+$ .

$$A_+ = \left\{ a \in \Omega \mid \tau(a) > \frac{1}{k} \right\}$$

Now, we separate the subtraction inside the sum.

$$\delta(\tau, \nu) = \sum_{a \in A_+} \tau(a) - \sum_{a \in A_+} \nu(a)$$

The first term  $\sum_{a \in A_+} \tau(a)$  is, by definition, the probability of  $\tau$  over the set  $A_+$ , which is  $\tau(A_+)$ . The second term  $\sum_{a \in A_+} \nu(a)$  is  $\frac{|A_+|}{k}$ , so combining these finally completes Equation (4.94).



Inequality (4.95) provides an inequality for the variation distance

$$\sum_{x \in \Omega} (\text{Prob}(X = x))^2 \geq \frac{1 + 4(\delta(\tau, \nu))^2}{|\Omega|} \quad ([\text{CFPZ09}]). \quad (4.95)$$

Let  $\Omega$  be a finite set and  $(\alpha_x)_{x \in \Omega}$  a sequence of real numbers. We have:

$$\frac{(\sum_{x \in \Omega} |\alpha_x|)^2}{|\Omega|} \leq \sum_{x \in \Omega} \alpha_x^2. \quad (1)$$

*Proof.* This inequality is a direct consequence of Cauchy-Schwarz inequality:

$$\sum_{x \in \Omega} |\alpha_x| = \sum_{x \in \Omega} |\alpha_x| \cdot 1 \leq \sqrt{\sum_{x \in \Omega} \alpha_x^2} \cdot \sqrt{\sum_{x \in \Omega} 1^2} \leq \sqrt{|\Omega|} \sqrt{\sum_{x \in \Omega} \alpha_x^2}.$$

The result can be deduced easily.  $\square$

If  $X$  is a random variable with values in  $\Omega$  and if we consider that  $\alpha_x = \text{Pr}[X = x]$ , then, since the sum of probabilities is equal to 1, and since  $\text{Col}(X) = \sum_{x \in \Omega} \text{Pr}[X = x]^2$ , we have:

$$\frac{1}{|\Omega|} \leq \text{Col}(X). \quad (2)$$

*Proof.* Let  $d_x = p_x - \frac{1}{|\Omega|}$  be the difference between the probability  $p_x$  of  $X$  assuming the value  $x$  and the uniform distribution  $1/|\Omega|$ . Then,

$$\begin{aligned} \text{Col}(X) &= \sum_{x \in \Omega} p_x^2 = \sum_{x \in \Omega} \left( \frac{1}{|\Omega|} + d_x \right)^2 \\ &= \sum_{x \in \Omega} \frac{1}{|\Omega|^2} + \frac{2}{|\Omega|} \sum_{x \in \Omega} d_x + \sum_{x \in \Omega} d_x^2 \\ &= \frac{1}{|\Omega|} + \sum_{x \in \Omega} d_x^2 \quad (\because \sum_{x \in \Omega} d_x = 0) \\ &\geq \frac{1}{|\Omega|} \end{aligned} \quad \square$$

Let  $X$  be a random variable with values in a set  $\Omega$  of size  $|\Omega|$ ,  $U_\Omega \stackrel{\text{def}}{=} \nu$  be a random variable uniformly distributed over  $\Omega$ , and  $\varepsilon = \delta(X, U_\Omega)$  be the statistical distance between  $X$  and  $U_\Omega$  a random variable uniformly distributed over  $\Omega$ . We have:

$$\sum_{x \in \Omega} (\text{Prob}(X = x))^2 = \text{Col}(X) \geq \frac{1 + 4\varepsilon^2}{|\Omega|}.$$

*Proof.* If  $\varepsilon = 0$  the result is an easy consequence of equation Equation (2). Let assume that  $\varepsilon$  is different from 0. Let define

$$q_x = \frac{\left| \Pr[X = x] - \frac{1}{|\Omega|} \right|}{2\varepsilon} = \frac{|d_x|}{2\varepsilon},$$

then, since

$$\delta = \frac{1}{2} \sum_{x \in \Omega} \left| p_x - \frac{1}{|\Omega|} \right| = \frac{1}{2} \sum_{x \in \Omega} |d_x|$$

by (4.93), we have  $\sum_x q_x = 1$ . According to equation Equation (1) and (2) (1)의  $\alpha_x$  자리에  $q_x$ 를 대입, we have:

$$\begin{aligned} \frac{1}{|\Omega|} &\leq \sum_{x \in \Omega} q_x^2 = \sum_{x \in \Omega} \frac{(\Pr[X = x] - 1/|\Omega|)^2}{4\varepsilon^2} \\ &= \frac{1}{4\varepsilon^2} \left( \sum_{x \in \Omega} \Pr[X = x]^2 - \frac{1}{|\Omega|} \right) \\ &\leq \frac{1}{4\varepsilon^2} \left( \text{Col}(X) - \frac{1}{|\Omega|} \right). \end{aligned}$$

The lemma can be deduced easily. □

615

For the uniform distribution  $\nu$  on  $\Omega$ , we have

$$\max_{\omega \in \Omega} \{\tau(\omega)\} \leq \frac{1}{|\Omega|} + \delta(\tau, \nu). \quad (4.96)$$

According to the equation (4.92) and (4.94), the statistical distance  $\delta(\tau, \nu)$  between the distribution  $\tau$  and  $\nu$  is

$$\delta(\tau, \nu) = \sum_{\omega \in A_+} \left( \tau(\omega) - \frac{1}{|\Omega|} \right)$$

where  $A_+$  is a set satisfying  $\tau(\omega) > \frac{1}{|\Omega|}$ .

Let  $\omega_*$  be the element in  $\Omega$  that has the highest probability under the distribution  $\tau$ .

$$\tau(\omega_*) = \max_{\omega \in \Omega} \{\tau(\omega)\}$$

If  $\tau(\omega_*) \leq \frac{1}{|\Omega|}$ , the highest probability value is less than or equal to the average, which means that all probabilities are the same as  $\frac{1}{|\Omega|}$ . In this case,  $\delta = 0$  and the inequality becomes  $\frac{1}{|\Omega|} \leq \frac{1}{|\Omega|} + 0$ , which is trivially.

If  $\tau(\omega_*) > \frac{1}{|\Omega|}$ , then  $\omega_*$  is included in the set  $A_+$ , which consists of elements whose proba-

bilities are greater than  $\frac{1}{|\Omega|}$ .

Then,

$$\delta(\tau, \nu) = \left( \tau(\omega_*) - \frac{1}{|\Omega|} \right) + \sum_{\omega \in A_+, \omega \neq \omega_*} \left( \tau(\omega) - \frac{1}{|\Omega|} \right).$$

By the definition of  $A_+$ , the probability of each element in  $\tau(\Omega) > \frac{1}{|\Omega|}$ , so  $\tau(\omega) - \frac{1}{|\Omega|} > 0$  for all  $\omega \in A_+$ .

Therefore,

$$\begin{aligned} \delta(\tau, \nu) &\geq \tau(\omega_*) - \frac{1}{|\Omega|} \\ \delta(\tau, \nu) + \frac{1}{|\Omega|} &\geq \tau(\omega_*) \\ \therefore \max_{\omega \in \Omega} \{\tau(\omega)\} &\leq \delta(\tau, \nu) + \frac{1}{|\Omega|} \end{aligned}$$

Inequality (4.96) implies an inequality for the min-entropy of  $\tau$

$$H_{\min}(\tau) \geq -\log_2 \left( \frac{1}{|\Omega|} + \delta(\tau, \nu) \right). \quad (4.97)$$

According to the equation (4.89), the min-entropy is defined as the  $-\log_2$  of the maximum probability value in the distribution  $\tau$ .

$$H_{\min}(\tau) = -\log_2 \left( \max_{\omega \in \Omega} \{\tau(\omega)\} \right).$$

By the equation (4.96),

$$\begin{aligned} \max_{\omega \in \Omega} \{\tau(\omega)\} &\leq \frac{1}{|\Omega|} + \delta(\tau, \nu) \\ -\log_2 \left( \max_{\omega \in \Omega} \{\tau(\omega)\} \right) &\geq -\log_2 \left( \frac{1}{|\Omega|} + \delta(\tau, \nu) \right). \end{aligned}$$

The left-hand side of the above inequality is exactly the definition of  $H_{\min}(\tau)$ , so by substituting it, the proof is completed.  $\square$

Note: Inequality (4.96) and thus also inequality (4.97) are sharp. The upper bound in (4.96) is attained if  $A_+ = \{\omega_*\}$  and  $\tau(\omega_*) = \frac{1}{|\Omega|} + \delta(\tau, \nu)$ .



616

[Renyi entropy of order  $\alpha$ ] For independent random variables  $X_1, \dots, X_m$ , (4.98) follows immediately from definition (4.87). Of course, this result matches with the intuition.

$$H_\alpha(X_1, \dots, X_m) = H_\alpha(X_1) + \dots + H_\alpha(X_m). \quad (4.98)$$

Pars. 630 to 640 consider min-entropy in the context of homogeneous Markov chains, improving the general inequality (4.109).

[Joint entropy]

R.V.  $Z := (X, Y) \sim p(z) = p(x, y) := P[X = x \text{ and } Y = y]$

$$H_\alpha(Z) = H_\alpha(X, Y) = \frac{1}{1-\alpha} \log_2 \sum_{x,y} (\Pr[X, Y])^\alpha$$

For independent  $X, Y$  (with simple notation),

$$\begin{aligned} H(X, Y) &= \frac{1}{1-\alpha} \log_2 \sum_{x,y} (\Pr[X, Y])^\alpha \\ &= \frac{1}{1-\alpha} \log_2 \sum_x \Pr[X]^\alpha \sum_y \Pr[Y]^\alpha \\ &= \frac{1}{1-\alpha} \log_2 \sum_x \Pr[X]^\alpha + \frac{1}{1-\alpha} \log_2 \sum_y \Pr[Y]^\alpha \\ &= H(X) + H(Y). \end{aligned}$$

WLOG,

$$H(X_1, X_2, \dots, X_m) = \sum_{i=1}^m H(X_i).$$



617

[Shannon entropy: joint entropy] Assume that  $X_1$  and  $X_2$  denote (not necessarily independent) random variables that assume values in  $\Omega_1$  and  $\Omega_2$ , respectively. Then the (joint) Shannon entropy of  $X_1$  and  $X_2$  is given by

$$\begin{aligned} H(X_1, X_2) &= - \sum_{i,j=1}^k \text{Prob}(X_1 = \omega_i, X_2 = \omega_j) \log_2(\text{Prob}(X_1 = \omega_i, X_2 = \omega_j)) \\ &= - \sum_{i,j=1}^k \text{Prob}(X_2 = \omega_j \mid X_1 = \omega_i) \text{Prob}(X_1 = \omega_i) \\ &\quad \cdot \log_2(\text{Prob}(X_2 = \omega_j \mid X_1 = \omega_i) \text{Prob}(X_1 = \omega_i)). \end{aligned} \quad (4.99)$$

Since  $\log_2(ab) = \log_2(a) + \log_2(b)$  (here,  $a = \text{Prob}(X_2 = \omega_j \mid X_1 = \omega_i)$  and  $b = \text{Prob}(X_1 = \omega_i)$ ), by rearranging the terms, we obtain the useful functional equation

$$H(X_1, X_2) = H(X_2 \mid X_1) + H(X_1) \quad \text{where} \quad (4.100)$$

$$H(X_2 \mid X_1) = - \sum_{i=1}^k \text{Prob}(X_1 = \omega_i) H(X_2 \mid X_1 = \omega_i). \quad (4.101)$$

$$\begin{aligned} H(X_1, X_2) &= - \sum_{x_1, x_2} \text{Pr}[X_2 = x_2 \mid X_1 = x_1] \text{Pr}[X_1 = x_1] \log_2(\text{Pr}[X_2 = x_2 \mid X_1 = x_1] \text{Pr}[X_1 = x_1]) \\ &= - \sum_{x_1, x_2} \text{Pr}[X_2 = x_2 \mid X_1 = x_1] \text{Pr}[X_1 = x_1] \log_2 \text{Pr}[X_2 = x_2 \mid X_1 = x_1] \\ &\quad - \sum_{x_1, x_2} \text{Pr}[X_2 = x_2 \mid X_1 = x_1] \text{Pr}[X_1 = x_1] \log_2 \text{Pr}[X_1 = x_1] \\ &= \sum_{x_1} \text{Pr}[X_1 = x_1] \left( - \sum_{x_2} \text{Pr}[X_2 = x_2 \mid X_1 = x_1] \log_2 \text{Pr}[X_2 = x_2 \mid X_1 = x_1] \right) \\ &\quad - \sum_{x_1} \text{Pr}[X_1 = x_1] \log_2 \text{Pr}[X_1 = x_1] \sum_{x_2} \text{Pr}[X_2 = x_2 \mid X_1 = x_1] \\ &= \sum_{x_1} \text{Pr}[X_1 = x_1] H(X_2 \mid X_1 = x_1) + H(X_1) \\ &= H(X_2 \mid X_1) + H(X_1) \quad \text{by eq.(4.101)} \end{aligned}$$

because the sum of marginal dist.  $\text{Pr}[X_2 \mid X_1]$  is 1.

Here,  $H(X_2 \mid X_1 = x_i)$  denotes the entropy of  $X_2$  under the condition that  $X_1$  assumes the value  $x_i$ . The term  $H(X_2 \mid X_1)$  is the conditional (Shannon) entropy of  $X_2$  und  $X_1$ . It quantifies the average entropy of  $X_2$  if  $X_1$  is known. Clearly,

$$\min_{\omega_1 \in \Omega_1} \{H(X_2 | X_1 = \omega_1)\} \leq H(X_2 | X_1) \leq \max_{\omega_1 \in \Omega_1} \{H(X_2 | X_1 = \omega_1)\}. \quad (4.102)$$

We focus on  $H(X_2 | X_1 = x_1)$  of  $H(X_2 | X_1) = \sum_{x_1} \Pr[X_1 = x_1] H(X_2 | X_1 = x_1)$ .

The following is trivial.

$$\min_{\omega_1 \in \Omega_1} \{H(X_2 | X_1 = \omega_1)\} \leq H(X_2 | X_1 = x_1) \leq \max_{\omega_1 \in \Omega_1} \{H(X_2 | X_1 = \omega_1)\}.$$

For probability  $\Pr[X_1 = x_1]$ ,

$$\begin{aligned} \Pr[X_1 = x_1] \min_{\omega_1 \in \Omega_1} \{H(X_2 | X_1 = \omega_1)\} &\leq \Pr[X_1 = x_1] H(X_2 | X_1 = x_1) \\ &\leq \Pr[X_1 = x_1] \max_{\omega_1 \in \Omega_1} \{H(X_2 | X_1 = \omega_1)\}, \end{aligned}$$

and

$$\begin{aligned} \sum_{x_1} \Pr[X_1 = x_1] \min_{\omega_1 \in \Omega_1} \{H(X_2 | X_1 = \omega_1)\} &\leq \sum_{x_1} \Pr[X_1 = x_1] H(X_2 | X_1 = x_1) \\ &\leq \sum_{x_1} \Pr[X_1 = x_1] \max_{\omega_1 \in \Omega_1} \{H(X_2 | X_1 = \omega_1)\}. \end{aligned}$$

Since  $\min_{\omega_1 \in \Omega_1} \{H(X_2 | X_1 = \omega_1)\}$  and  $\max_{\omega_1 \in \Omega_1} \{H(X_2 | X_1 = \omega_1)\}$  are independent with  $\Pr[X_1 = x_1]$ ,

$$\begin{aligned} \min_{\omega_1 \in \Omega_1} \{H(X_2 | X_1 = \omega_1)\} \sum_{x_1} \Pr[X_1 = x_1] &\leq \sum_{x_1} \Pr[X_1 = x_1] H(X_2 | X_1 = x_1) \\ &\leq \max_{\omega_1 \in \Omega_1} \{H(X_2 | X_1 = \omega_1)\} \sum_{x_1} \Pr[X_1 = x_1], \end{aligned}$$

moreover

$$\min_{\omega_1 \in \Omega_1} \{H(X_2 | X_1 = \omega_1)\} \leq \sum_{x_1} \Pr[X_1 = x_1] H(X_2 | X_1 = x_1) \leq \max_{\omega_1 \in \Omega_1} \{H(X_2 | X_1 = \omega_1)\}.$$

618

The functional equation (4.100) generalizes to several random variables. More precisely,

$$H(X_1, \dots, X_{m+1}) = H(X_1, \dots, X_m) + H(X_{m+1} \mid X_1, \dots, X_m). \quad (4.103)$$

Formula (4.103) is well-known and very useful for the evaluation of physical RNGs.

For the r.v.s  $X_1, X_2, \dots, X_n$

$$H(X_1, \dots, X_n) = \sum_{i=1}^n H(X_i \mid X_{i-1}, \dots, X_1).$$

*Proof.*

$$\begin{aligned} p(x_1, x_2) &= p(x_1)p(x_2 \mid x_1) \\ p(x_1, x_2, x_3) &= p(x_1, x_2)p(x_3 \mid x_1, x_2) = p(x_1)p(x_2 \mid x_1)p(x_3 \mid x_1, x_2) \\ &\vdots \\ p(x_1, x_2, \dots, x_n) &= \prod_{i=1}^n p(x_i \mid x_{i-1}, \dots, x_1) \end{aligned}$$

Therefore,

$$\begin{aligned} H(X_1, \dots, X_n) &= - \sum_{x_1, \dots, x_n} p(x_1, \dots, x_n) \log p(x_1, \dots, x_n) \\ &= - \sum_{x_1, \dots, x_n} p(x_1, \dots, x_n) \log \prod_{i=1}^n p(x_i \mid x_{i-1}, \dots, x_1) \\ &= - \sum_{x_1, \dots, x_n} p(x_1, \dots, x_n) \sum_{i=1}^n \log p(x_i \mid x_{i-1}, \dots, x_1) \\ &= - \sum_{i=1}^n \sum_{x_1, \dots, x_n} p(x_1, \dots, x_n) \log p(x_i \mid x_{i-1}, \dots, x_1) \\ &= - \sum_{i=1}^n \sum_{x_1, \dots, x_i} p(x_1, \dots, x_i) \log p(x_i \mid x_{i-1}, \dots, x_1) \\ &= \sum_{i=1}^n H(X_i \mid X_{i-1}, \dots, X_1) \\ H(X_1, \dots, X_{m+1}) &= \sum_{i=1}^{m+1} H(X_i \mid X_{i-1}, \dots, X_1) \\ &= \sum_{i=1}^m H(X_i \mid X_{i-1}, \dots, X_1) + H(X_{m+1} \mid X_m, \dots, X_1) \\ &= H(X_1, \dots, X_m) + H(X_{m+1} \mid X_m, \dots, X_1) \quad \square \end{aligned}$$



We can prove this to another way using the (4.100), or

$$H(X, Y | Z) = H(X | Z) + H(X | Y, Z)$$

so

$$\begin{aligned} H(X_1, \dots, X_n) &= H(X_1) + H(X_2, \dots, X_n | X_1) \\ &= H(X_1) + H(X_2 | X_1) + \dots + H(X_n | X_{n-1}, \dots, X_1) \\ &= \sum_{i=1}^n H(X_i | X_{i-1}, \dots, X_1). \end{aligned}$$

619

Depending on the distribution of the random variables  $X_1, X_2, \dots$ , the formula for the conditional entropy in (4.103) may simplify considerably. In particular,

$$H(X_{m+1} | X_1, \dots, X_m) \leq H(X_{m+1}), \quad (4.104)$$

$$H(X_{m+1} | X_1, \dots, X_m) = H(X_{m+1}) \quad \text{if } X_1, X_2, \dots \text{ are independent,} \quad (4.105)$$

$$H(X_{m+1} | X_1, \dots, X_m) = H(X_{m+1} | X_m) \quad \text{if } X_1, X_2, \dots \text{ are Markovian.} \quad (4.106)$$

According to (4.103),

$$H(X_1, \dots, X_{m+1}) = H(X_1, \dots, X_m) + H(X_{m+1} | X_1, \dots, X_m).$$

Therefore,

$$H(X_{m+1} | X_1, \dots, X_m) \leq H(X_1, \dots, X_{m+1}) \leq H(X_{m+1}),$$

Jensen inequality로도 증명 가능: If  $\phi(t) = -\log t$ ,  $\phi''(t) < 0$  for all  $t$ , then  $E(\phi(Z)) \geq \phi(E(Z))$ .

and for independent  $X_1, X_2, \dots$ ,

$$\begin{aligned} H(X_{m+1} | X_1, \dots, X_m) &= H(X_1) \cdots H(X_m) + H(X_{m+1}) \quad (\text{by (4.98)}) \\ &= H(X_1, \dots, X_m) + H(X_{m+1} | X_1, \dots, X_m) \quad (\text{by (4.103)}) \\ \therefore H(X_{m+1}) &= H(X_{m+1} | X_1, \dots, X_m). \end{aligned}$$

Moreover, (4.106) is trivial.

620

Assume that  $X_1, X_2, \dots$  defines a homogeneous ergodic Markov chain with limiting distribution  $\nu$ . If the Markov chain has already (at least almost) reached equilibrium, then (4.106) simplifies to

$$H(X_{m+1} \mid X_m) = \sum_{i=1}^k \nu(\omega_i) H(X_{m+1} \mid X_m = \omega_i) \quad (4.107)$$

where  $H(X_{m+1} \mid X_m = \omega_i)$  depends on the  $i^{th}$  row of the transition matrix.

621

If the sequence of random variables  $X_1, X_2, \dots$  is stationary, for each shift parameter  $t$  and for each  $h \leq m$

$$H(X_{m+t+1} \mid X_{m+t-h+1}, \dots, X_{m+t}) = H(X_{m+1} \mid X_{m-h+1}, \dots, X_m). \quad (4.108)$$